

The Dissection of Expression Quantitative Trait Locus Hotspots

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ABSTRACT Studies of the genetic loci that contribute to variation in gene expression frequently identify loci with broad effects on gene expression: expression quantitative trait locus hotspots. We describe a set of exploratory graphical methods as well as a formal likelihood-based test for assessing whether a given hotspot is due to one or multiple polymorphisms. We first look at the pattern of effects of the locus on the expression traits that map to the locus: the direction of the effects and the degree of dominance. A second technique is to focus on the individuals that exhibit no recombination event in the region, apply dimensionality reduction (e.g., with linear discriminant analysis), and compare the phenotype distribution in the nonrecombinant individuals to that in the recombinant individuals: if the recombinant individuals display a different expression pattern than the nonrecombinant individuals, this indicates the presence of multiple causal polymorphisms. In the formal likelihood-based test, we compare a two-locus model, with each expression trait affected by one or the other locus, to a single-locus model. We apply our methods to a large mouse intercross with gene expression microarray data on six tissues.

KEYWORDS eQTL; pleiotropy; multivariate analysis; data visualization; gene expression

THERE is a long history of efforts to map the genetic loci [called quantitative trait loci (QTL)] that contribute to variation in quantitative traits in experimental organisms, particularly to learn about the etiology of disease (Broman 2001; Jansen 2007). But it remains difficult to identify the genes underlying QTL (Nadeau and Frankel 2000). There has been much interest recently in measuring gene expression in disease-relevant tissues in QTL experiments as a way to speed the process from QTL to gene (Jansen and Nap 2001; Albert and Kruglyak 2015). The genetic control of gene expression is itself of great interest.

Expression quantitative trait loci (eQTL) analysis attempts to find the genomic locations that influence variation in gene expression levels [messenger RNA (mRNA) abundances].

eQTL near the genomic location of the influenced gene are called *local eQTL*, and eQTL far away from the influenced gene are called *trans-eQTL*. When a genomic region influences the expression of many genes, the region is called a *trans-eQTL hotspot*.

eQTL hotspots have been observed in many genetic studies (e.g., Brem *et al.* 2002; Schadt *et al.* 2003; Yvert *et al.* 2003; Chesler *et al.* 2005), and they are of particular interest because gene expressions mapping to the same location may indicate the existence of a genetic regulator.

Batch effects (artifacts arising from technical or environmental factors) are common in microarray experiments. This has led to the development of a number of methods to control for underlying confounding factors (Leek and Storey 2007; Kang *et al.* 2008; Listgarten *et al.* 2010; Stegle *et al.* 2010; Fusi *et al.* 2012; Gagnon-Bartsch and Speed 2012). However, these methods generally cannot distinguish *trans-eQTL* hotspots from batch effects. There is some controversy about whether *trans-eQTL* hotspots are themselves artifacts and whether one should control for them, as one does for batch effects, in eQTL analysis (e.g., Breitling *et al.* 2008; Kang *et al.*

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2008). In many cases, however, the associations between genotype and expression phenotypes are extremely strong (with a LOD score > 100), which largely precludes the possibility of a batch-effect artifact because the strength of association between batch and genotype in the region would have to be even stronger.

In Tian *et al.* (2015), we considered a large mouse intercross between the strains C57BL/6J (abbreviated B6) and BTBR $T^+ tf/J$ (abbreviated BTBR), with gene expression microarray data on six tissues (*i.e.*, adipose, gastrocnemius muscle, hypothalamus, pancreatic islets, kidney, and liver), and mapped a *trans*-eQTL hotspot to a 298-kb region containing just three genes. This sort of fine-mapping approach is meaningful only if the hotspot is due to polymorphisms in a single gene.

This raises an important question about *trans*-eQTL hotspots: are they the result of polymorphisms in a single gene, or are there multiple underlying genes? In other words, is there complete pleiotropy, or are there multiple linked eQTL? Methods for testing pleiotropy *vs.* tight linkage of multiple QTL (Jiang and Zeng 1995; Knott and Haley 2000) do not scale well to the case of the very large number of expression traits that map to a *trans*-eQTL hotspot. We developed a likelihood-based test that is a variation on the method of Knott and Haley (2000), as well as a number of exploratory data visualizations, to test whether multiple eQTL underlie a hotspot. We apply our approaches to the data considered in Tian *et al.* (2015).

Materials and Methods

We focus on the case of an intercross between two inbred strains, B and R (these labels were chosen to match the strains used in the application later). We assume dense marker genotype data and genome-wide gene expression phenotype data (*e.g.*, from microarrays or RNA-Seq). We first perform a genome scan to identify QTL, considering each expression trait individually. We use Haley-Knott regression (Haley and Knott 1992) for this purpose, for the sake of speed. For each expression trait and each chromosome, we consider the location of the single largest LOD score, provided that it exceeds a significance threshold that adjusts for the genome scan but not the search across expression traits.

We count the number of expression traits that show a *trans*-eQTL within a sliding window (*e.g.*, of 10 cM) across the genome and use peaks in these counts to define *trans*-eQTL hotspots. We then focus on one such hotspot and on the set of expression traits that map to an interval centered at the peak count.

We then ask: are the multiple expression traits that map to this *trans*-eQTL hotspot all affected by a common eQTL, or could there be multiple causal polymorphisms in the region? We have developed a set of exploratory data visualizations to address this question, as well as a formal likelihood-based test. We prefer to exclude expression traits whose genomic position is near (or even on the same chromosome as) the

hotspot of interest because these may be driven by separate local eQTL.

Exploratory data visualizations

We first consider the pattern of effects of the locus on the expression traits that map to the region: the direction of the effects and the degree of dominance. We use a pair of data visualizations: a plot of the signed LOD score (with the sign taken from the estimated additive effect) *vs.* the estimated eQTL location for each expression trait analyzed separately. That is, for each expression trait, we find the largest LOD score on the chromosome, multiply it by ± 1 according to the sign of the estimated additive effect of the locus, and plot this signed LOD score *vs.* the location at which that maximum LOD score was attained. If there are two nearby loci with effects in opposite directions, they may be revealed by this plot.

In addition, we plot the estimated dominance effect against the estimated additive effect for each expression trait. Let B and R denote the two alleles in the cross, and let $\hat{\mu}_{BB}$, $\hat{\mu}_{BR}$, and $\hat{\mu}_{RR}$ denote the average expression levels for genotypes BB, BR, and RR, respectively. We estimate the additive effect as half the difference between the two homozygotes, that is, $\hat{a} = (\hat{\mu}_{RR} - \hat{\mu}_{BB})/2$, and the dominance effect as the difference between the heterozygote and the midpoint between the two homozygotes, that is, $\hat{d} = \hat{\mu}_{BR} - (\hat{\mu}_{BB} + \hat{\mu}_{RR})/2$. We then plot $\hat{d}$ *vs.* $\hat{a}$ for all expression traits mapping to the hotspot. If there are two nearby loci with different inheritance patterns (*e.g.*, one has additive allele effects and the other has an allele that is dominant), they may be revealed by this plot.

As a second technique, we consider the individuals that have no recombination event in the region. For these individuals, we know their eQTL genotype. We apply linear discriminant analysis (LDA) (Hastie *et al.* 2009, Chapter 4) to the top 100 traits with the largest LOD scores and make a scatter plot of the first and second linear discriminants; this should show three distinct clusters (or, for a fully dominant locus, two clusters). We calculate the linear discriminants for individuals that show a recombination event in the region and add them as points to the plot. If the recombinant individuals fall within the clusters defined by the nonrecombinant individuals, this is consistent with there being a single causal locus. If, however, the recombinants look distinctly different from the nonrecombinants, then multiple polymorphisms are indicated.

The basic idea underlying this visualization is that the nonrecombinant individuals can be used to derive an estimate of the conditional distribution of the multivariate expression phenotype given the eQTL genotype. We use LDA as a dimension-reduction technique. The goal of the visualization is to compare the expression pattern in the recombinant and nonrecombinant individuals. If there is a single eQTL, the recombinant individuals should look no different from the nonrecombinant individuals; if there is a difference, we can conclude that there are multiple eQTL.

Formal statistical test

To formally assess evidence of multiple linked loci *vs.* complete pleiotropy at a *trans*-eQTL hotspot, we developed a likelihood-based test to compare the null hypothesis of a single eQTL affecting all expression traits to the hypothesis of two eQTL, with each expression trait affected by one or the other eQTL (but not both). The approach can handle only a limited number of expression traits, so we focus on the 50 traits with the largest LOD scores (when considered individually) in the interval centered at the hotspot.

We assume that the traits follow a multivariate normal distribution, conditional on eQTL genotype, and apply the multivariate QTL analysis method of Knott and Haley (2000). For a given QTL model, we have $Y = X\beta + E$, where Y is an $n \times p$ matrix of phenotypes, with n as the number of F_2 individuals and p as the number of traits, X is an $n \times q$ matrix of covariates (including additive covariates, interactive covariates, genotype probabilities for the position under investigation, and the interactive covariates times the genotype probabilities), and β is a $q \times p$ matrix of coefficients. We obtain $\hat{\beta} = (X'X)^{-1}X'Y$, calculate the matrix of estimated residuals $\hat{E} = Y - X\hat{\beta}$, and calculate the residual sum of squares matrix $RSS = \hat{E}'\hat{E}$. The LOD score is $(n/2)\log_{10}\{|RSS_0|/|RSS|\}$, where $|RSS|$ denotes the determinant of the RSS matrix, and RSS_0 is the residual sum of squares matrix for the null model (with additive covariates but no genotype probabilities or interactive covariates).

We perform a QTL scan over the interval; at each putative QTL location, denoted λ , we calculate the LOD score $LOD_1(\lambda)$, comparing this single-QTL model to the null model of no QTL. Let $M_1 = \max_{\lambda} LOD_1(\lambda)$.

We compare this to a two-QTL model in which each expression trait is affected by one or the other QTL but not both. In principle, one would need to consider, with p expression traits, 2^{p-1} possible assignments of the expression traits to the left and right QTL. This is a prohibitively large number, so we make an approximation: we sort the expression traits according to their estimated QTL location when considered individually, and we consider only the $p - 1$ cut points of this list. We randomly order any expression traits that map to the same position. For each cut point, we perform a two-dimensional (2D) scan over possible two-QTL models and calculate $LOD_2^{(c)}(\lambda_1, \lambda_2)$, comparing the two-QTL model, with QTL at positions λ_1 and λ_2 and with the first c expression traits affected by the QTL at λ_1 and the last $p - c$ traits affected by the QTL at λ_2 to the null model of no QTL. Let $M_2^{(c)} = \max_{\lambda_1, \lambda_2} LOD_2^{(c)}(\lambda_1, \lambda_2)$, and let $M_2 = \max_c M_2^{(c)}$. The estimated cut point is $\hat{c} = \arg\max_c M_2^{(c)}$, and the estimated QTL positions are $(\hat{\lambda}_1, \hat{\lambda}_2) = \arg\max_{\lambda_1, \lambda_2} LOD_2^{(\hat{c})}(\lambda_1, \lambda_2)$. As evidence for the presence of two QTL, we consider the $\log_{10}$ likelihood ratio $LOD_{2v1} = M_2 - M_1$.

The exhaustive 2D scan of $LOD_2^{(c)}(\lambda_1, \lambda_2)$ is computationally intensive. We can accelerate this calculation by iteratively searching for the maximum on each of the two dimensions. There is no guarantee that this will converge

to the overall maximum, especially when there are multiple modes in the two-QTL likelihood surface, but in practice, we have found that this algorithm works well. As a starting point of this iterative search, we can use either the estimated QTL location under the single-QTL model or a randomly selected position.

Statistical significance: To assess the statistical significance of the result, we need an approximation of the distribution of the test statistic under the null hypothesis of a single QTL. We consider two approaches: a parametric bootstrap and a stratified permutation test.

In the parametric bootstrap test, we simulate new phenotype data using the estimated single-QTL model. In the stratified permutation test, we randomly permute the rows in the phenotype data relative to the rows in the genotype data within each QTL genotype group. When there are unmeasured genotypes at the inferred QTL, we infer the QTL genotype for each individual to be that with maximum probability, conditional on the observed marker data. These conditional QTL genotype probabilities are calculated by a hidden Markov model (Broman and Sen 2009, Appendix D).

For each procedure, we generate 1000 data sets, perform the full likelihood analysis (the scan for the single-QTL model and the 2D scan for the two-QTL model for each possible partition of the traits) and calculate the test statistic. The P -value for the test is taken to be the proportion of simulated or permuted data sets with a test statistic that is greater than or equal to the observed test statistic.

Visualizations: To visualize the results for the two-QTL model, we plot profile LOD score curves for the left and right QTL using the estimated cut point $\hat{c}$ for the expression traits into those mapping to the left QTL and those mapping to the right QTL. For the left QTL, we plot the slice $LOD_2^{(\hat{c})}(\lambda_1, \hat{\lambda}_2)$ against λ_1 for varying values of λ_1 . Similarly, for the right QTL, we plot $LOD_2^{(\hat{c})}(\hat{\lambda}_1, \lambda_2)$ against λ_2 for varying values of λ_2 . These sorts of profile LOD score curves follow from an innovation of Zeng *et al.* (2000).

As a further diagnostic plot, we consider the statistic $LOD_{2v1}^{(c)} = M_2^{(c)} - M_1$ as a function of the cut point c . This is evidence for two *vs.* one QTL, for a given cut point of the expression traits into those that map to the left QTL and those that map to the right QTL. This displays the evidence for two *vs.* one QTL as well as the evidence for a particular split of the expression traits.

Application

To illustrate our methods for the dissection of *trans*-eQTL hotspots, we consider a large mouse F_2 intercross (Tian *et al.* 2015) with gene expression microarray data on six tissues. The experiment was carried out to identify genes and pathways that contribute to obesity-induced type II diabetes. Greater than 500 offspring were generated from an F_2 intercross between diabetes-resistant (C57BL/6J, abbreviated B6) and diabetes-susceptible (BTBR $T^+ tf/J$, abbreviated

BTBR) mouse strains. All mice were genetically obese through introgression of the leptin mutation (*Lep^{ob/ob}*) and were killed at 10 weeks of age, the age when essentially all BTBR *ob/ob* mice are diabetic.

Mice were genotyped with the Affymetrix 5K GeneChip system. After data cleaning, there were 519 F₂ mice genotyped at 2057 informative markers. Gene expression was assayed with custom two-color ink-jet microarrays manufactured by Agilent Technologies (Palo Alto, CA). Six tissues from each F₂ mouse were used for expression profiling: adipose, gastrocnemius muscle (abbreviated gastroc), hypothalamus (abbreviated hypo), pancreatic islets (abbreviated islet), kidney, and liver. Tissue-specific messenger RNA (mRNA) pools were used for the reference channel, and gene expression was quantified as the ratio of the mean log₁₀ intensity (mlratio). For further details, see Keller *et al.* (2008). In the final data set, there were 519 mice with gene expression data on at least one tissue (487 for adipose, 490 for gastroc, 369 for hypo, 491 for islet, 474 for kidney, and 483 for liver). The microarray included 40,572 total probes; we focused on the 37,797 probes with known location on one of the autosomes or the X chromosome.

QTL analysis

For QTL analysis, we first transformed the gene expression measures for each microarray probe in each of the six tissues to normal quantiles, taking $\Phi^{-1}[(R_i - 0.5)/n]$, where Φ is the cumulative distribution function for the standard normal distribution, and R_i is the rank in $\{1, \dots, n\}$ for mouse i . We then performed single-QTL genome scans, separately for each probe in each tissue, by Haley-Knott regression (Haley and Knott 1992) with microarray batch as an additive covariate and with sex as an interactive covariate (*i.e.*, allowing the effects of QTL to be different in the two sexes). Calculations were performed at the genetic markers and at a set of pseudomarkers inserted into marker intervals, selected so that adjacent positions were separated by ≤ 0.5 cM. We calculated conditional genotype probabilities given observed multipoint marker genotype data using a hidden Markov model assuming a genotyping error rate of 0.2% and with genetic distances converted to recombination fractions with the Carter-Falconer map function (Carter and Falconer 1951). Calculations were performed with R/qtl (Broman *et al.* 2003), an add-on package to the general statistical software R (R Core Team 2015).

For each probe in each tissue, we focused on the single largest LOD score peak on each chromosome and on LOD score peaks ≥ 5 (corresponding to genome-wide significance at the 5% level for a single probe in a single tissue, as determined by computer simulations under the null hypothesis of no QTL).

Data availability

The data are available at the QTL Archive, now part of the Mouse Phenome Database, at <http://phenome.jax.org/db/q?rtn=projects/projdet&reqprojid=532>.

Results

The inferred eQTL for all genes with a LOD score ≥ 5 are displayed in Figure 1, with the y-axis corresponding to the genomic position of the microarray probe and the x-axis corresponding to the estimated eQTL position. As expected, we see a large number of local eQTL along the diagonal for each tissue-specific panel. These local eQTL correspond to genes for which expression or mRNA abundance is strongly associated with genotype near their genomic position.

In addition to the local eQTL, there are a number of prominent vertical bands: genomic loci that influence the expression of genes located throughout the genome. These are the *trans*-eQTL hotspots. Overall, we detected many more *trans*-eQTL than local eQTL. The *trans*-eQTL hotspots can show either remarkable tissue specificity or be observed in multiple tissues. For example, a locus near the centromere of chromosome 17, at 11.7 cM, shows effects in all tissues. In contrast, the *trans*-eQTL hotspot located at the distal end of chromosome 6 was observed only in pancreatic islets.

To define *trans*-eQTL hotspots of potential interest, we focused on a more conservative threshold for eQTL: LOD score ≥ 10 . We further excluded local eQTL, defined here to be those for which the distance between the gene's genomic position and its inferred eQTL position was < 10 cM. We then counted the number of expression traits with a *trans*-eQTL in a sliding interval of length 10 cM (Supporting Information, Figure S1).

For each *trans*-eQTL of interest, we widened the interval to be considered beyond that initial 10-cM window to consider the interval in which the count of expression traits with eQTL was > 50 and then padded this further by adding 5 cM on either end. We will focus on a set of six hotspots: adipose chromosome 1 at 39 cM, adipose chromosome 10 at 48 cM, islet chromosome 2 at 75 cM, islet chromosome 6 at 91 cM, kidney chromosome 13 at 68 cM, and liver chromosome 17 at 18 cM. Results for additional hotspots are displayed in File S1.

Visualization of QTL effects

We first consider the estimated effects of a locus on the expression traits that map to the region (Figure 2). In the left panels, we display the signed LOD score (with positive values indicating that the BTBR allele is associated with larger average expression and negative values indicating that the B6 allele is associated with larger average expression) vs. the estimated eQTL location. In the right panels, we plot the estimated dominance effect vs. the estimated additive effect for all transcripts mapping to the hotspot. The key value in these visualizations is for the case that two linked QTL show distinct inheritance patterns.

The islet chromosome 6 hotspot, at 92 cM, shows approximately equal numbers of expression traits for which the BTBR allele causes an increase or decrease in gene expression (Figure 2A), and the allele effects are approximately additive (Figure 2B), with estimated dominance effect near 0. These

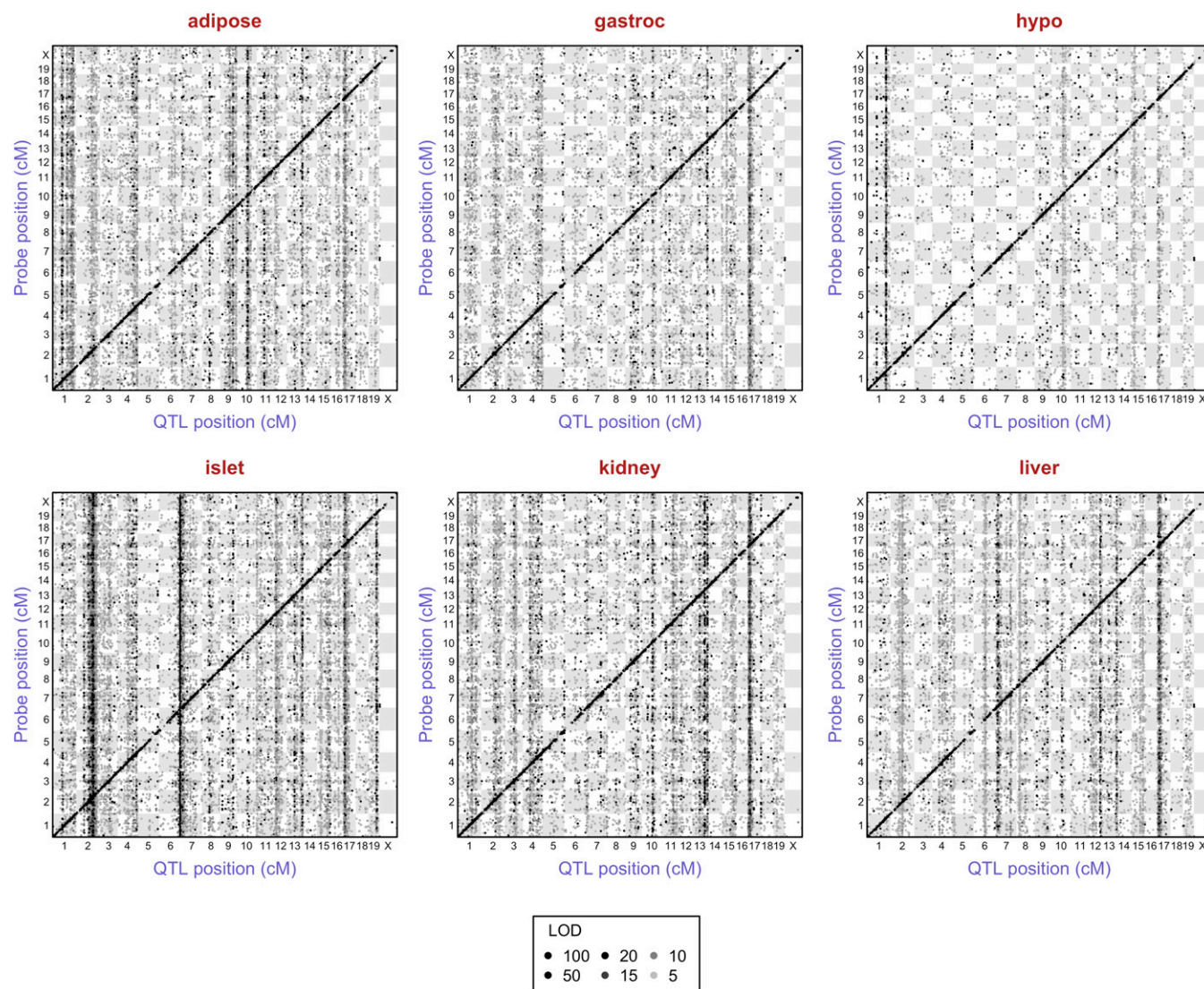


Figure 1 Inferred eQTL with LOD score ≥ 5 by tissue. Points correspond to peak LOD scores from single-QTL genome scans with each microarray probe with known genomic position. The y-axis is the position of the probe, and the x-axis is the inferred QTL position. Points are shaded according to the corresponding LOD score, although we threshold at 100: all points with a LOD score ≥ 100 are black. [A version of this figure appeared as figure 1 in Tian *et al.* (2015).]

results are consistent with there being a single QTL. In Tian *et al.* (2015), this locus was resolved to a 298-kb interval containing just three genes, with good evidence for *Slco1a6* as the causal gene.

The kidney chromosome 13 hotspot, at 68 cM, shows clear evidence for two QTL. In Figure 2C, we see that for expression traits mapping to ~ 57 cM, the BTBR allele is associated with a decrease in expression, while for traits mapping to ~ 68 cM, the BTBR allele is predominantly associated with an increase in expression, although with some traits having effects in the opposite direction. From Figure 2D, we can infer that for the traits mapping to ~ 57 cM, the B6 allele is nearly dominant ($d \approx -a$, along the line with slope -1), while for the traits mapping to ~ 68 cM, the BTBR allele is dominant ($d \approx a$, along the line with slope $+1$).

The islet chromosome 2 hotspot, at 75 cM, shows expression traits with high LOD scores across a broad region (Figure 2E). For traits mapping to 70–75 cM, the B6 allele is associated with increased expression, while for traits mapping to 55–60 cM, the effect is in the opposite direction. The allele effects are nearly additive for all expression traits (Figure 2F).

The liver chromosome 17 hotspot, at 11 cM, has approximately equal numbers of traits with effects in each direction (Figure 2G), and the B6 alleles appears to be nearly dominant in most cases (Figure 2H). The adipose chromosome 10 hotspot, at 48 cM, is similar, with effects in both directions (Figure 2I) and with the B6 allele being nearly dominant (Figure 2J).

The adipose chromosome 1 hotspot, at 43 cM, again shows evidence for two QTL. For the expression traits mapping to 38–40 cM, the B6 allele is associated with increased

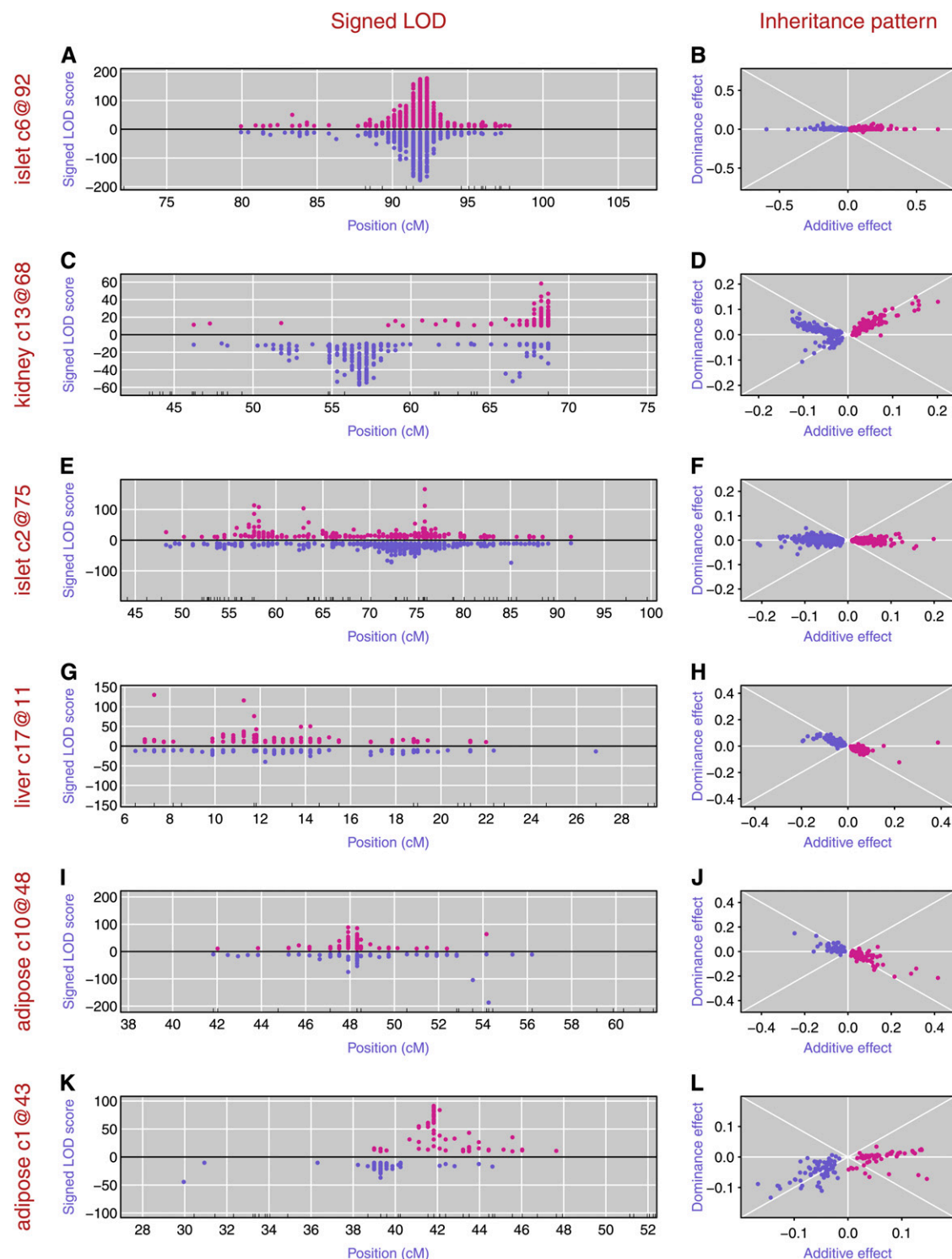


Figure 2 Visualizations of the QTL effects on the multiple expression traits that map with LOD score ≥ 10 to a *trans*-eQTL hotspot. Each row is a hotspot. The left panels are scatter plots of signed LOD scores (with positive values indicating that the BTBR allele is associated with larger average gene expression and negative values indicating that the B6 allele is associated with larger average gene expression) vs. the estimated QTL location. Each point is a single expression trait. Tick marks at the bottom indicate the locations of the genetic markers. The right panels are scatter plots of the estimated dominance effects vs. the estimated additive effects.

expression (Figure 2K), but the BTBR allele appears dominant (Figure 2L). For traits mapping to 42–46 cM, however, the BTBR allele is associated with increased expression, and the allele effects appear additive.

In summary, for two of these six hotspots, these visualizations of the estimated QTL effects provide good evidence for two QTL. In one case (kidney chromosome 13), the two QTL are well separated, but in the other

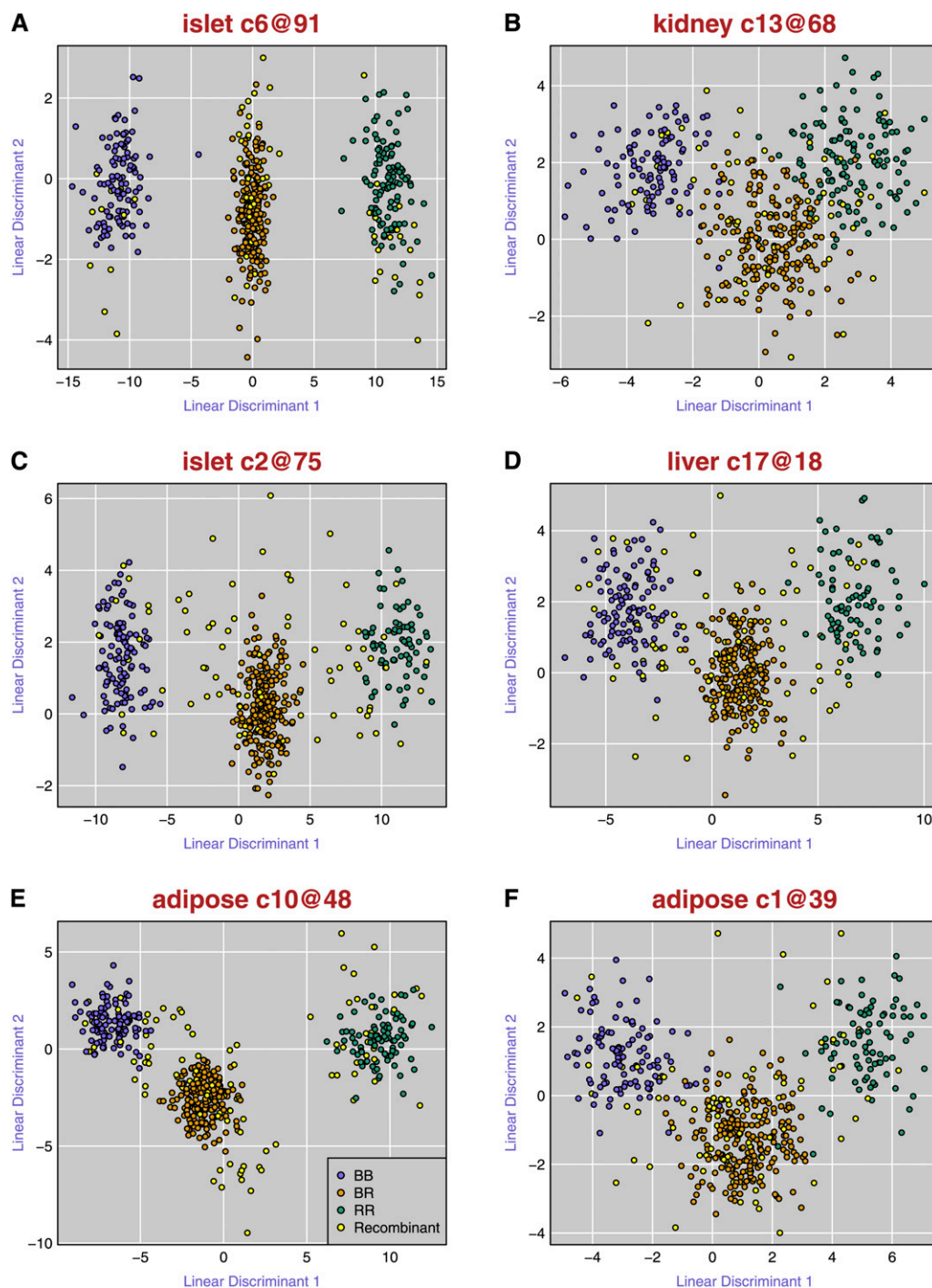


Figure 3 Scatter plots of the first two linear discriminants from application of linear discriminant analysis to data on the mice that show no recombination event in the region of a trans-eQTL hotspot, using the 100 expression traits that map to the region with the highest LOD score. Blue, orange, and green points correspond to the nonrecombinant mice with genotype BB, BR, and RR, respectively, at the eQTL. Yellow points correspond to recombinant mice.

case (adipose chromosome 1), the two loci are tightly linked.

Comparison of recombinants and nonrecombinants

Our second graphical technique is to consider the individuals exhibiting no recombination event in the region of a *trans*-eQTL hotspot (for these individuals, we know their eQTL genotype), apply LDA using the top 100 expression traits that map to the region, and make a scatter plot of the first two linear discriminants. Superposing points for the recombinant

individuals, we can make a direct comparison of the recombinants and nonrecombinants (Figure 3). If there is a single eQTL in the region, the recombinant individuals should reside within the clusters defined by the nonrecombinant individuals. If the recombinant individuals appear different from the nonrecombinant individuals, this indicates the presence of a second QTL.

For the islet chromosome 6 hotspot, the nonrecombinant mice form three distinct clusters, and the recombinant mice (in yellow) fit reasonably well within those

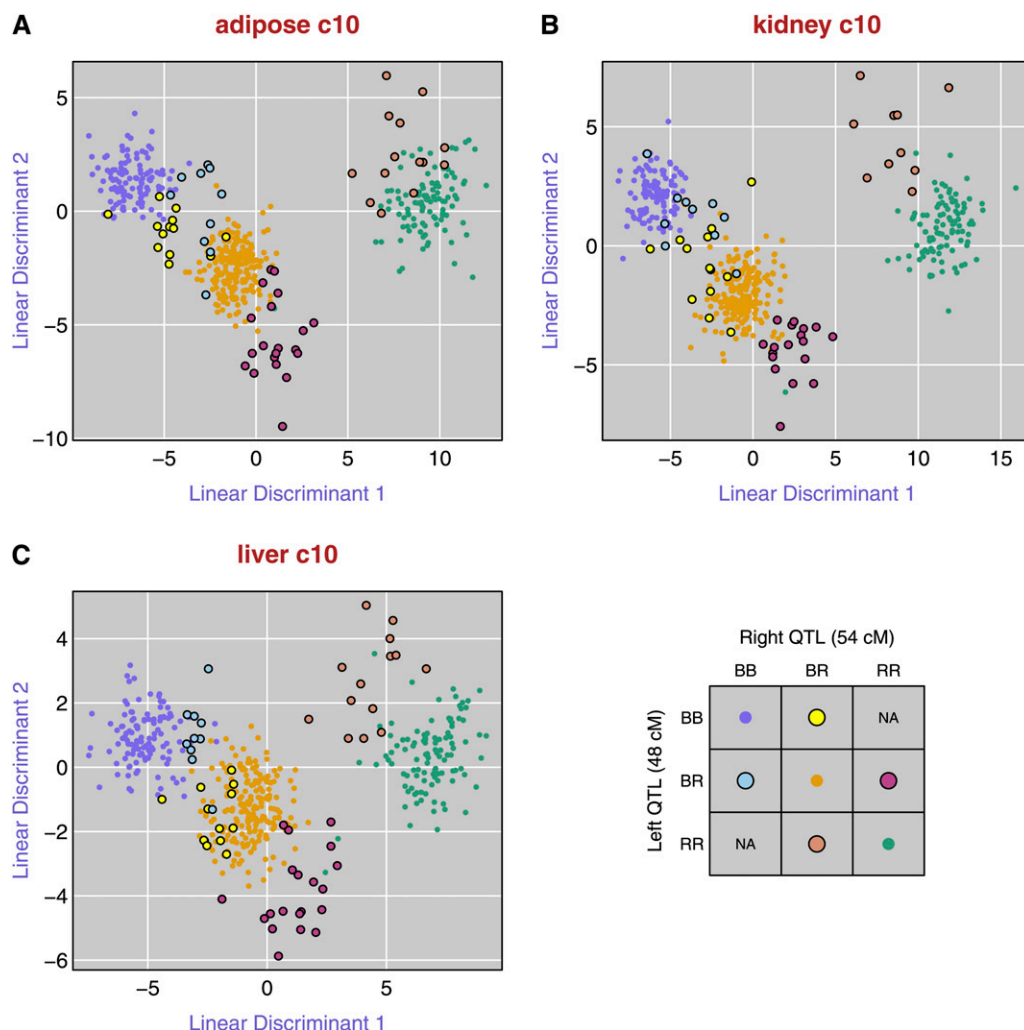


Figure 4 Scatter plots of the first two linear discriminants, as in Figure 3E, for the *trans*-eQTL hotspot on chromosome 10, here considering three tissues: adipose, kidney, and liver. Points correspond to mice, and they are colored according to their two-locus genotypes, for the inferred two QTL model, with one locus at ~48 cM and the other at ~54 cM.

clusters (Figure 3A). This is consistent with there being a single eQTL.

For the islet chromosome 2 (Figure 3C) and adipose chromosome 10 (Figure 3E) hotspots, the nonrecombinant mice again form tight clusters, but the recombinant mice fall clearly outside those clusters. This is evidence for the presence of more than one eQTL.

In the other three cases, kidney chromosome 13 (Figure 3B), liver chromosome 17 (Figure 3D), and adipose chromosome 1 (Figure 3F), the clusters of nonrecombinant mice are not so tight, and the recombinant mice are not obviously different from the recombinant mice. However, for the liver chromosome 17 hotspot (Figure 3D), one might make the case that the majority of recombinant mice are at the boundaries between the clusters, so multiple eQTL may be indicated.

Returning to the adipose chromosome 10 hotspot (Figure 3E), note how the recombinant mice form tight clusters that are distinct from the nonrecombinant mice. If there are two eQTL in the region, perhaps these clusters correspond to different two-locus recombinant genotypes. Via the fit of a two-QTL model (in the next section), we estimate the two QTL to be at 48 and 54 cM. If we color the points by the two-locus

genotypes for these two positions, we see that the clusters of nonrecombinant mice do share a common two-locus genotype (Figure 4A).

This chromosome 10 hotspot also shows effect in kidney and liver, so we applied this technique for this same region, with expression data for these tissues (Figure 4, B and C). The three tissues give consistent results. Mice that are heterozygous at one QTL and homozygous BB at the other (yellow and light blue) sit between the nonrecombinant mice that are homozygous BB (dark blue) and those that are heterozygous (orange). Mice that are homozygous RR at the left QTL and heterozygous at the right QTL (brown) sit above the nonrecombinant RR mice (green), while mice that are heterozygous at the left QTL and homozygous RR at the right QTL (red) sit below the nonrecombinant heterozygotes (orange).

There is one green point (nonrecombinant RR) sitting among the red points (BR at the left QTL and RR at the right QTL). This mouse (ID 3117) has a recombination event just to the left of the left QTL; if we moved that QTL slightly to the left, it would become a red point (BR at the left QTL and RR at the right QTL). In principle, a series of graphs of this form, with varying locations for the left and right QTL, could be

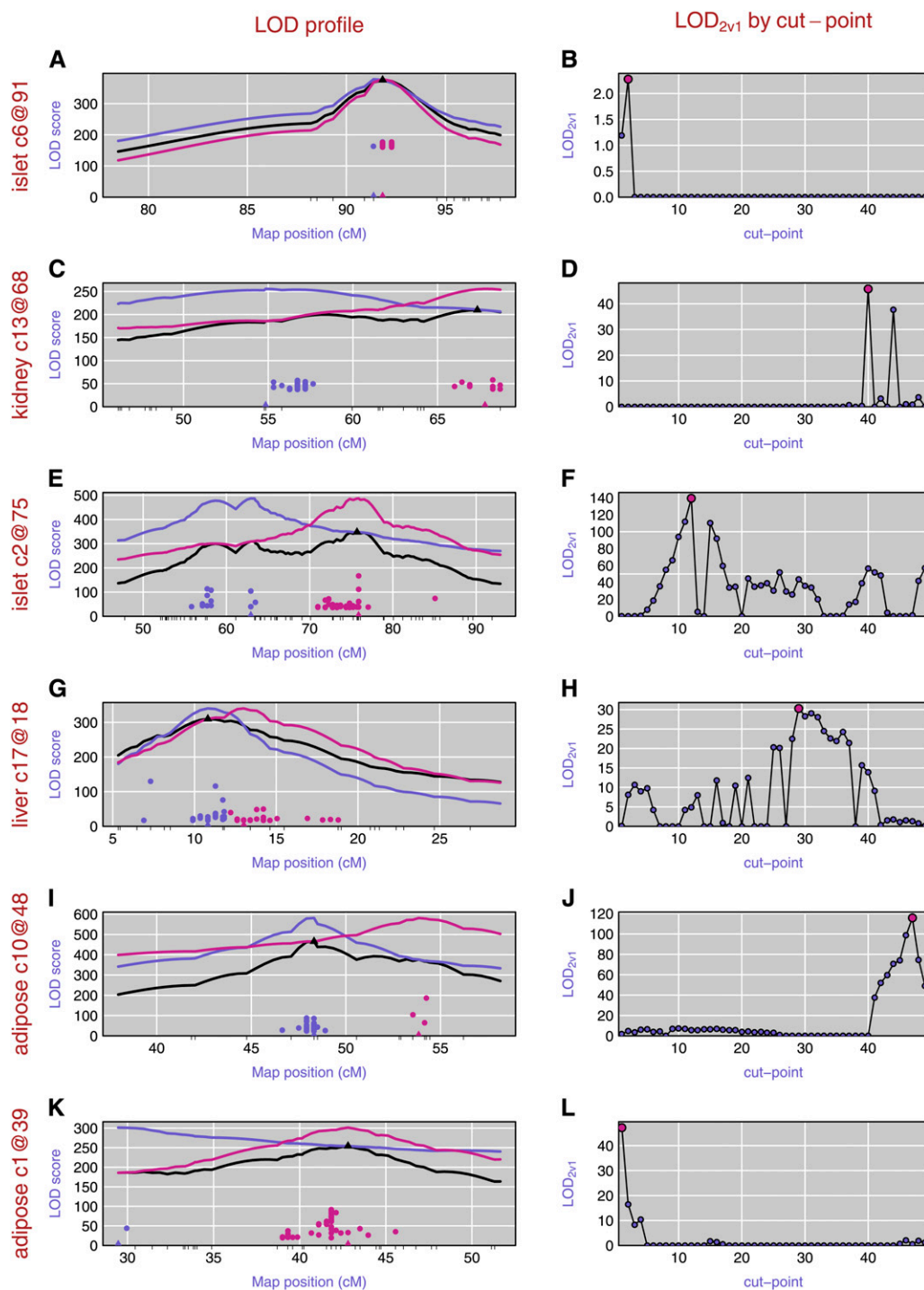


Figure 5 Results of a test of one vs. two QTL at a *trans*-eQTL hotspot, considering the top 50 traits, in terms of LOD score, that map to the region. Each row is a hotspot. In the left panels, the black curve is the LOD score curve for the single-QTL model, with estimated QTL location indicated by a black triangle. The blue and pink curves are profile LOD score curves for the left and right QTL, respectively, for the estimated two-QTL model (with the estimated cut point). Points indicate the LOD score and estimated QTL position for the 50 expression traits, analyzed separately. The points are colored according to whether they are estimated to be affected by the left QTL (blue) or the right QTL (pink). The right panels show the $\text{LOD}_{2v1}^{(c)}$ score, indicating evidence for two vs. one QTL, for each possible cut point c of the list of expression traits in those that map to the left QTL and those that map to the right QTL.

used to define the QTL intervals in the context of this two-QTL model.

There is one additional green point among the red points in Figure 4C (liver). This mouse (ID 3317) sits at the center of the cluster of green points in Figure 4, A and B, and shows no recombination event in the region of these two QTL.

This example illustrates that consideration of the two-locus genotypes can help to strengthen evidence for two loci underlying a *trans*-eQTL hotspot. However, as we will describe

later, in this particular case, the right eQTL appears to affect just three of the expression traits. Just one single trait, if affected by a separate locus, can have a great deal of leverage on these sorts of plots.

Formal tests for two QTL

To supplement these visualization techniques, we developed a formal statistical test for whether a *trans*-eQTL hotspot harbors one vs. two eQTL. The results of this approach

for the six hotspots under consideration are displayed in Figure 5.

Let's begin by considering the kidney chromosome 13 hotspot (Figure 5, C and D). For these multivariate likelihood analyses, we focus on the top 50 expression traits mapping to the region in terms of their LOD scores when considered individually. In the left panel (Figure 5C), the black curve is the LOD score curve for the multivariate QTL analysis with a single-QTL model. The estimated QTL location is at 67.4 cM. The blue and pink curves are LOD score profiles for the estimated two-QTL model, for which the estimated QTL locations are at 54.8 and 67.8 cM. The blue and pink points indicate the LOD scores and estimated QTL locations for the individual expression traits, with blue points affected by the left QTL and pink points affected by the right QTL. The right panel (Figure 5D) shows the evidence for two vs. one QTL as a function of the choice of cut point for the list of expression traits into those affected by the left and right QTL. The inferred cut point has 40 traits affected by the left QTL and 10 traits affected by the right QTL and a LOD_{2v1} score of 45.8, indicating very strong evidence for two QTL and for this particular cut point.

The results for the islet chromosome 6 hotspot are displayed in Figure 5, A and B. The inferred two-QTL model has QTL at 91.4 and 91.8 cM, with only the two expression traits affected by the left QTL. And $\text{LOD}_{2v1} = 2.3$, indicating weak evidence for two QTL.

The results for the islet chromosome 2 hotspot (Figure 5, E and F) indicate strong evidence for two QTL, with $\text{LOD}_{2v1} = 139$. The estimated QTL are at 62.9 and 75.7 cM. The left QTL is inferred to affect 12 of the 50 expression traits.

The liver chromosome 17 hotspot (Figure 5, G and H) has strong evidence for two QTL, with $\text{LOD}_{2v1} = 30$ and the estimated QTL locations at 10.8 and 13.0 cM. The choice of cut point of the expression traits is not so clear. We estimate that 29 of the expression traits are affected by the left eQTL, but a model with 32 traits affected by the left eQTL gives a similar likelihood.

For the adipose chromosome 10 (Figure 5, I and J) and the adipose chromosome 1 (Figure 5, K and L) hotspots, the evidence for two QTL is strong, but only three eQTL are inferred to be affected by the right eQTL at the chromosome 10 hotspot, and only one trait is inferred to be affected by the left eQTL at the chromosome 1 hotspot. If we trim off these expression traits and apply the procedure again, we find that for the adipose chromosome 10 locus (Figure S2), there is little evidence for more than one eQTL affecting the remaining traits. Further, the analysis of two-locus genotypes in the LDA plot in Figure 4 is largely driven by these three expression traits that map to 54 cM. Similarly, if we trim off the first expression trait for the adipose chromosome 1 hotspot (Figure S3), there is limited evidence for multiple QTL affecting the remaining traits.

In summary, the formal statistical test provides strong evidence for two eQTL in five of these six cases, but in two of the cases, most of the traits are affected by a single eQTL.

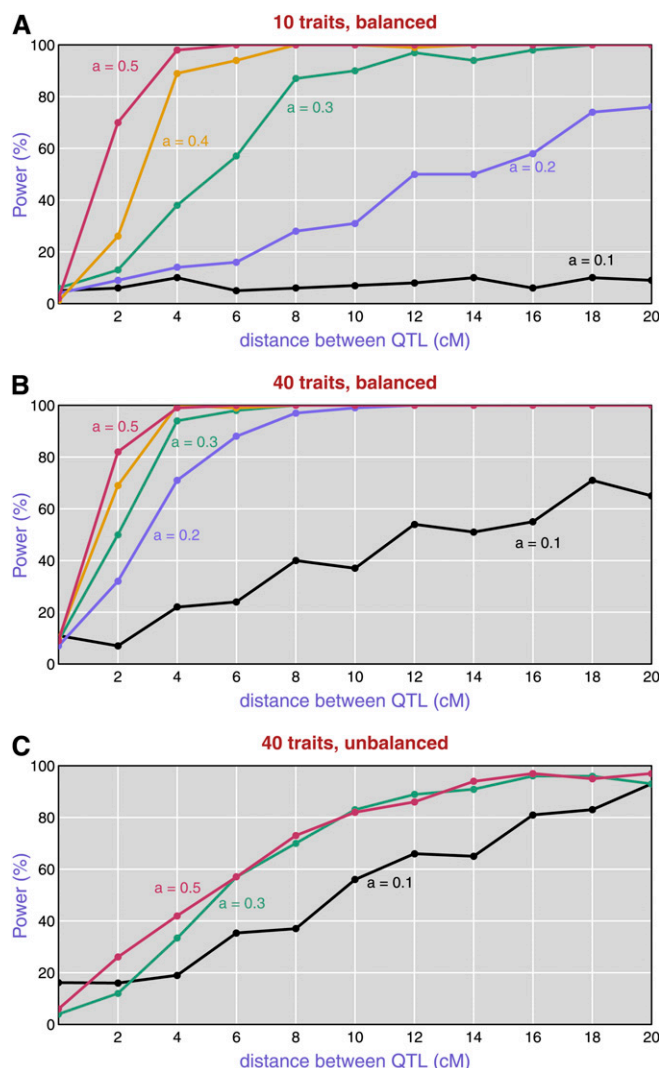


Figure 6 Power to detect two QTL as a function of the distance between the QTL for varying QTL effects. (A) Ten traits, with each QTL affecting five traits. (B) Forty traits, with each QTL affecting 20 traits. (C) Forty traits, with the left QTL affecting 5 traits and the right QTL affecting 35 traits.

Simulations

To further assess the performance of the proposed likelihood-based test for whether a *trans*-eQTL hotspot harbors more than one eQTL, we performed a set of computer simulation studies. We generated 500 intercross offspring with 100 markers on a 100-cM chromosome and then simulated $p = 10$ or 40 traits, with half the traits affected by a QTL at 50 cM and the other traits affected by a QTL 0–20 cM away (at 50–70 cM). We also considered an unbalanced case with 5 traits affected by the left QTL and 35 traits affected by the right QTL. We assumed additive allele effects, with the additive effect of each QTL being $a = 0.1, 0.2, 0.3, 0.4$, or 0.5 . Residual variation followed a normal distribution with mean 0 and SD 1, with traits conditionally independent given the QTL genotypes. We used 100 simulation replicates for each situation and calculated P -values by a parametric bootstrap with 1000 simulation replicates.

The estimated power to detect two linked QTL, as a function of the distance between the QTL, is shown in Figure 6. When the QTL effect is <0.3 , the power to distinguish two QTL within a distance of 10 cM is low for $p = 10$. For any fixed effect, the power to detect two QTL is higher for $p = 40$ than for $p = 10$. When the QTL effect is >0.4 , the power to distinguish two QTL separated by more than 5 cM is almost 100%. The power to detect two QTL in the unbalanced case, with the left QTL affecting 5 of 40 traits (Figure 6C), is considerably lower than for the balanced case (Figure 6B).

The case of distance = 0 corresponds to the null hypothesis of a single QTL affecting all traits. The power in this case is the type I error rate, and the parametric bootstrap method is seen to have a somewhat inflated type I error rate relative to the nominal 5%.

Discussion

In this paper we have proposed exploratory methods and a formal inference method for dissecting *trans*-eQTL hotspots. We applied these approaches to data on a large mouse intercross with gene expression microarray data on six tissues, and we performed a simulation study to investigate the performance of the formal inference method. Both the exploratory methods and the formal inference method are helpful in dissecting *trans*-eQTL hotspots and can give improved estimates of the eQTL positions.

The exploratory methods have the advantage of providing insight into the underlying evidence for multiple eQTL: the multiple eQTL may show distinct inheritance patterns, or the recombinant and nonrecombinant individuals may show differences in expression. However, while the visualization methods can be strongly informative, they will not necessarily reveal the presence of two eQTL because the inheritance pattern of the two linked eQTL may be the same, or the first two linear discriminants may not be revealing of the difference between the recombinants and nonrecombinants.

In forming a multivariate test statistic, we chose to follow the method of Knott and Haley (2000), but other multivariate analysis of variance (MANOVA) statistics also could be used, including Pillai's trace, Lawley-Hotelling's test, and Roy's lambda (Anderson 2003). Similarly, in the exploratory data visualization based on linear discriminant analysis, other dimension-reduction techniques could be used. A supervised (*i.e.*, classification) method, which makes use of the known eQTL genotypes of the nonrecombinant individuals, is preferred.

The main issue in the formal statistical test is the choice of expression traits because we cannot handle a very large number of expression traits. Our choice to focus on the 50 traits with the highest LOD score was arbitrary and deserves further investigation. Regularized methods (see Hastie *et al.* 2009, Section 5.8) or a Bayesian approach might have an advantage in this context. Such approaches also could be used to relax some of our modeling assumptions. For example, one might consider a model with two eQTL in which each expression trait can be affected by both.

We considered two methods to calculate P -values: a parametric bootstrap test and a stratified permutation test. The simulations to investigate power used the parametric bootstrap test and showed a somewhat inflated type I error rate. Moreover, neither approach takes account of the selection of hotspots, which may introduce further bias.

We considered a single tissue at a time. The joint consideration of multiple tissues could provide additional power to dissect *trans*-eQTL hotspots that are in common across tissues. We ignored the effects of eQTL elsewhere in the genome and considered just one region in isolation. In so doing, the effects of any other eQTL become part of the residual variation. Local eQTL are a particularly important case because they are quite common and often have large effects. Controlling for the effect of local eQTL could give better precision in the dissection of a *trans*-eQTL hotspot.

We have implemented our methods in an R package (R Core Team 2015), *qtlpvl*, available at <https://github.com/jianan/qtlpvl>.

Acknowledgments

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The Dissection of Expression Quantitative Trait Locus Hotspots

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The dissection of expression quantitative trait locus hotspots

SUPPLEMENT

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Brian S. Yandell^{*,§}, Alan D. Attie[†] Karl W. Broman^{‡,1}

Departments of ^{*}Statistics, [†]Biochemistry, [‡]Biostatistics and Medical Informatics, and [§]Horticulture,
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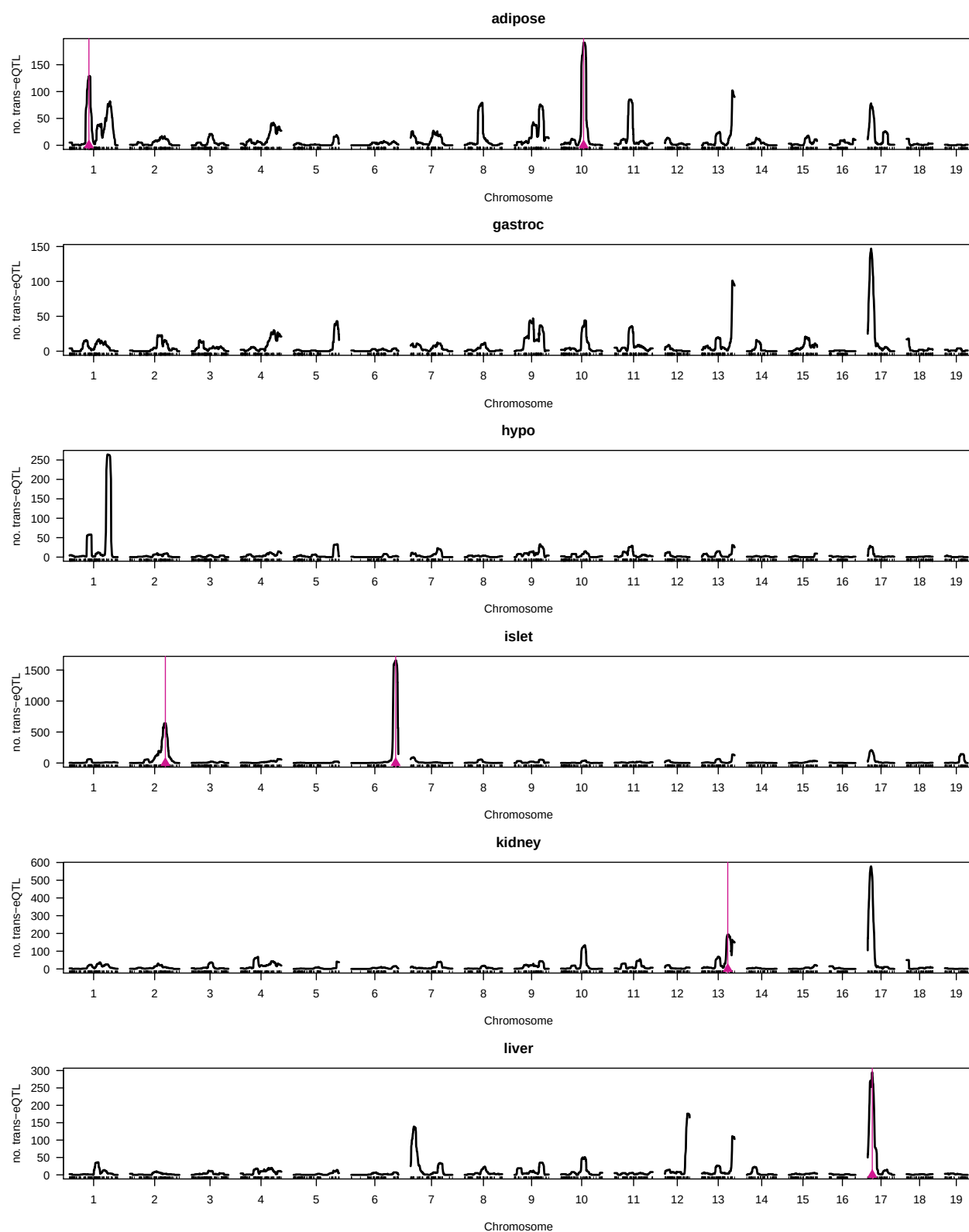
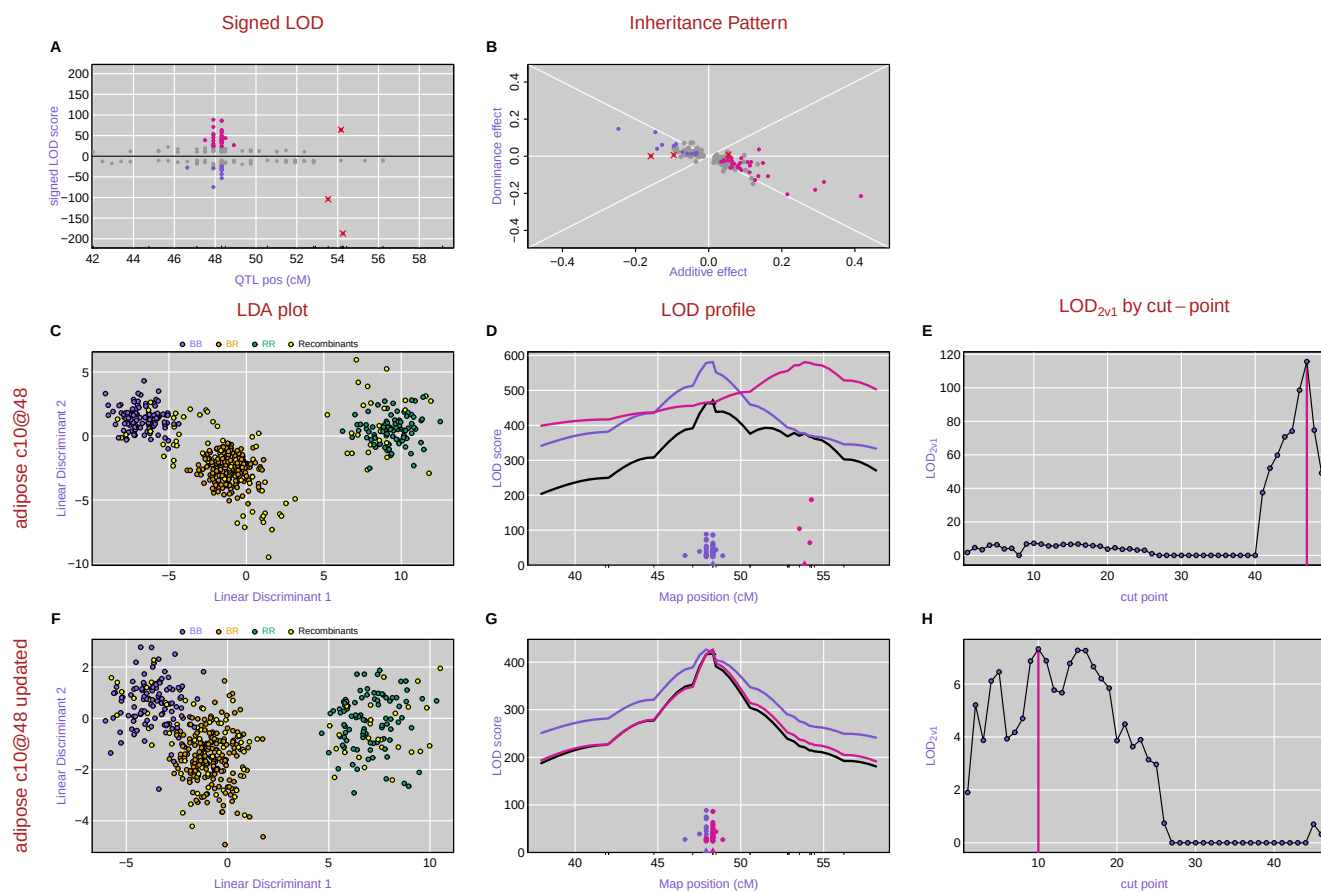
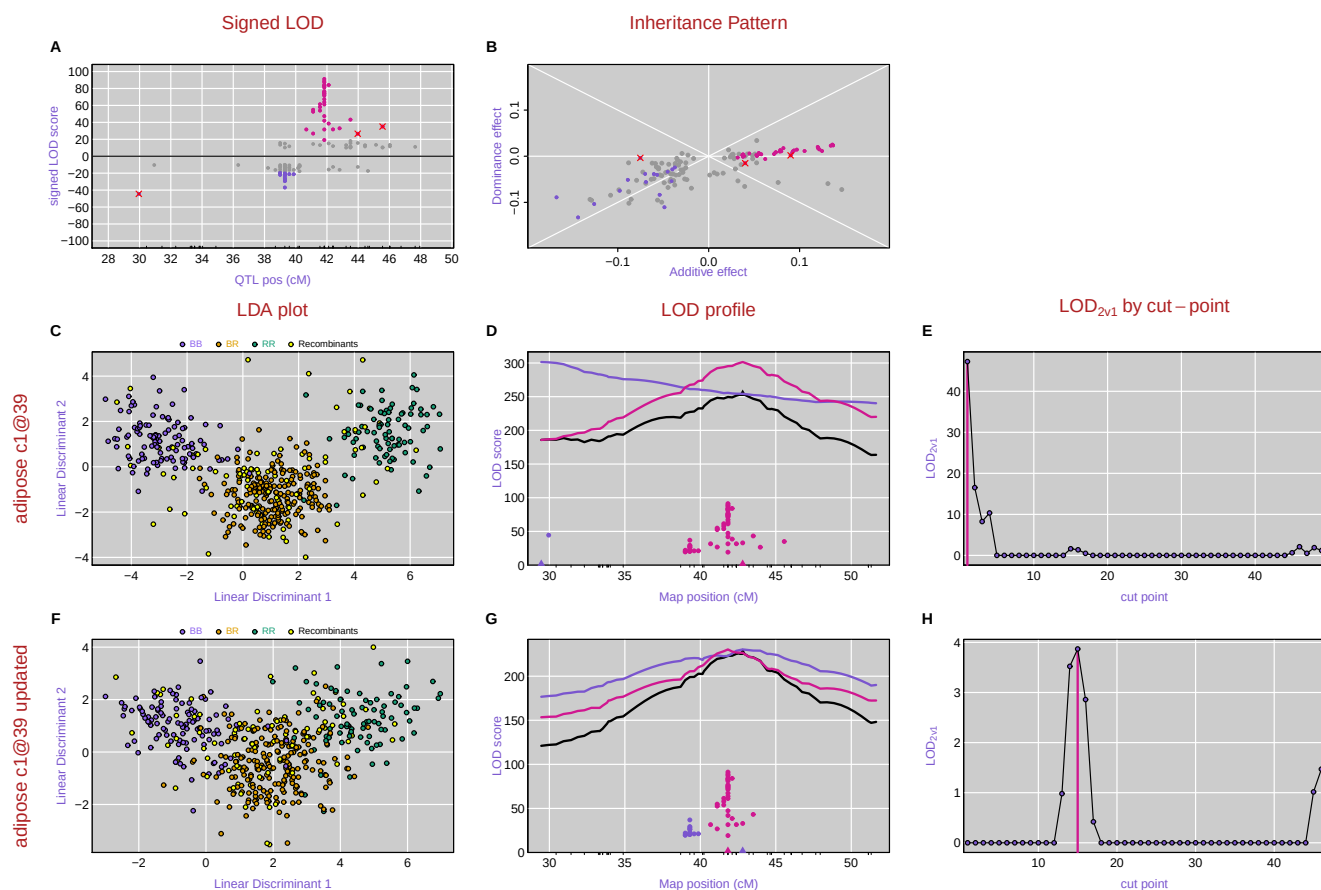


Figure S1 Number of expression traits with $\text{LOD} \geq 10$ in a 10 cM sliding window across the genome. Red triangles indicate the six *trans*-eQTL hotspots used as examples in Figures 2, 3, and 5.

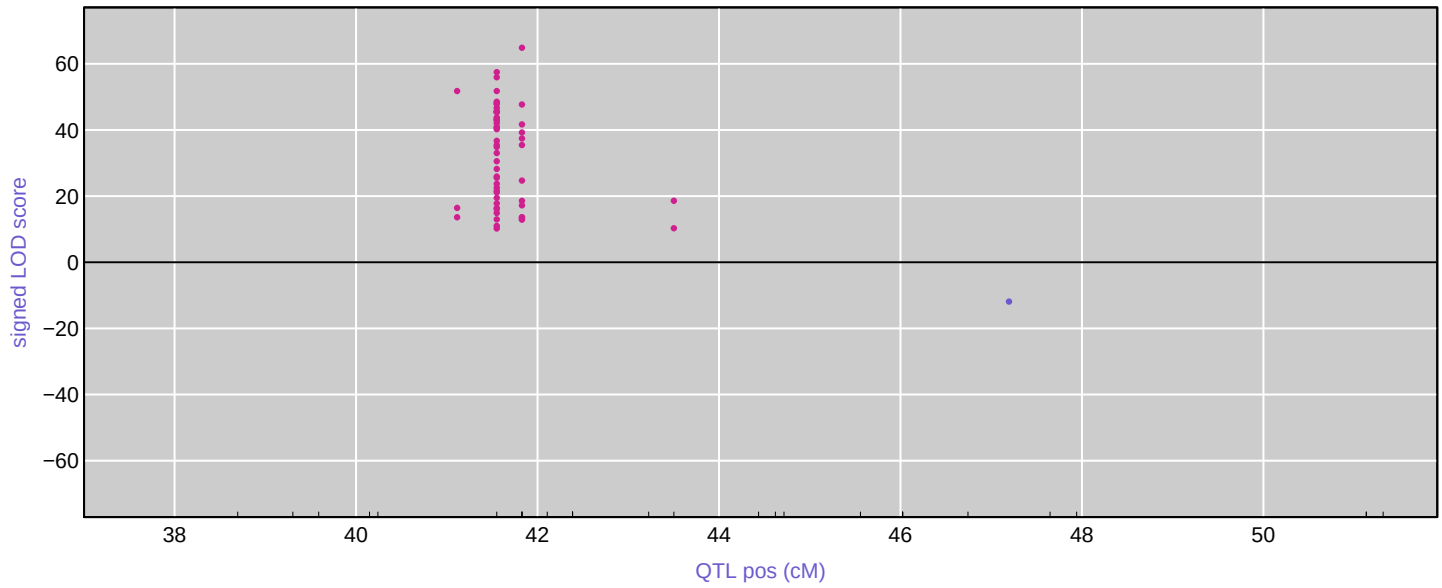




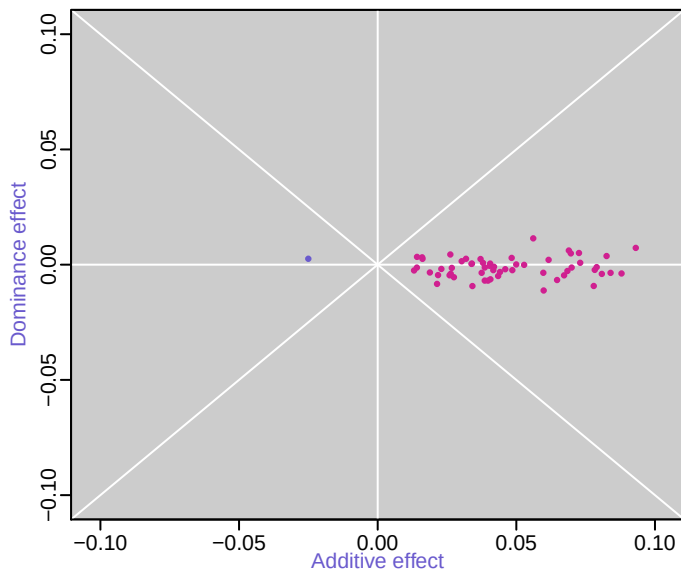
FileS1. A 35-page PDF with the results as in Figures 2, 3, and 4, for all 35 *trans*-eQTL hotspots identified.

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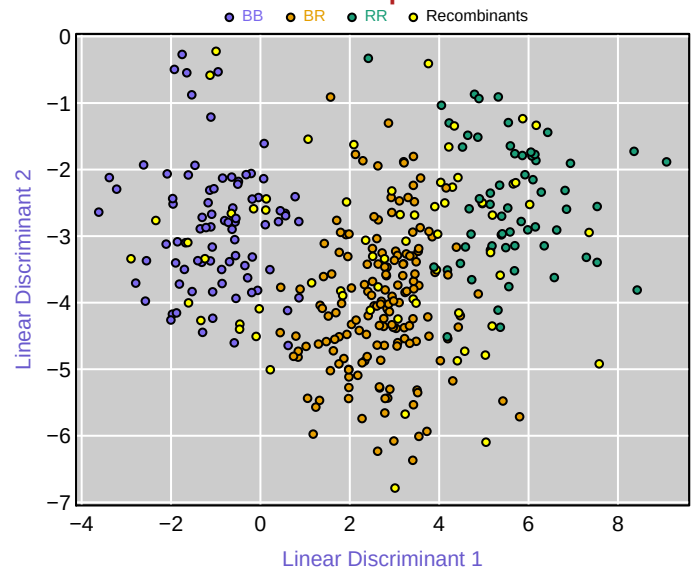
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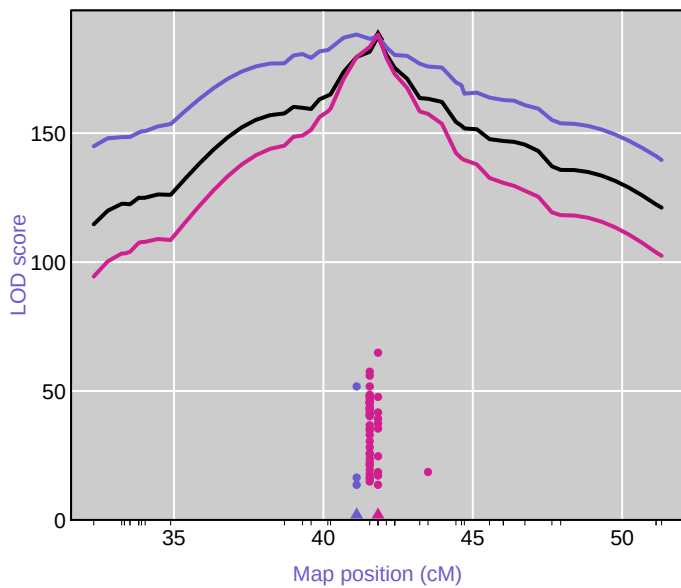
Inheritance Pattern



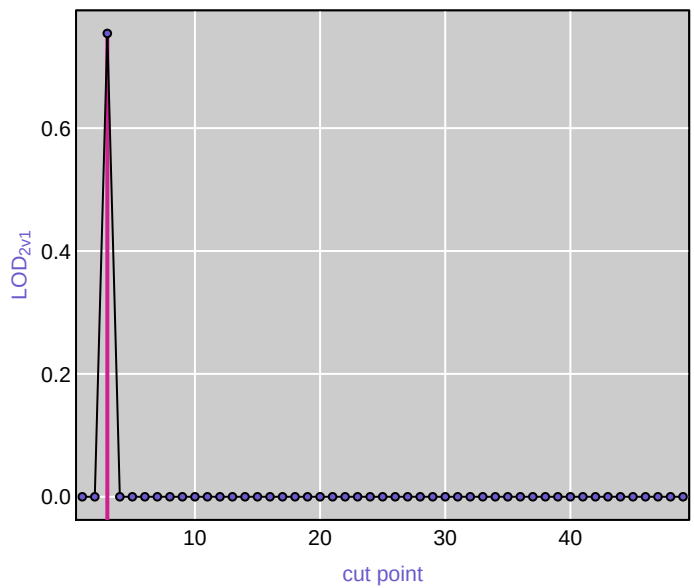
LDA plot



LOD profile

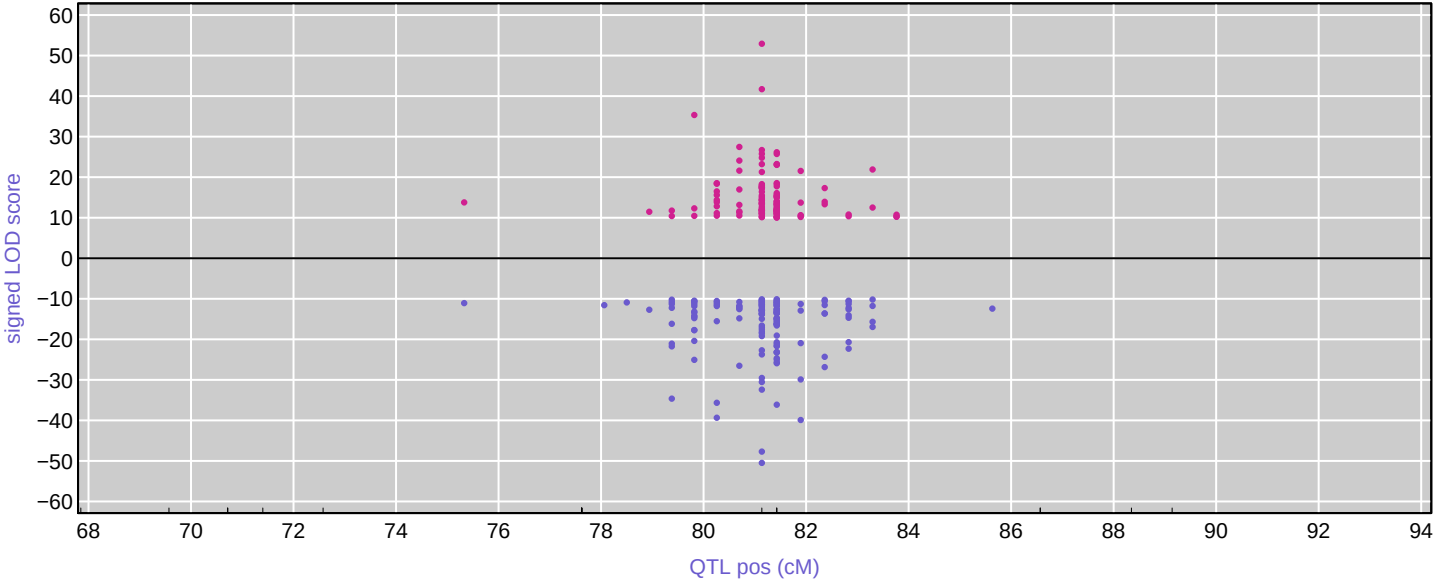


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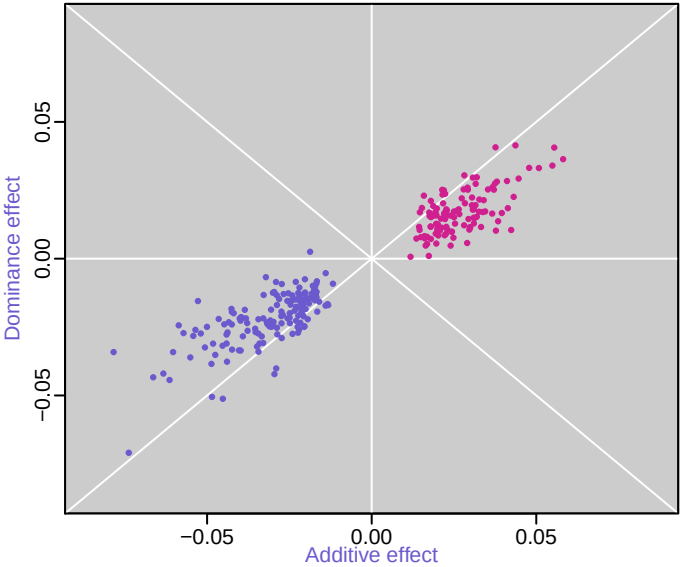


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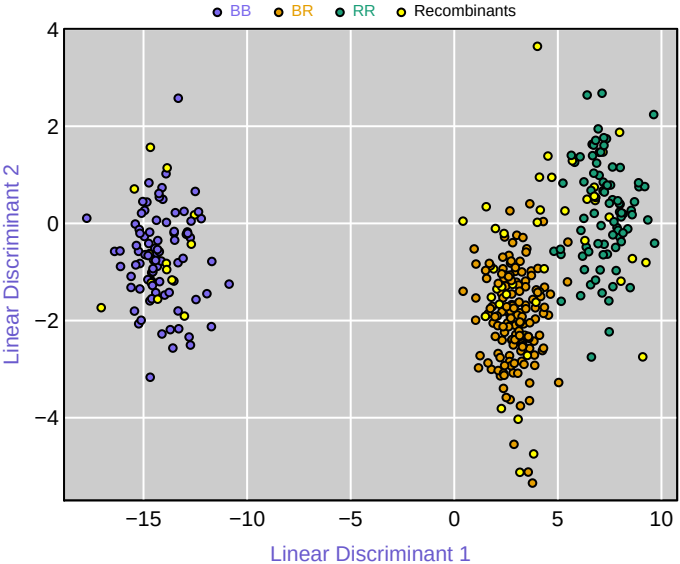
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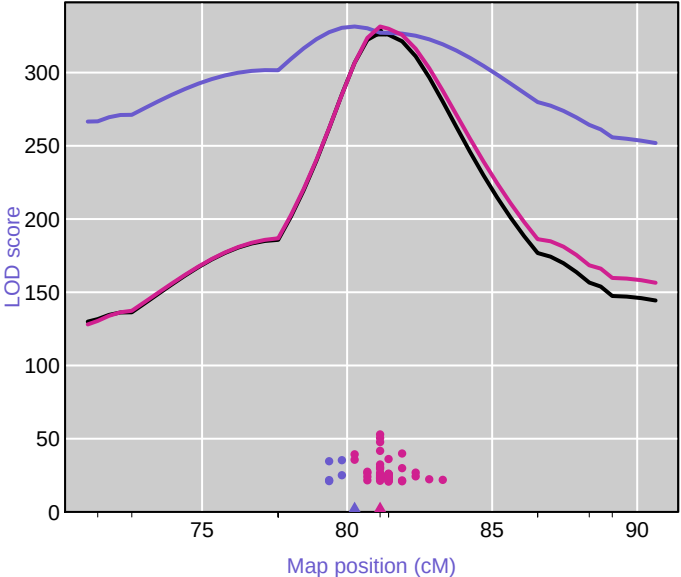
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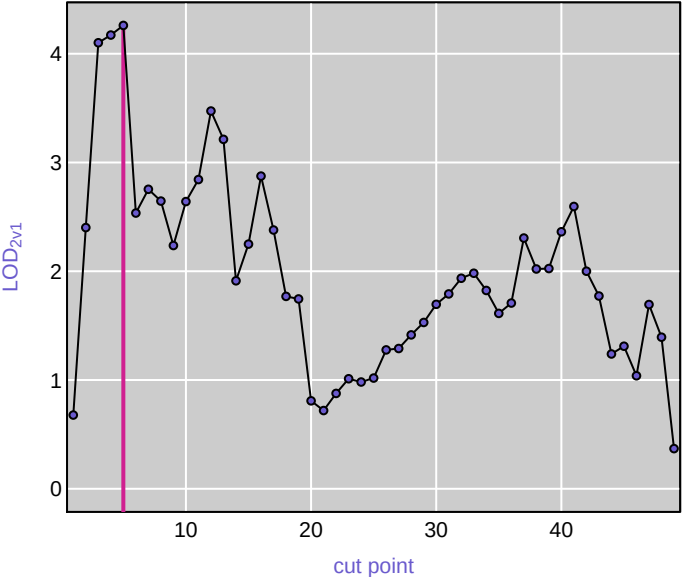
LDA plot



LOD profile

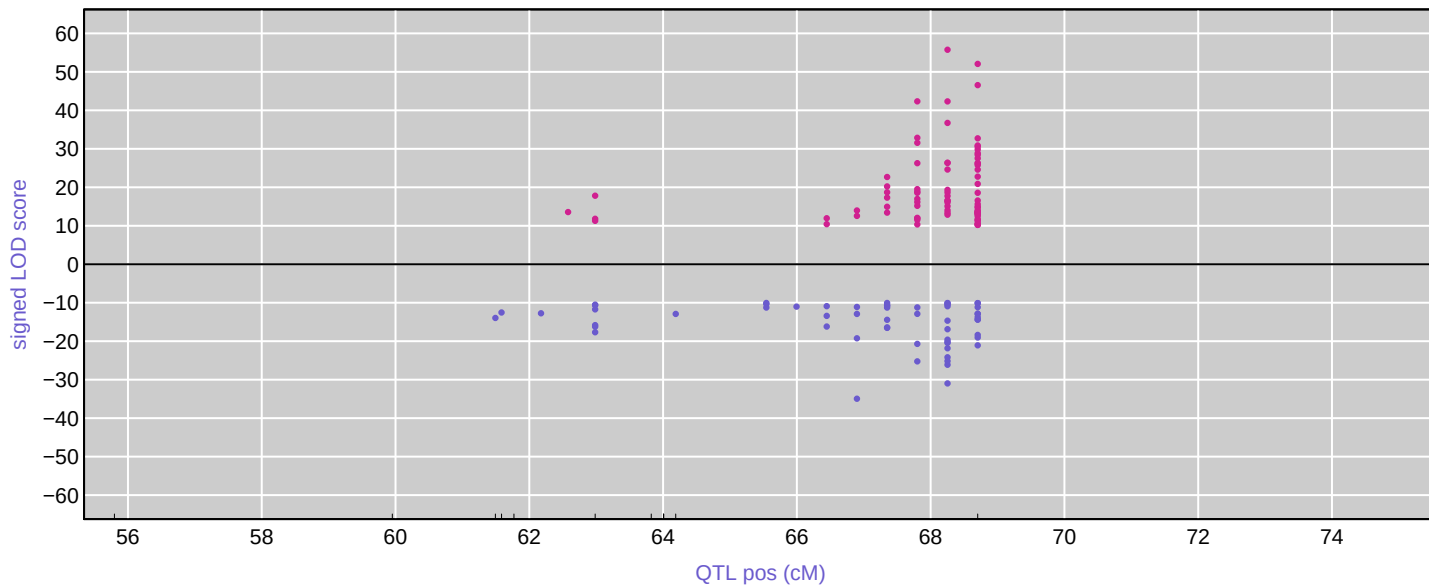


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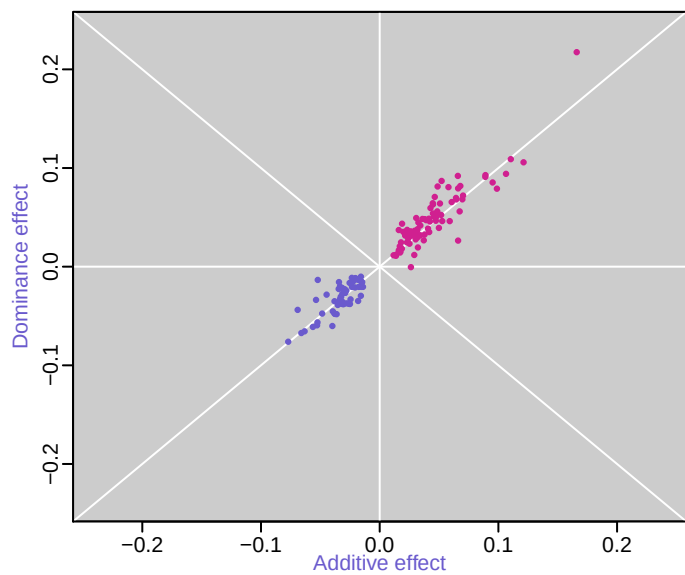


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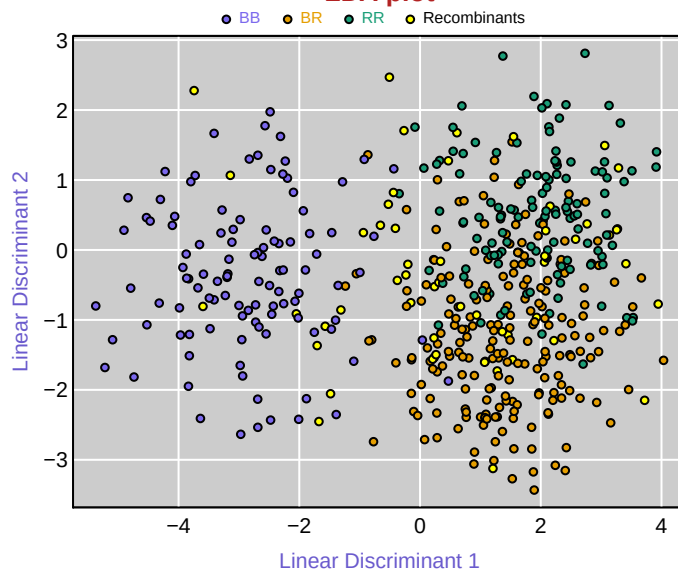
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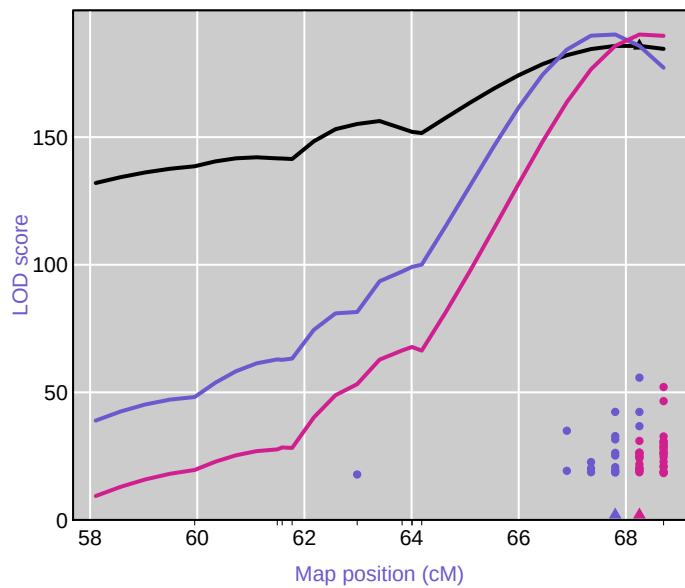
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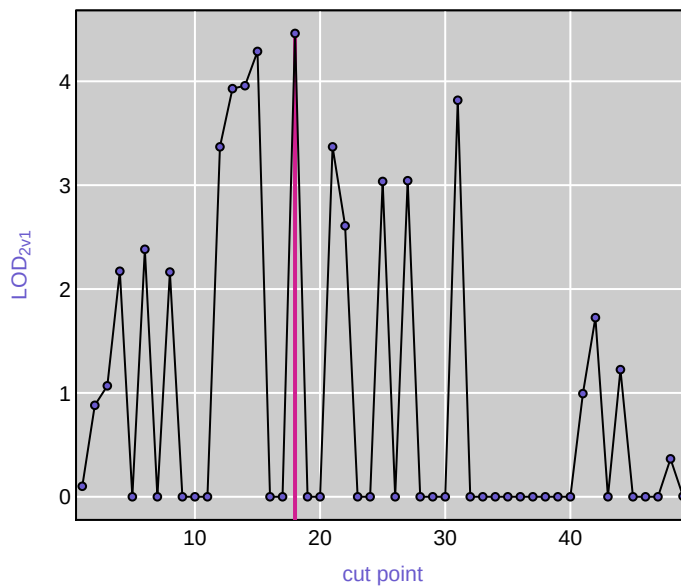
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LOD profile

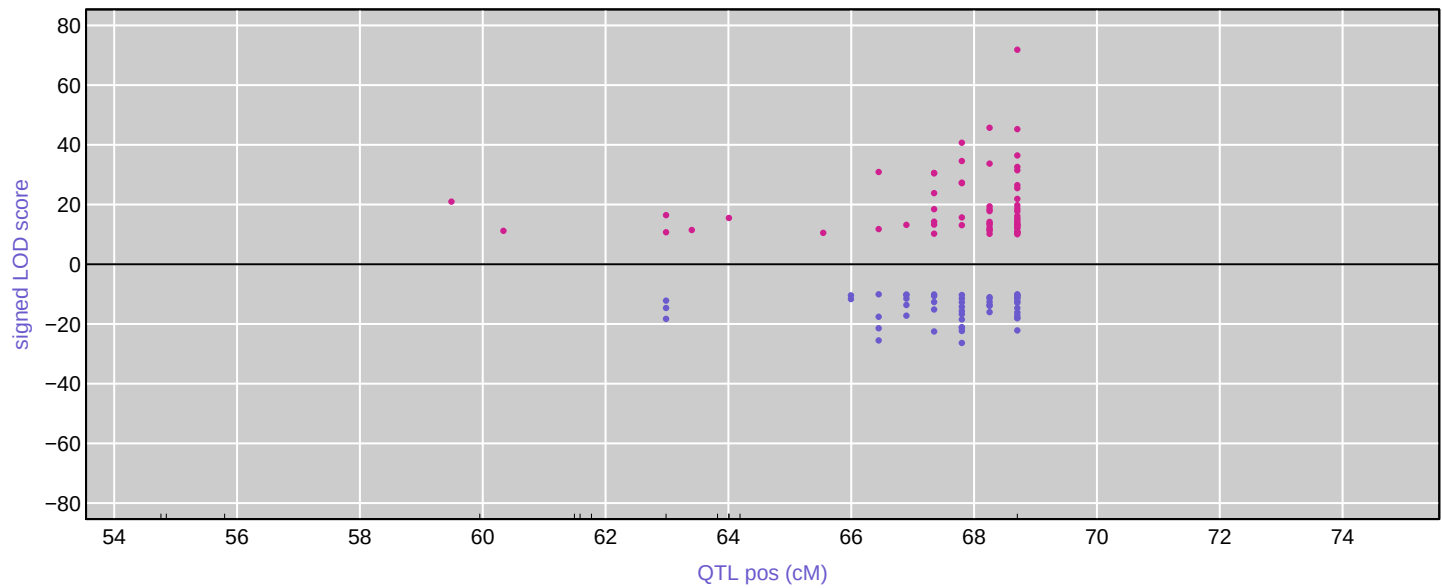


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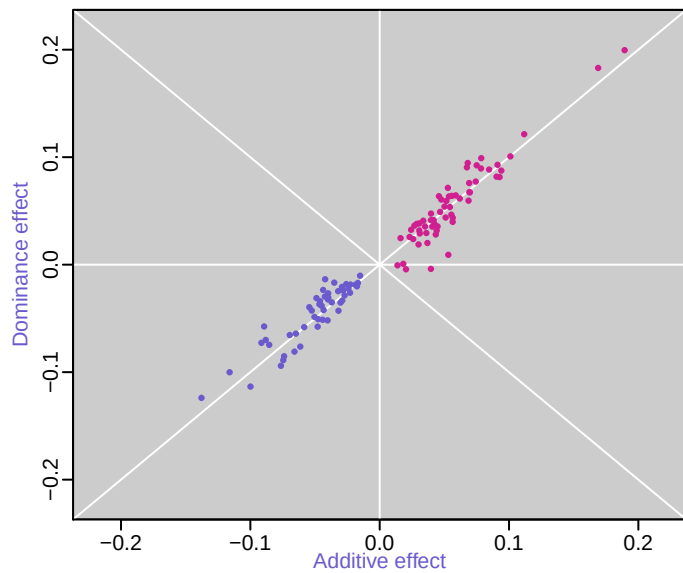


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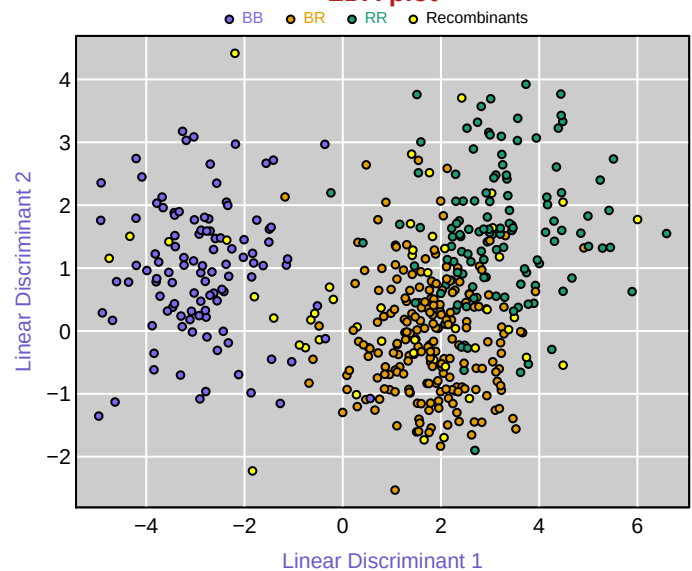
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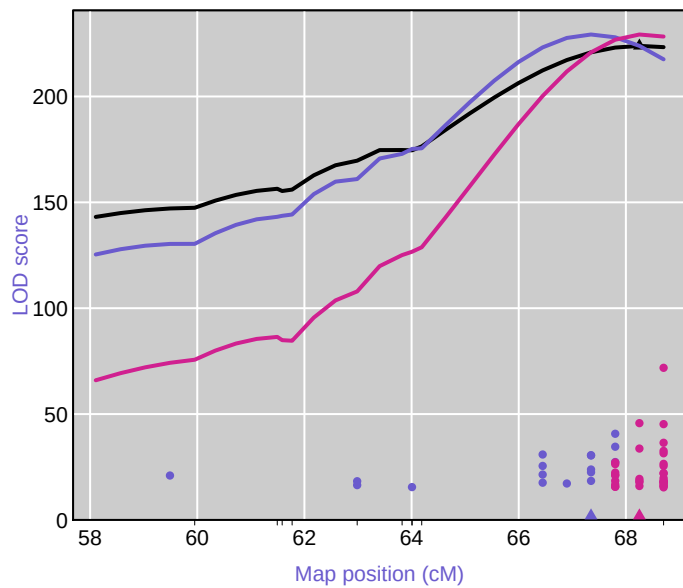
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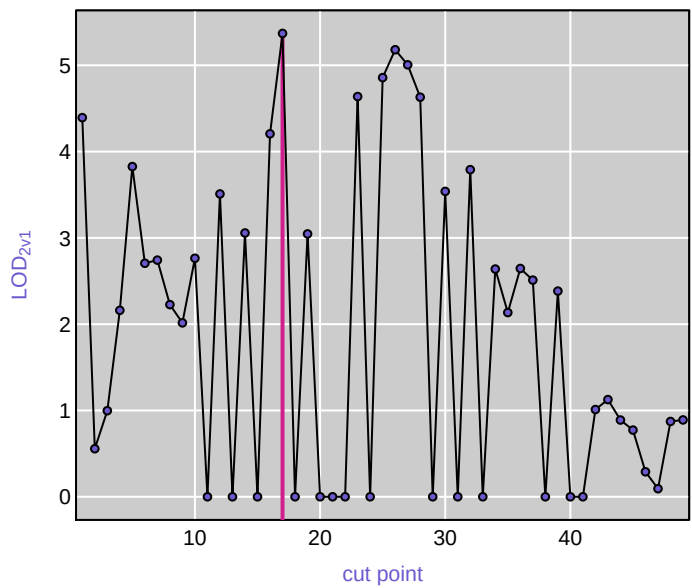
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LOD profile

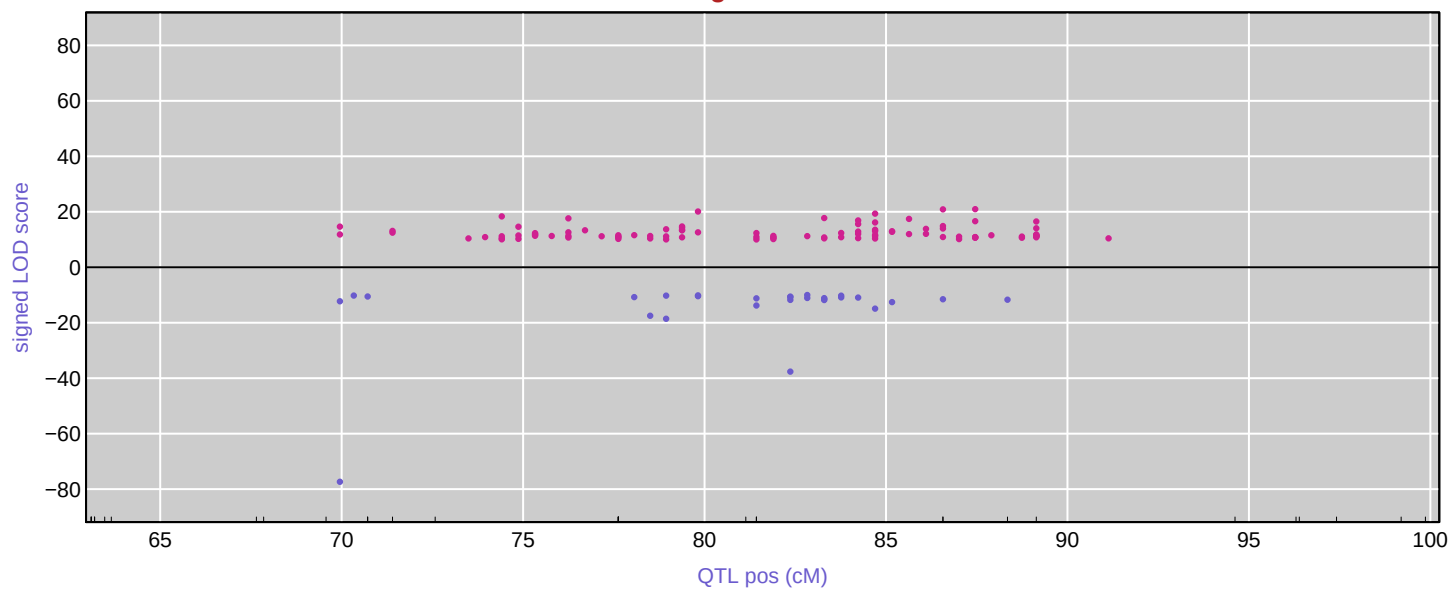


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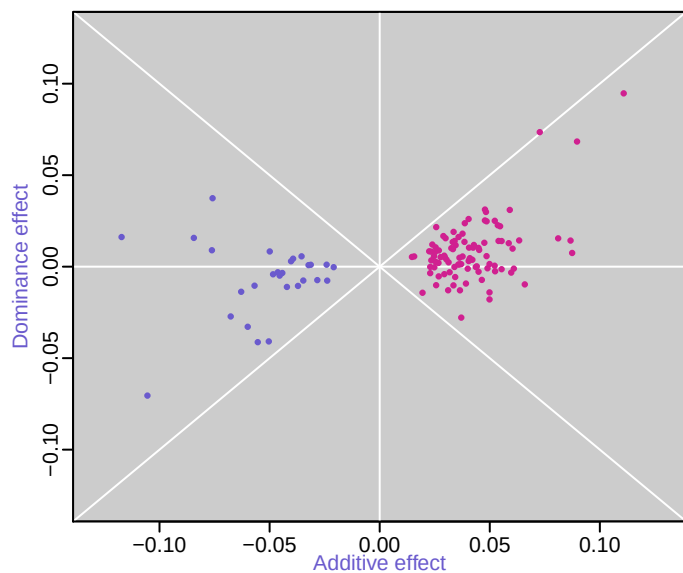


adipose chr1 @ 84 p-value=0.043

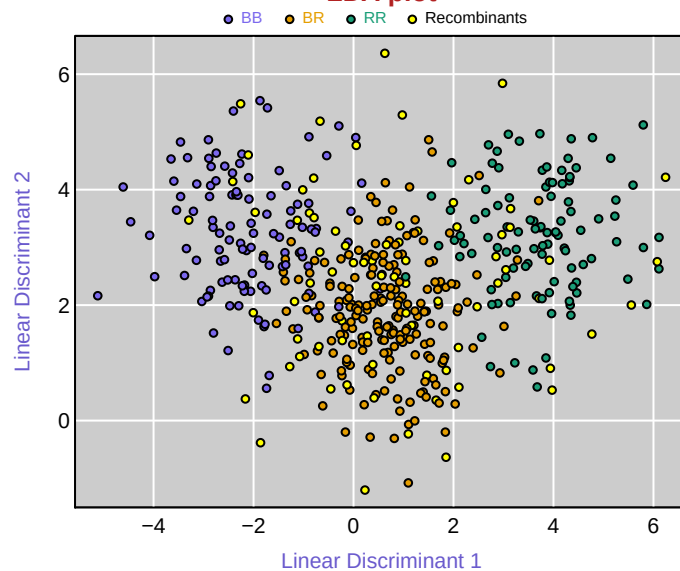
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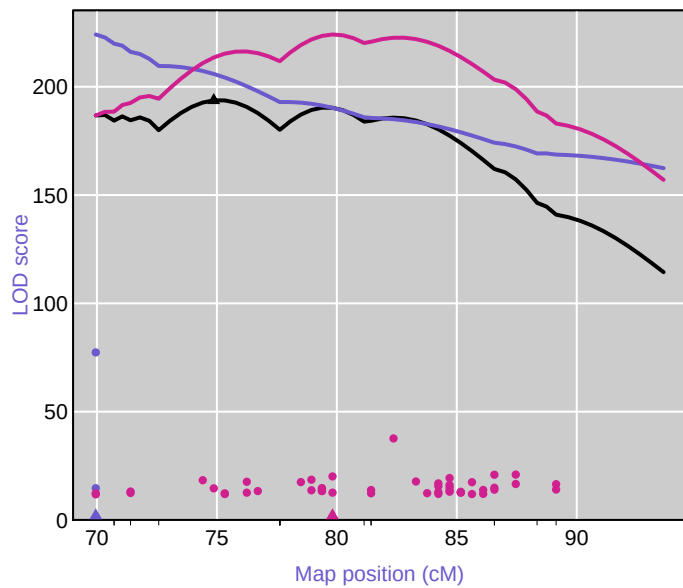
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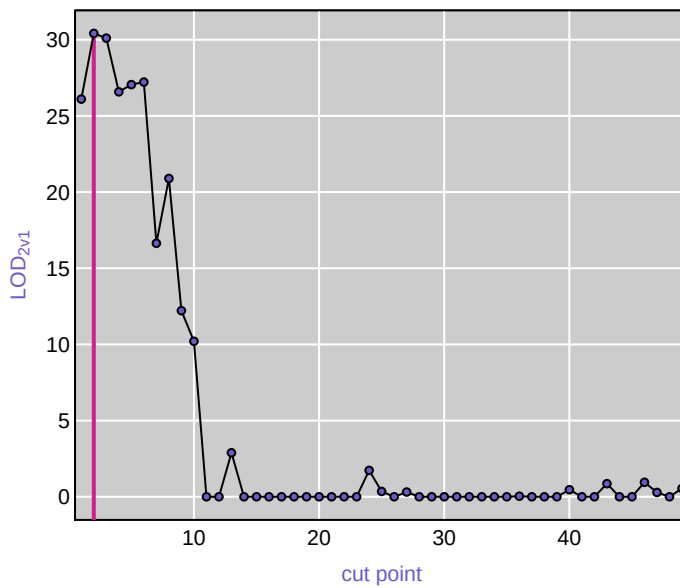
LDA plot



LOD profile

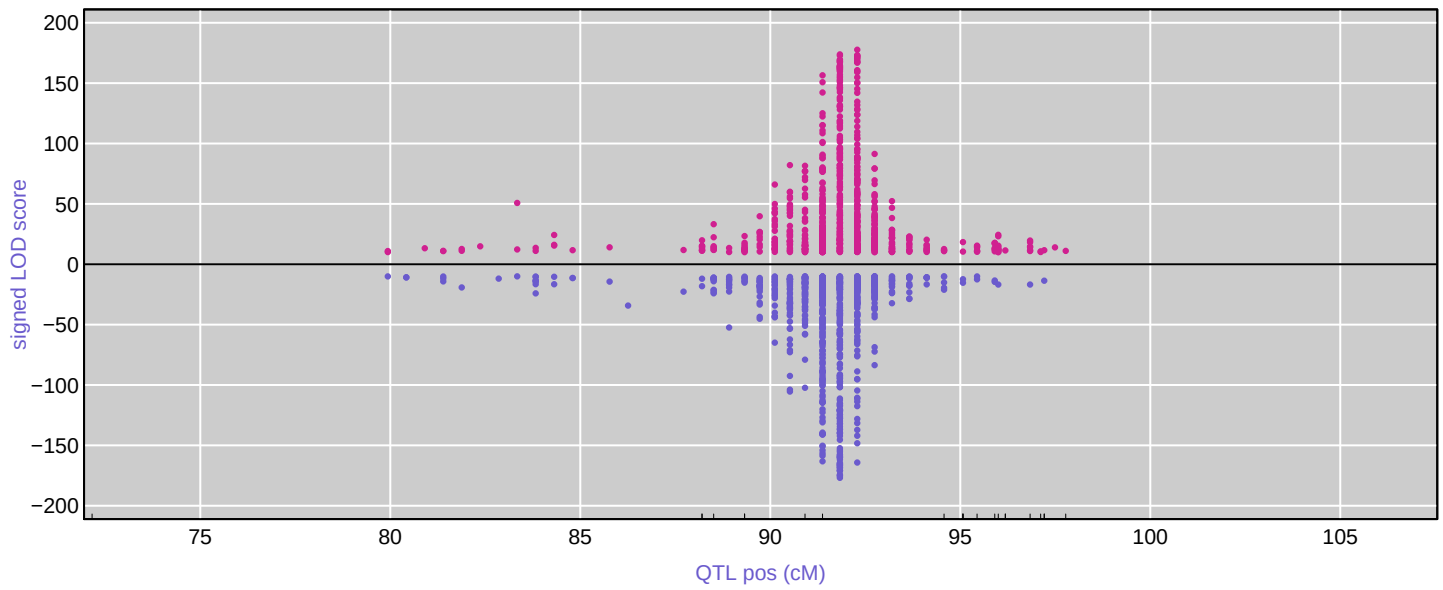


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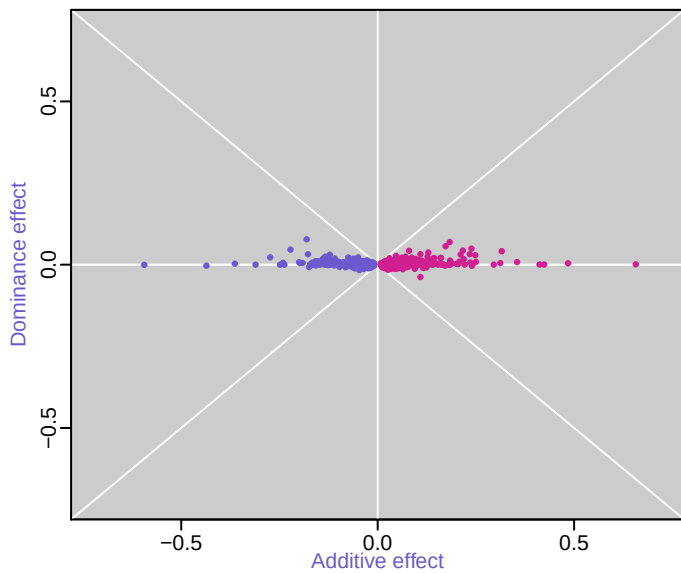


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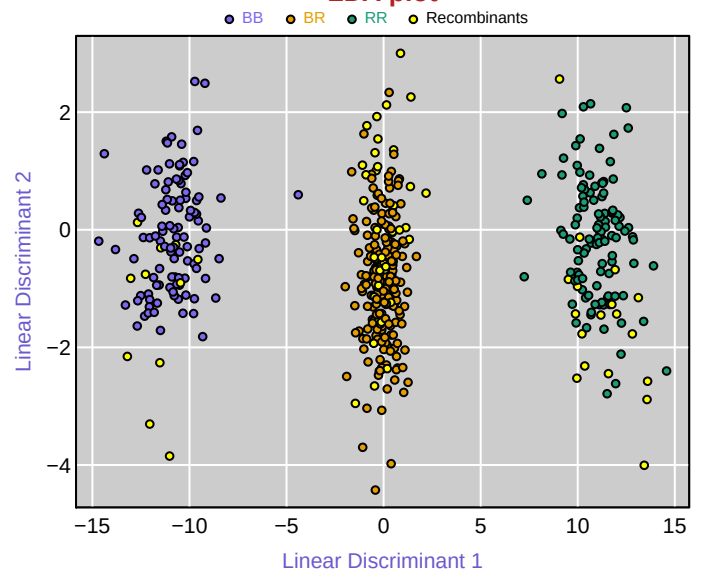
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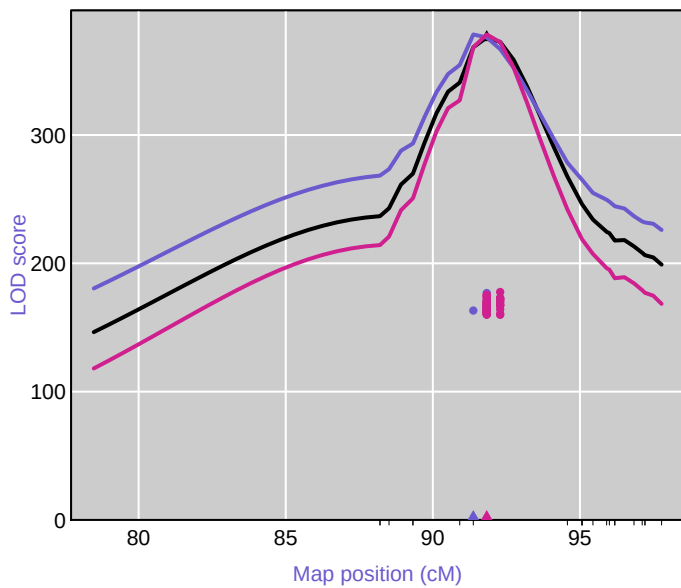
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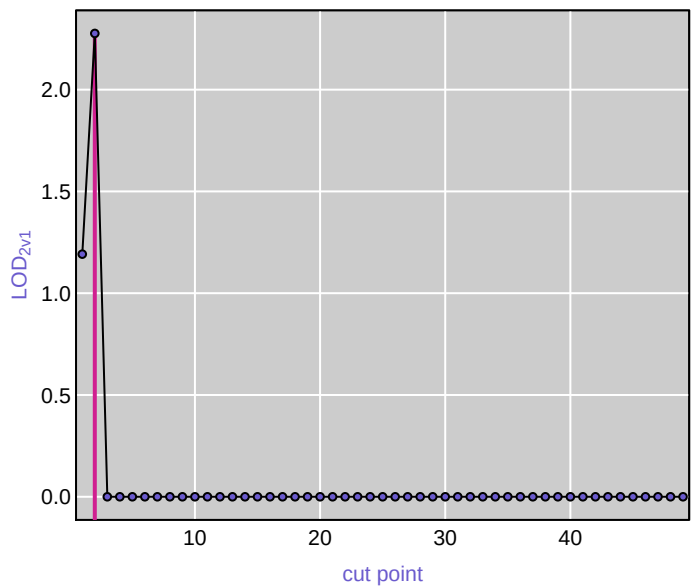
LDA plot



LOD profile

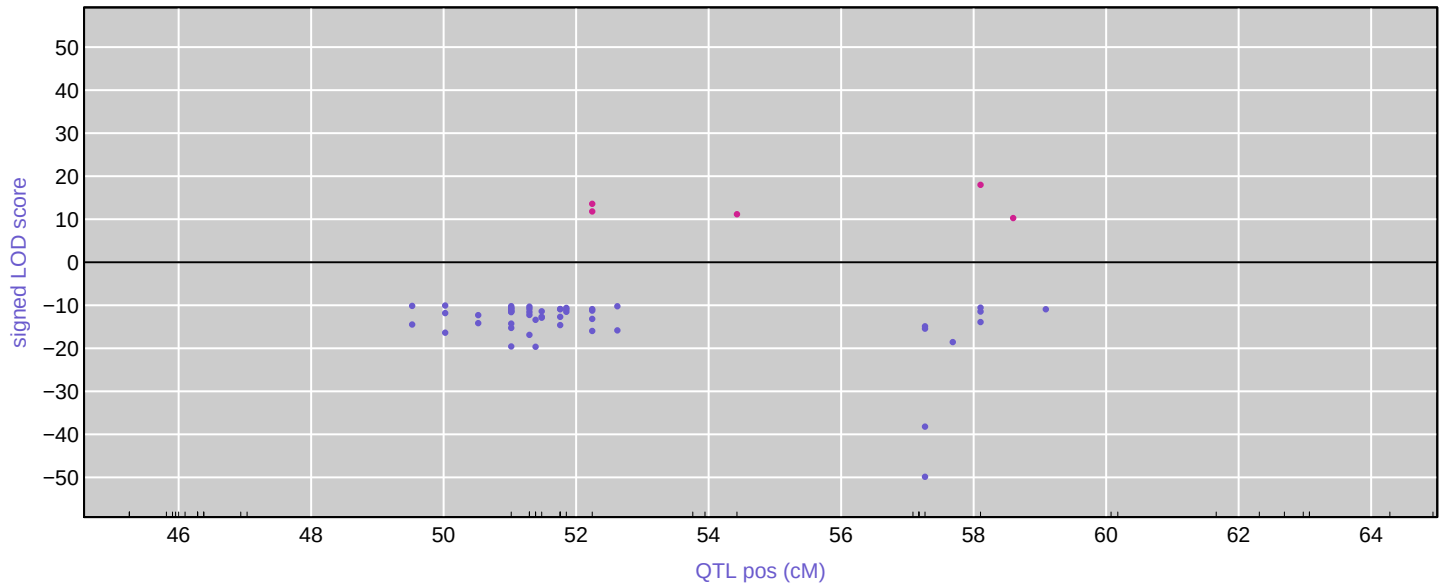


LOD_{2v1} by cut – point

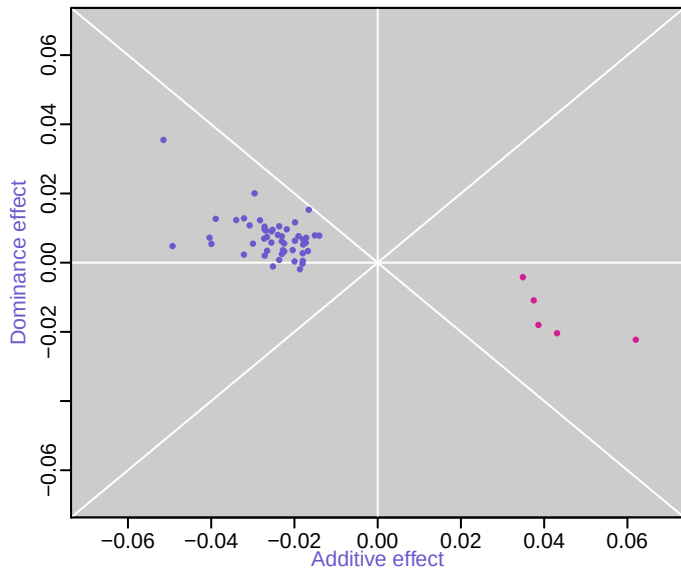


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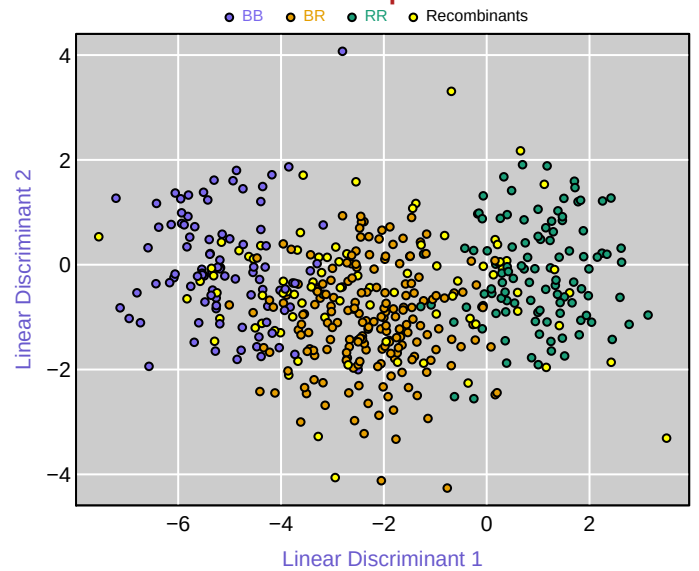
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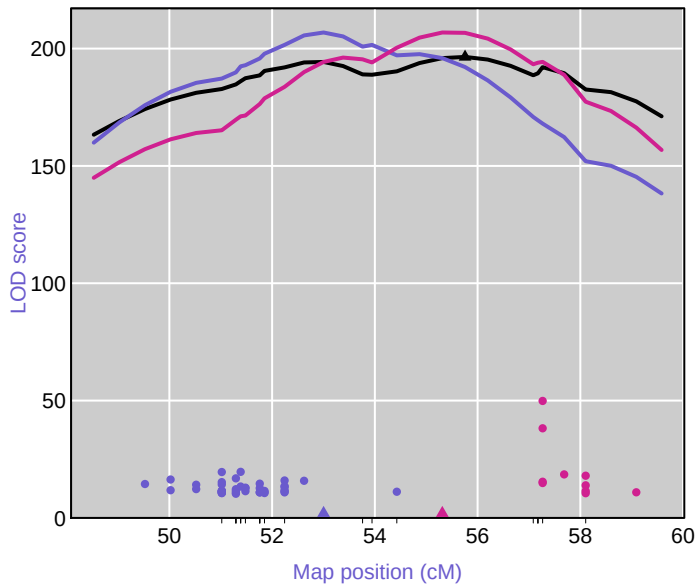
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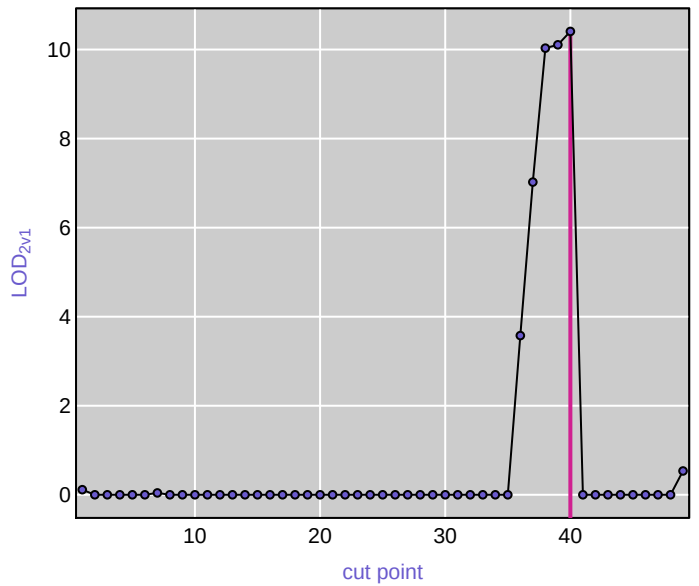
LDA plot



LOD profile

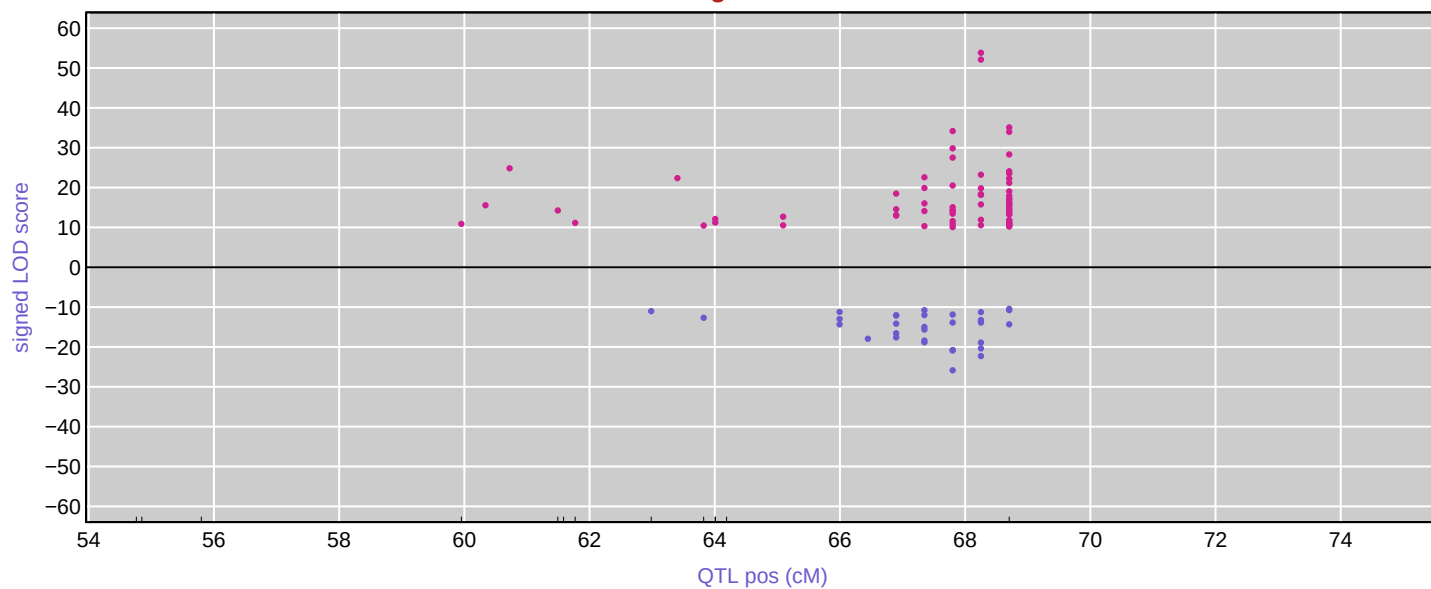


LOD_{2v1} by cut - point

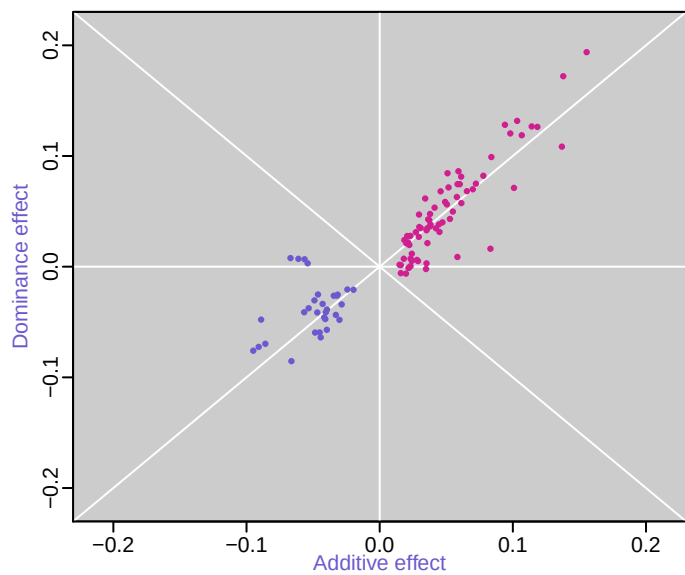


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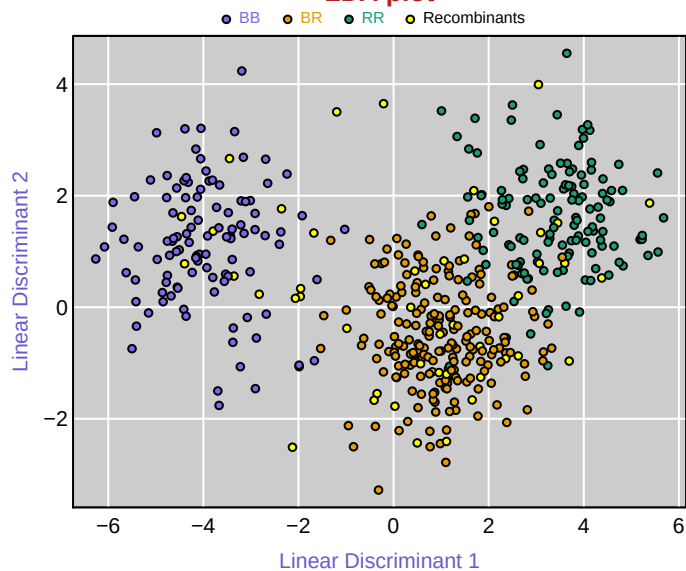
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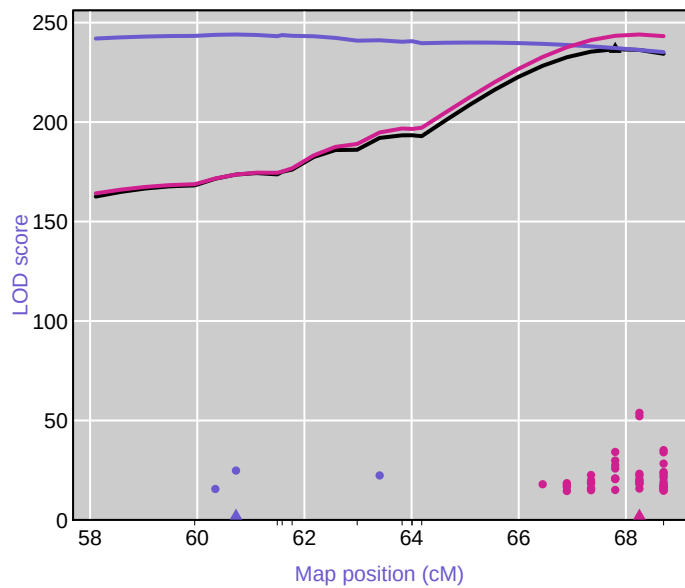
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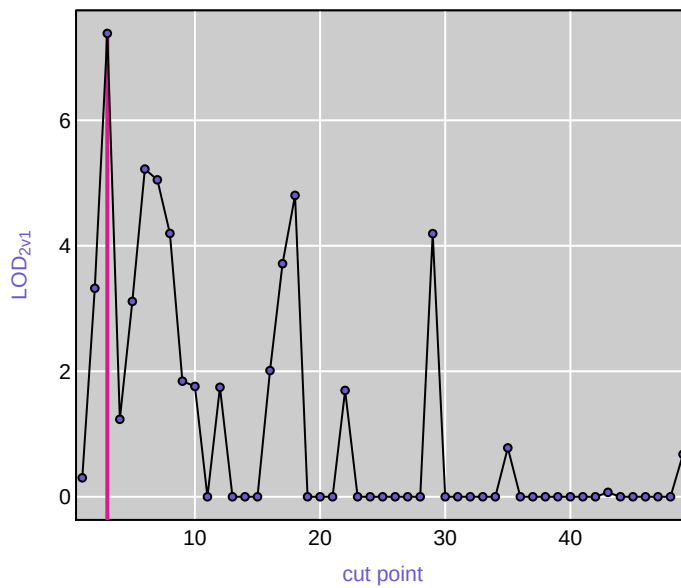
LDA plot



LOD profile

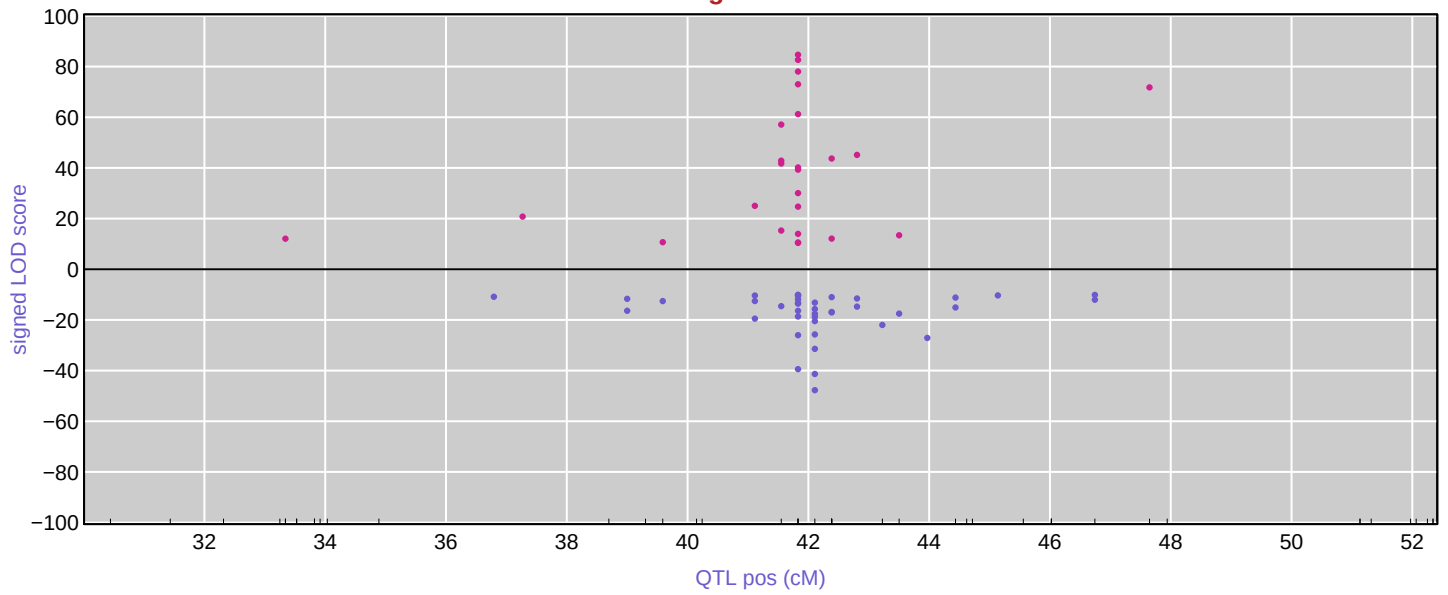


LOD_{2v1} by cut – point

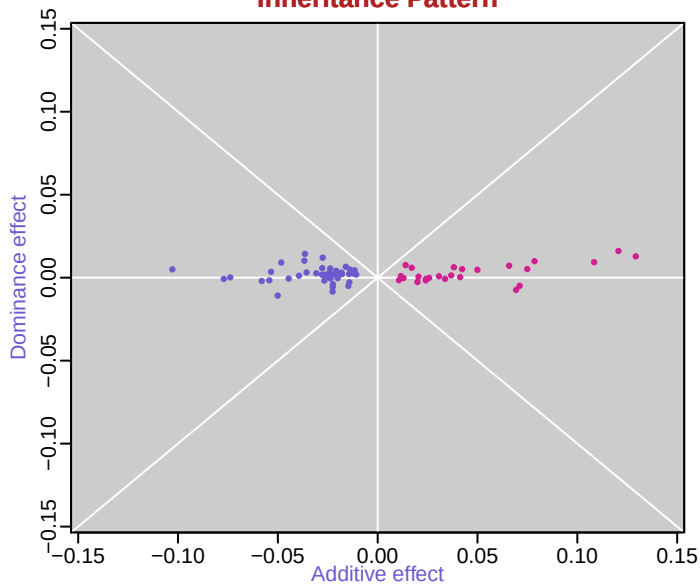


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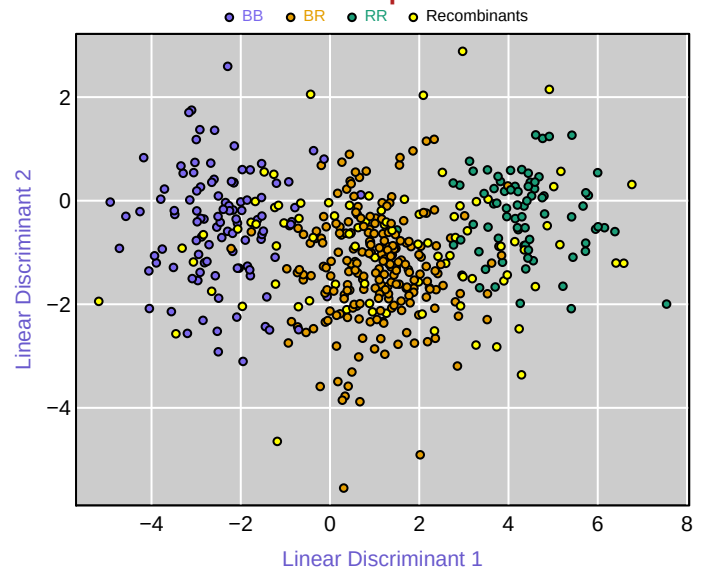
signed LOD



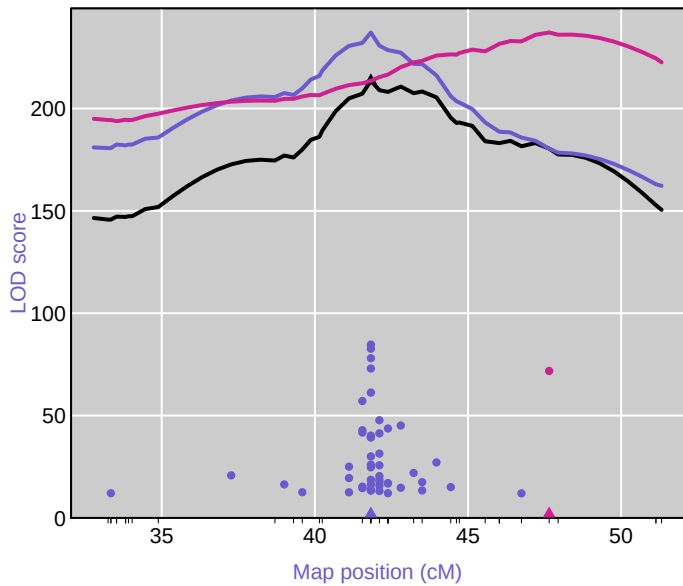
Inheritance Pattern



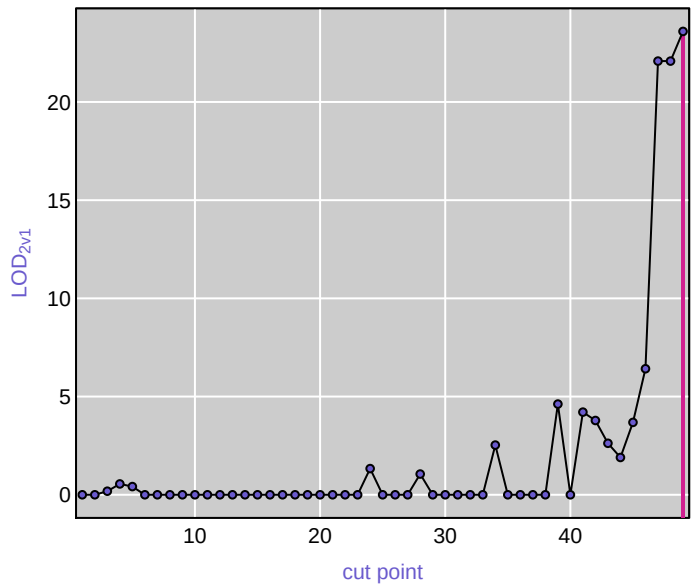
LDA plot



LOD profile

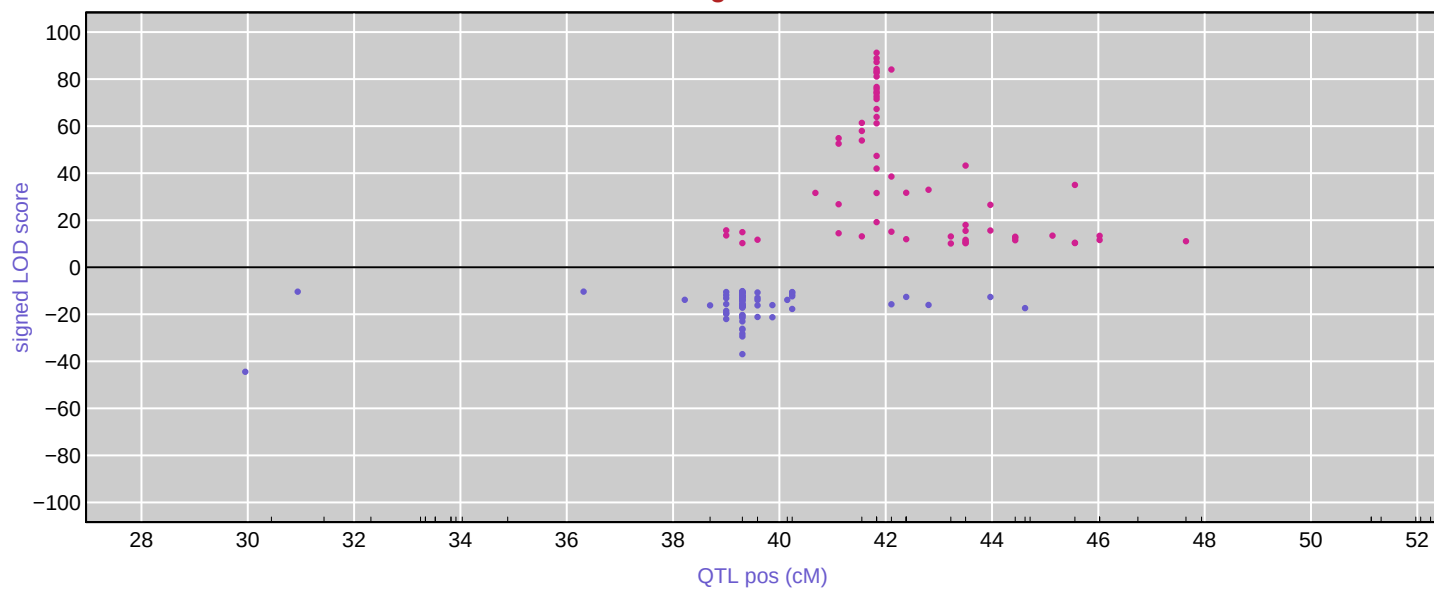


LOD_{2v1} by cut - point

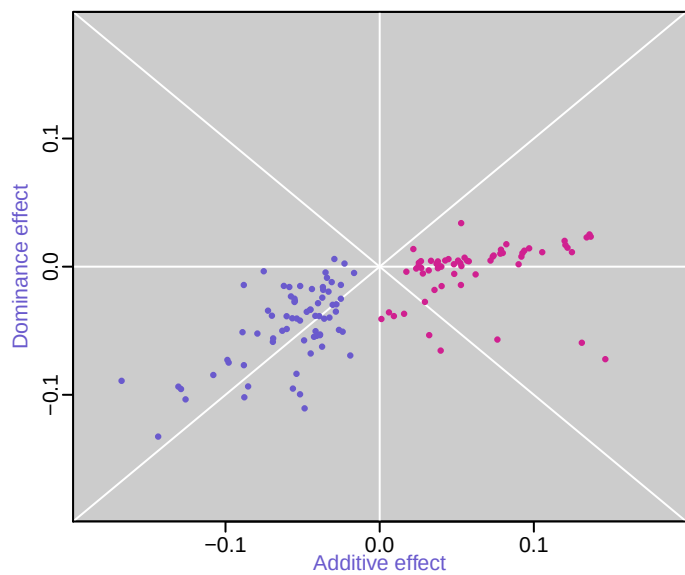


adipose chr1 @ 39 p-value=0

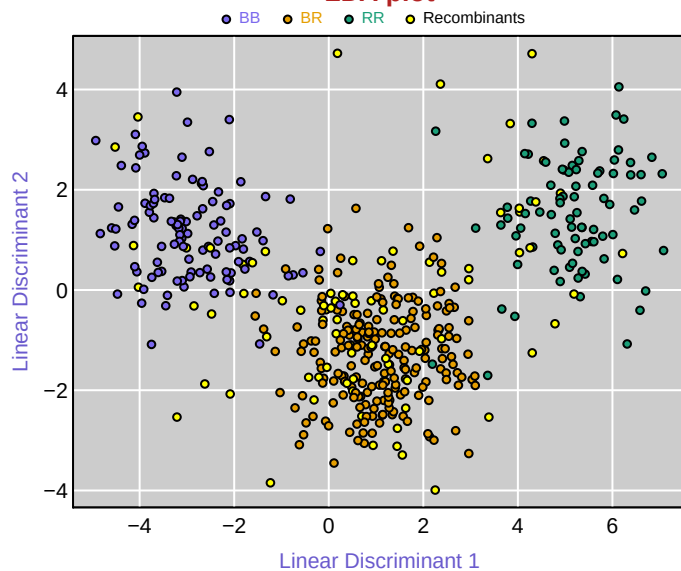
signed LOD



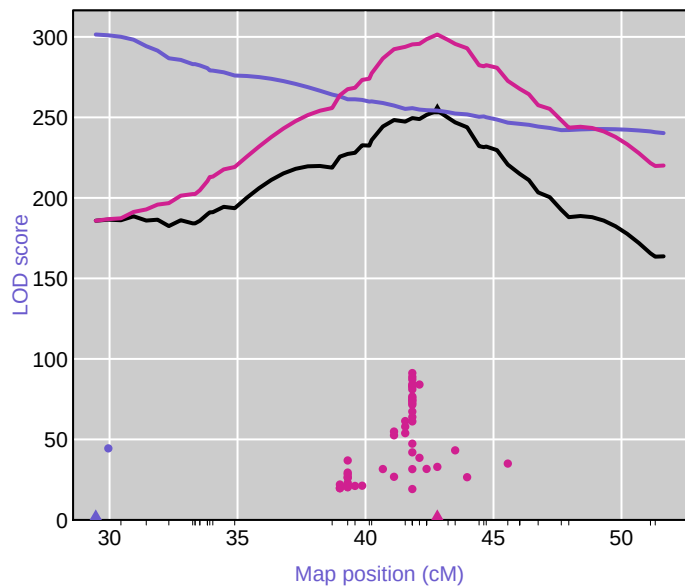
Inheritance Pattern



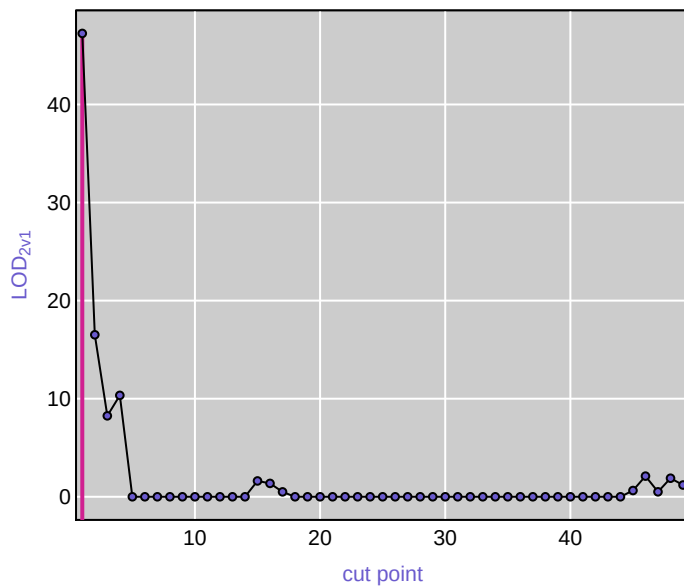
LDA plot



LOD profile

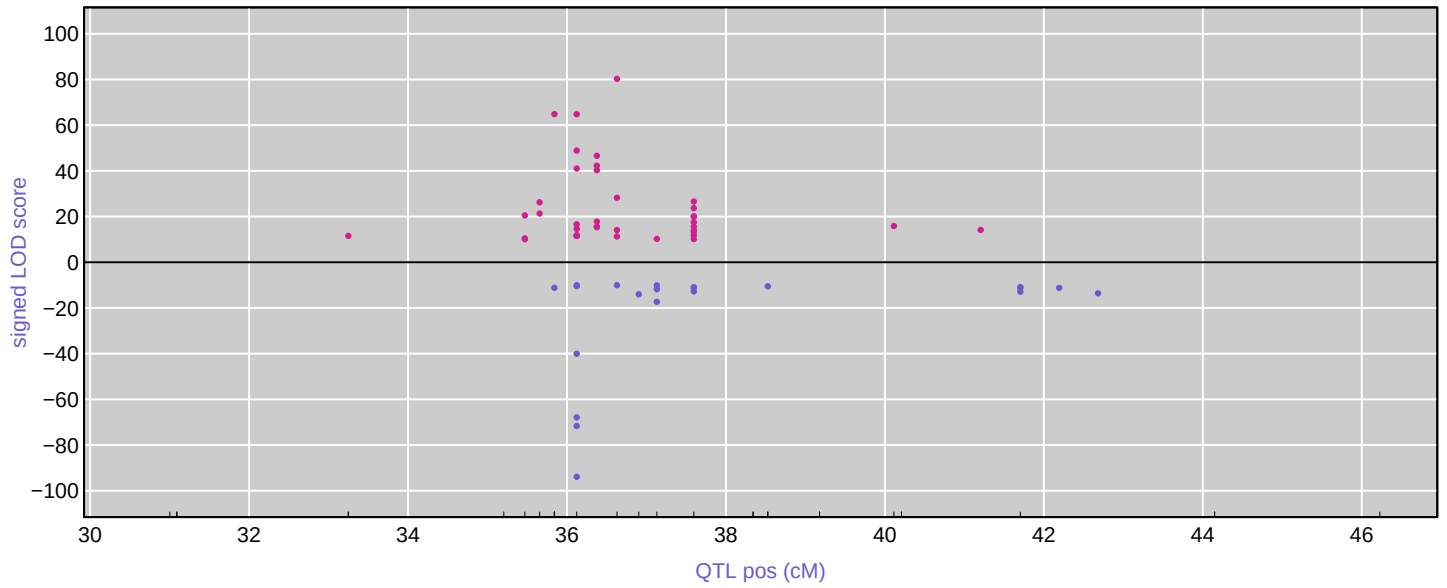


LOD_{2v1} by cut – point

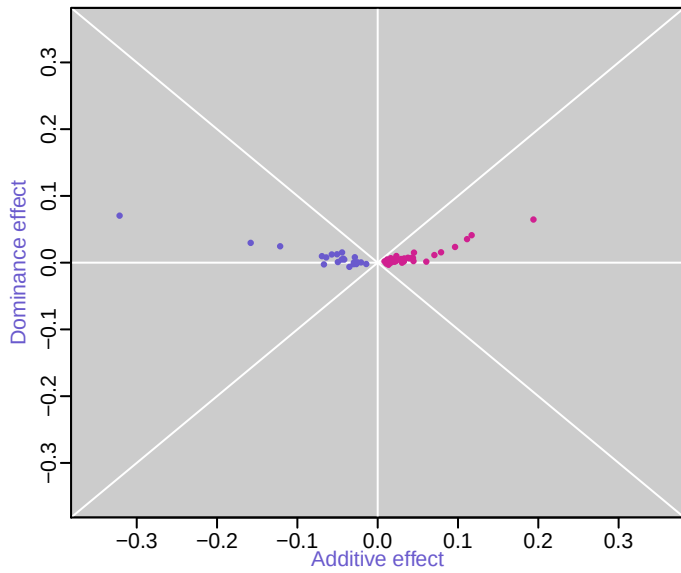


islet chr2 @ 36 p-value=0

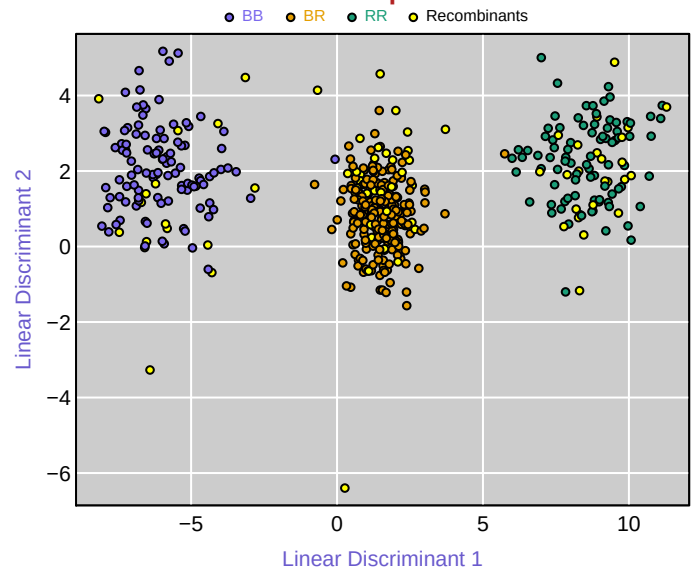
signed LOD



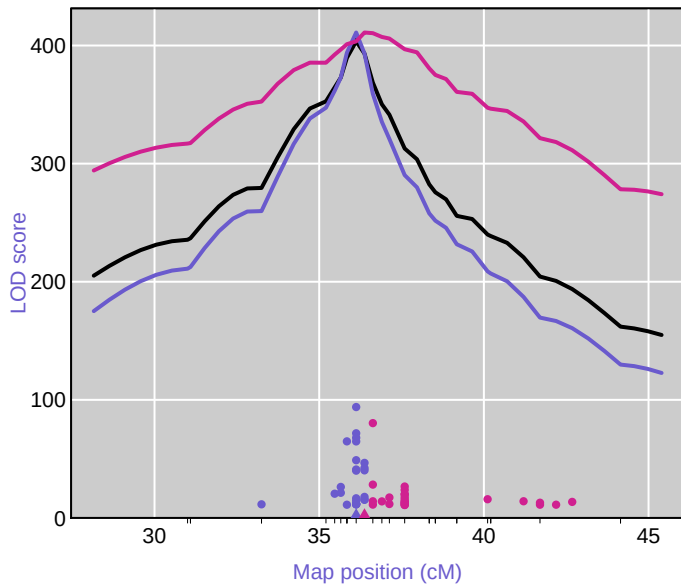
Inheritance Pattern



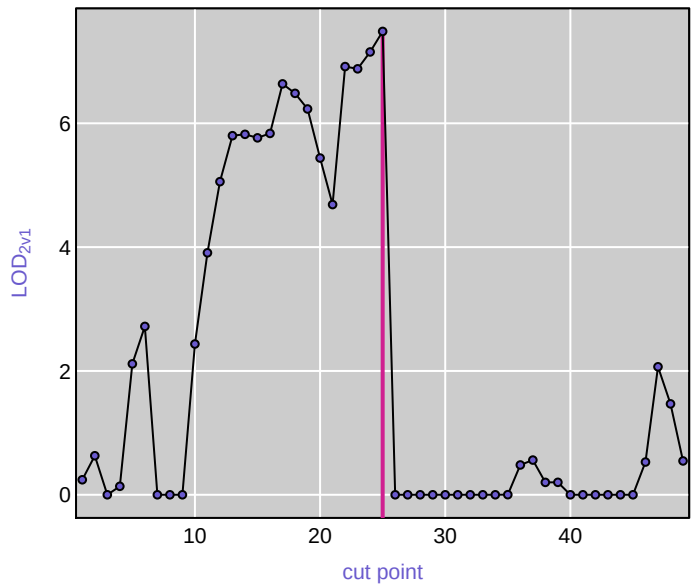
LDA plot



LOD profile

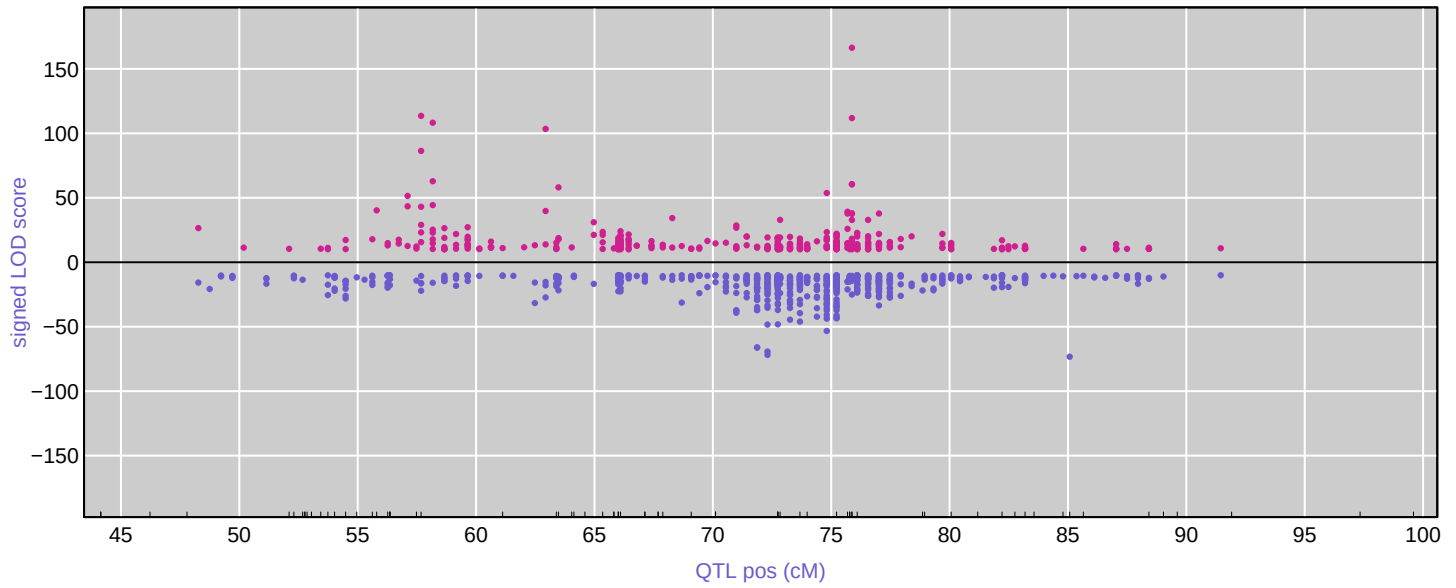


LOD_{2v1} by cut – point

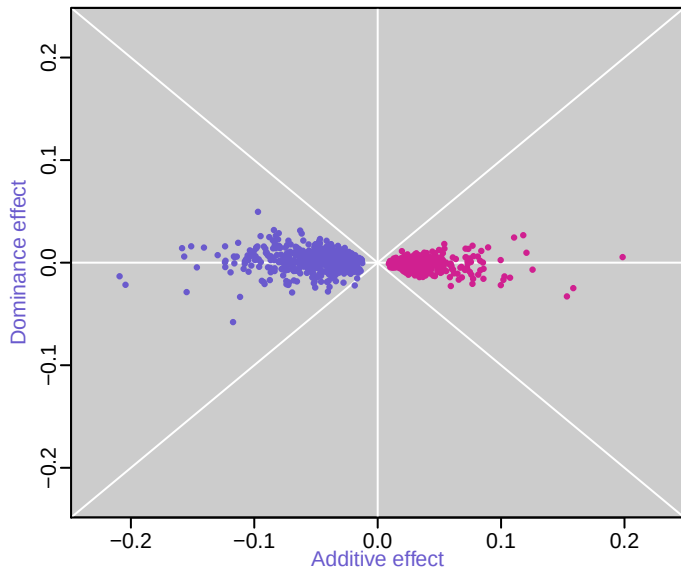


islet chr2 @ 75 p-value=0

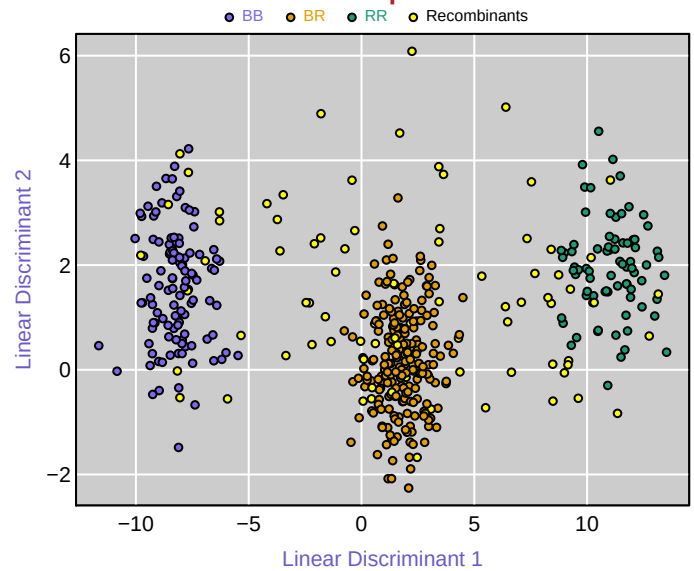
signed LOD



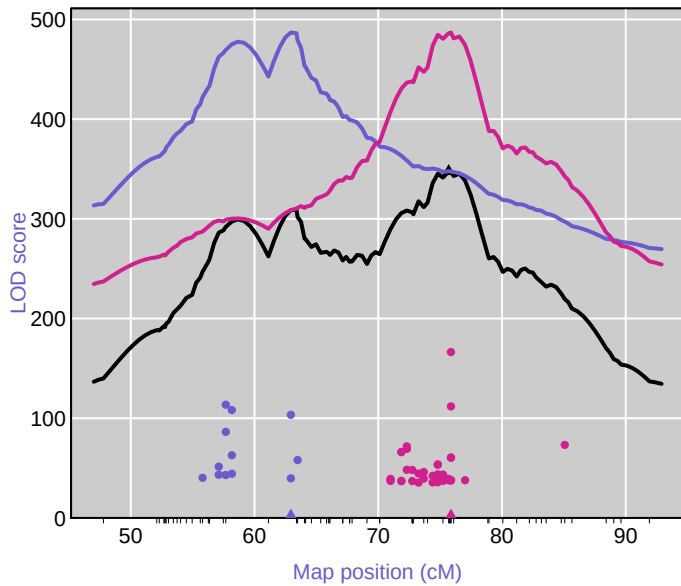
Inheritance Pattern



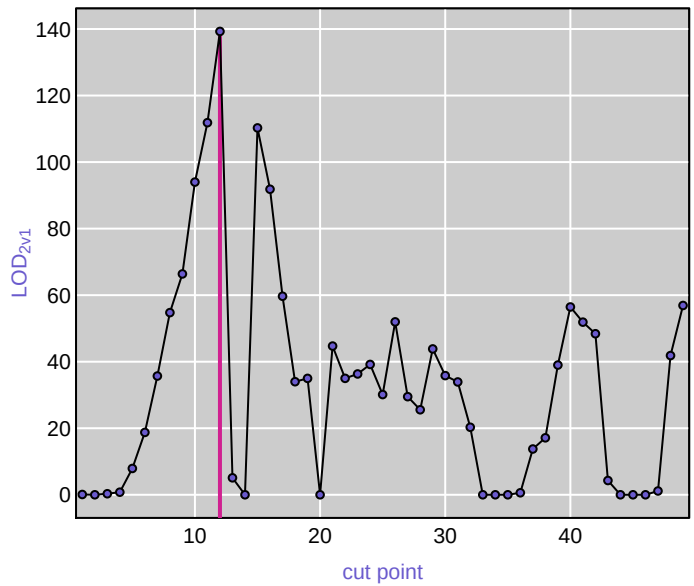
LDA plot



LOD profile

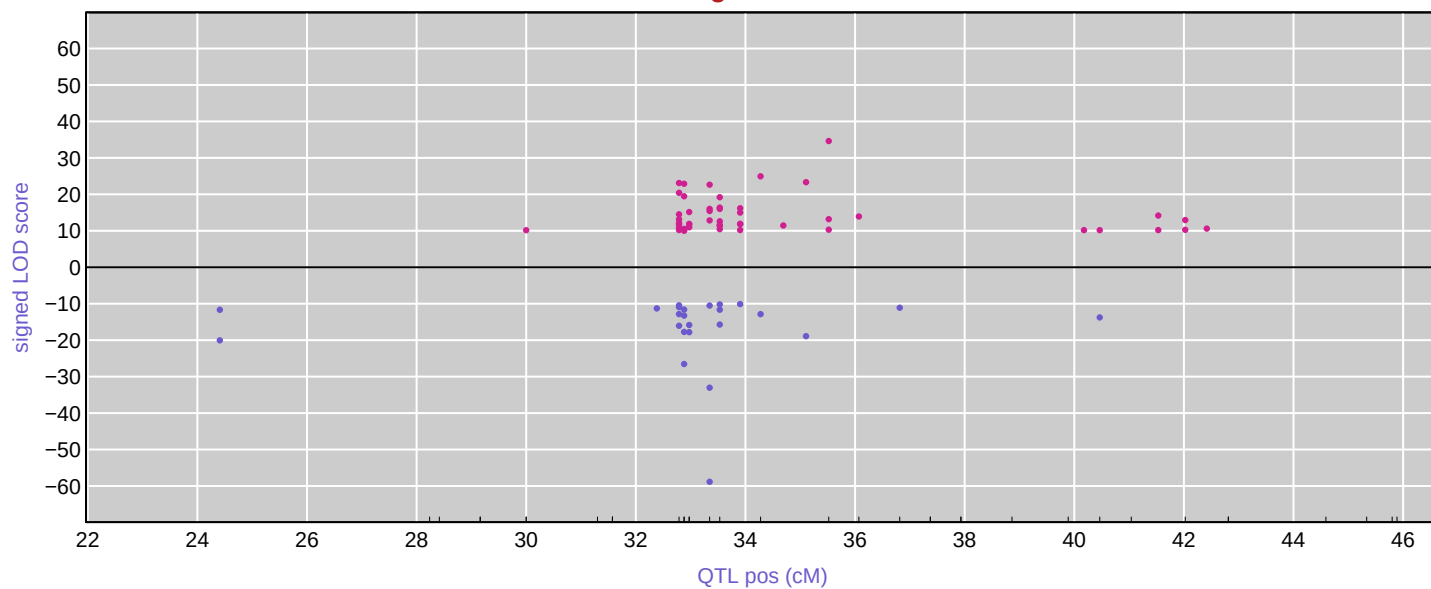


LOD_{2v1} by cut – point

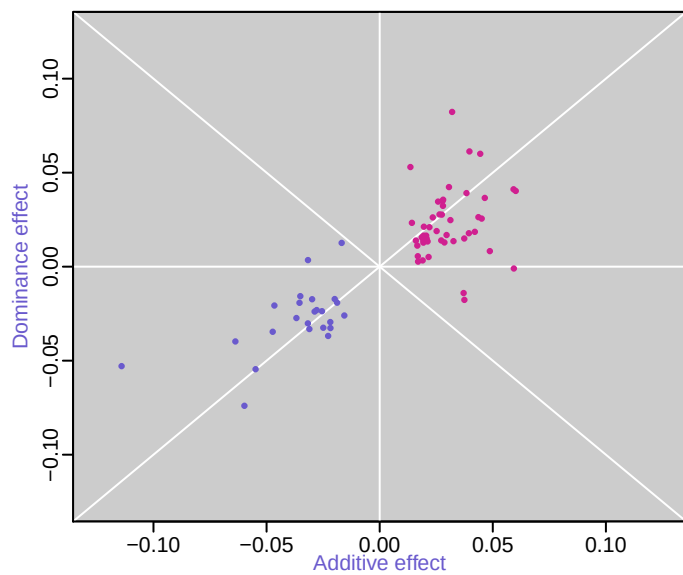


kidney chr4 @ 32 p-value=0

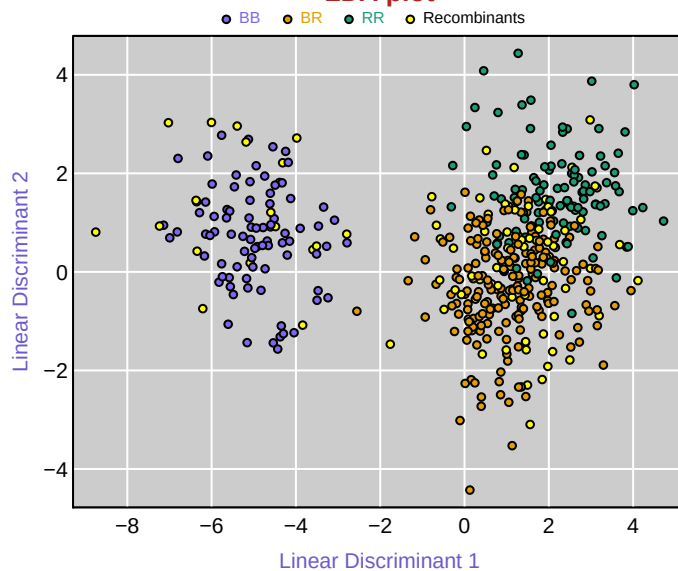
signed LOD



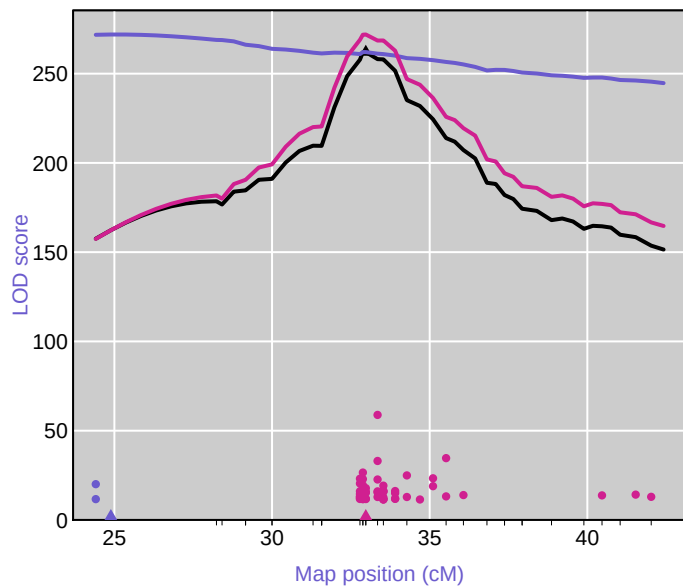
Inheritance Pattern



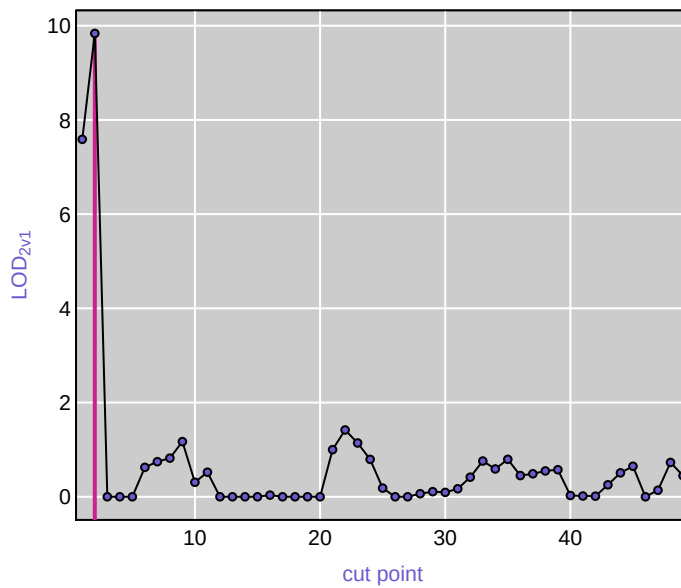
LDA plot



LOD profile

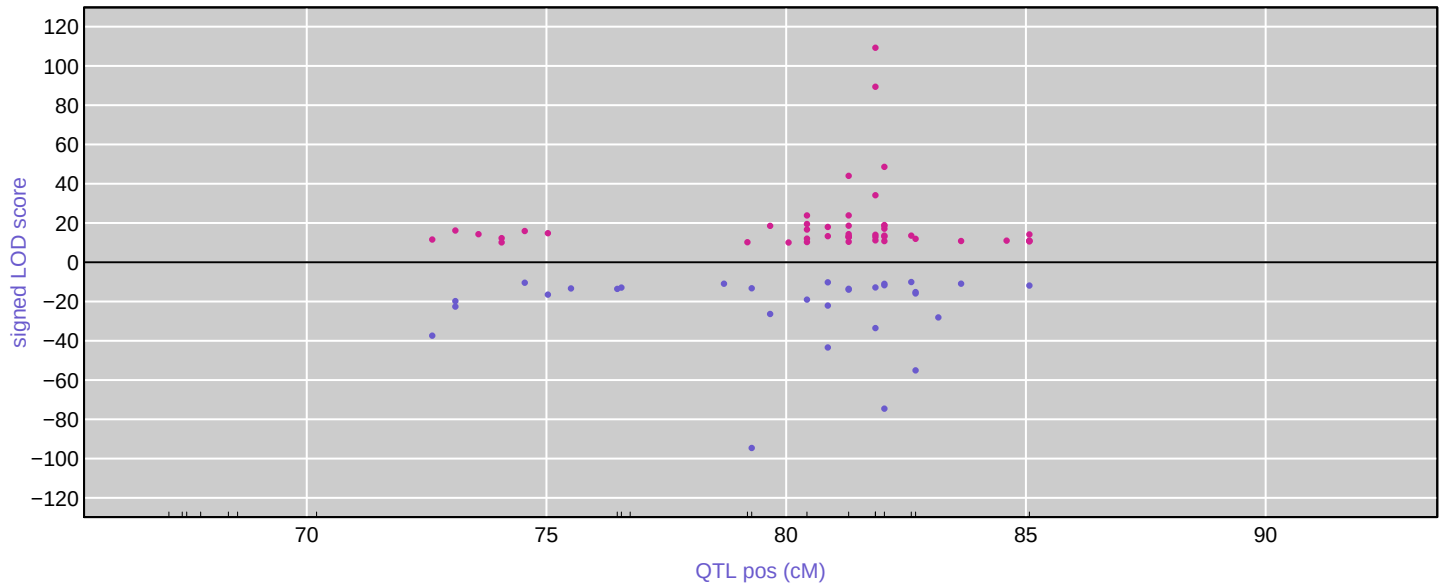


LOD_{2v1} by cut - point

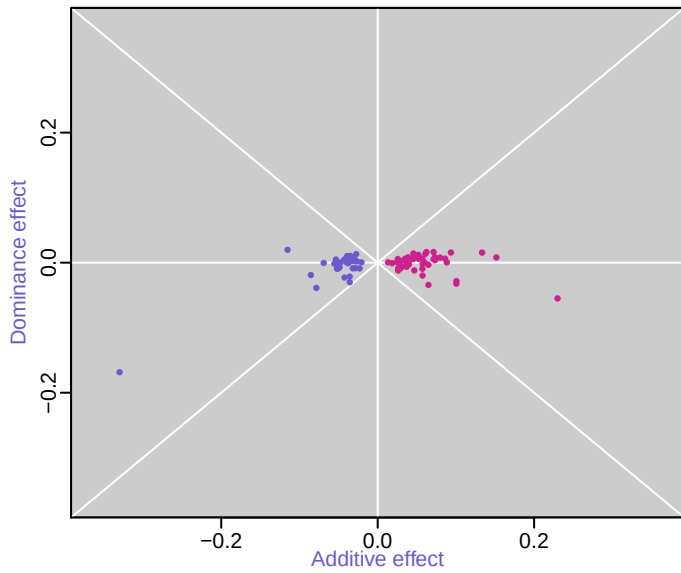


islet chr4 @ 81 p-value=0

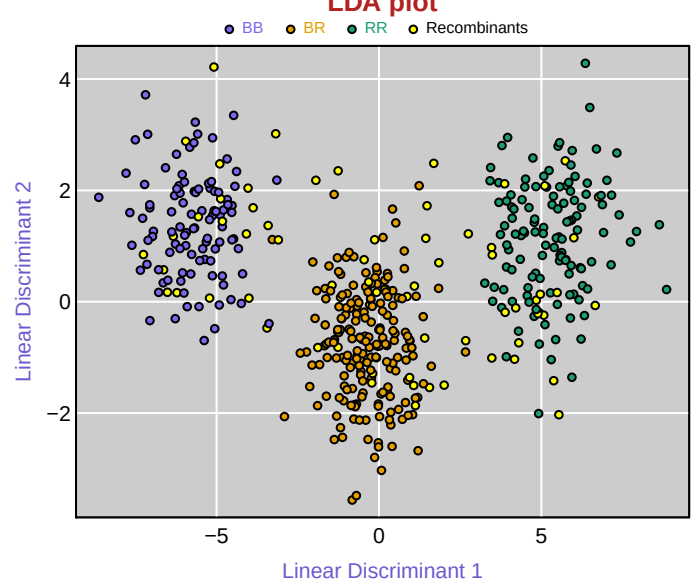
signed LOD



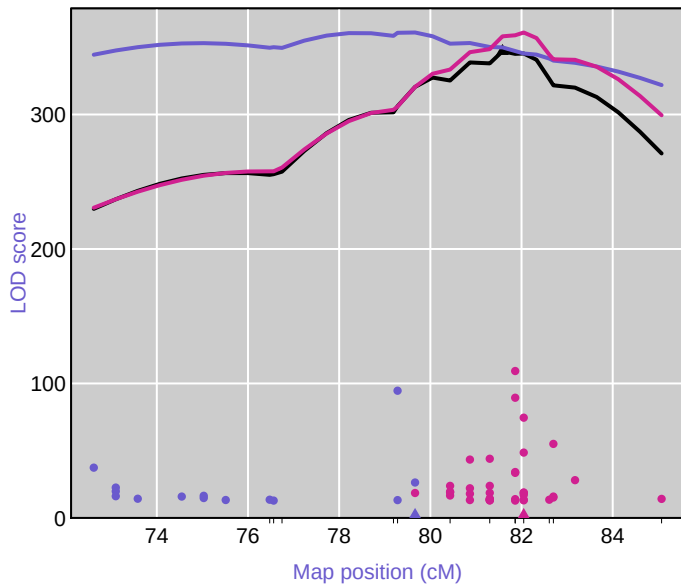
Inheritance Pattern



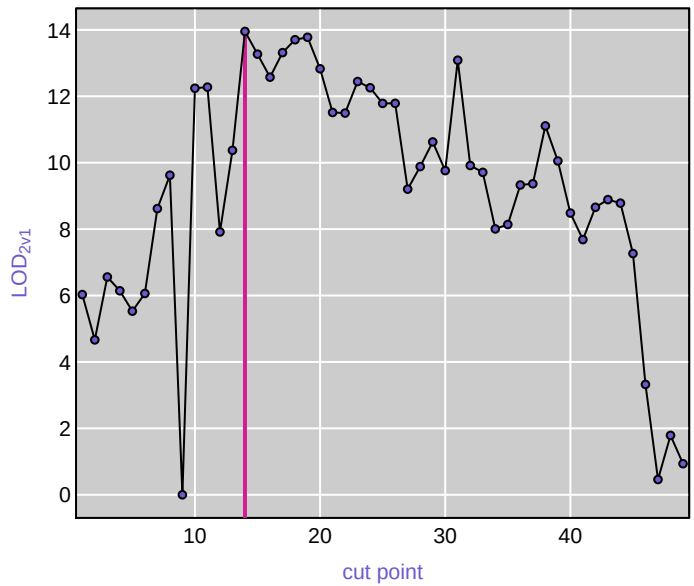
LDA plot



LOD profile

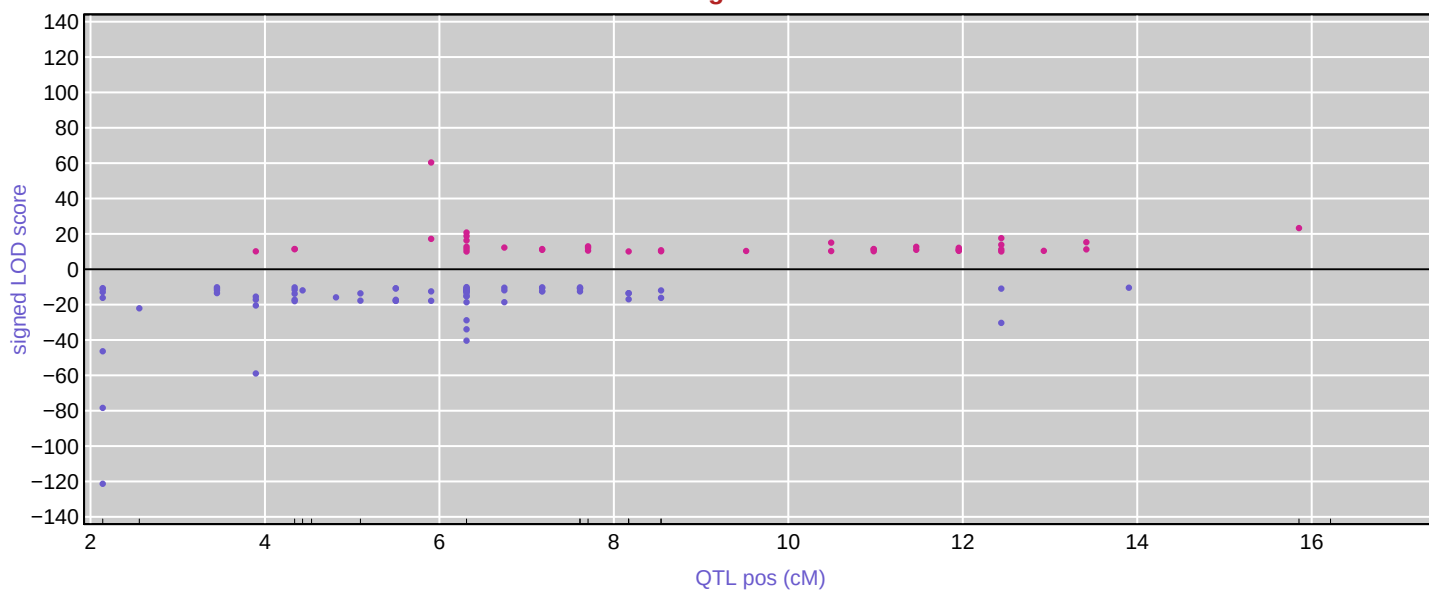


LOD_{2v1} by cut - point

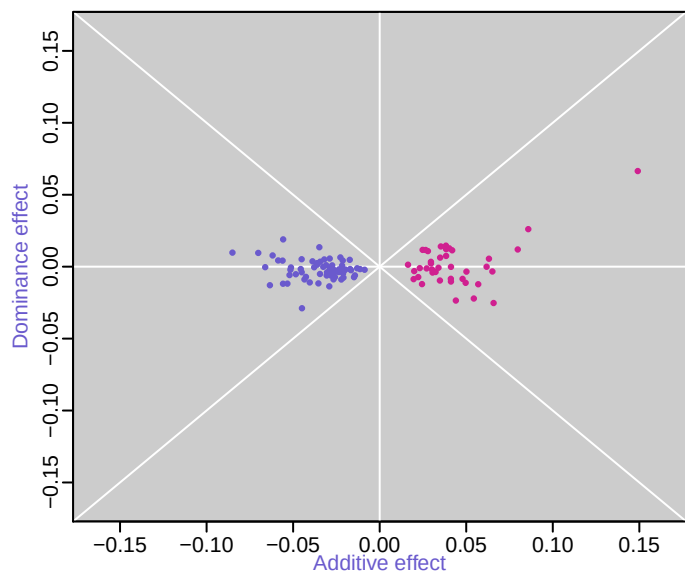


islet chr7 @ 6 p-value=0

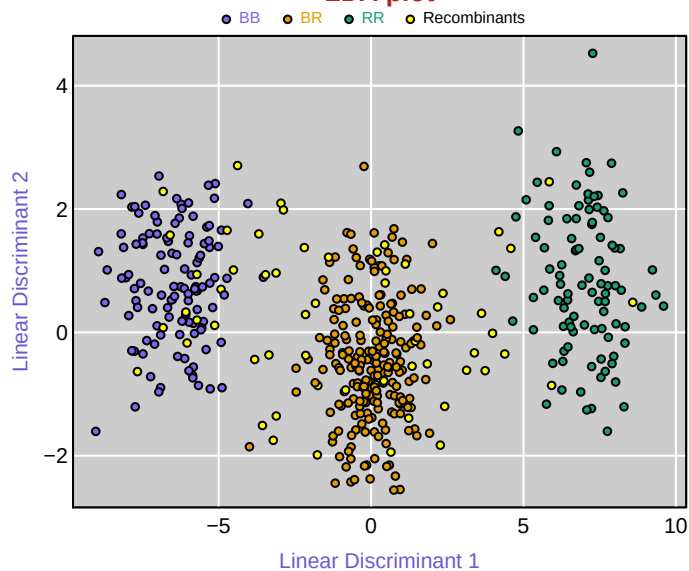
signed LOD



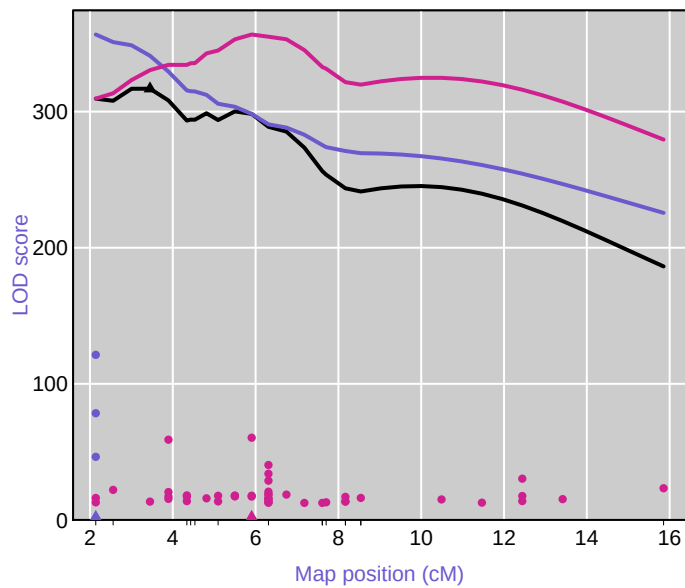
Inheritance Pattern



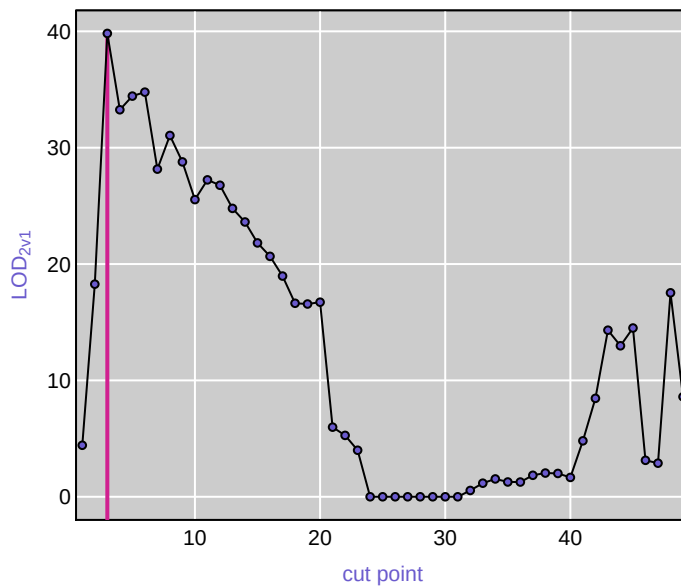
LDA plot



LOD profile

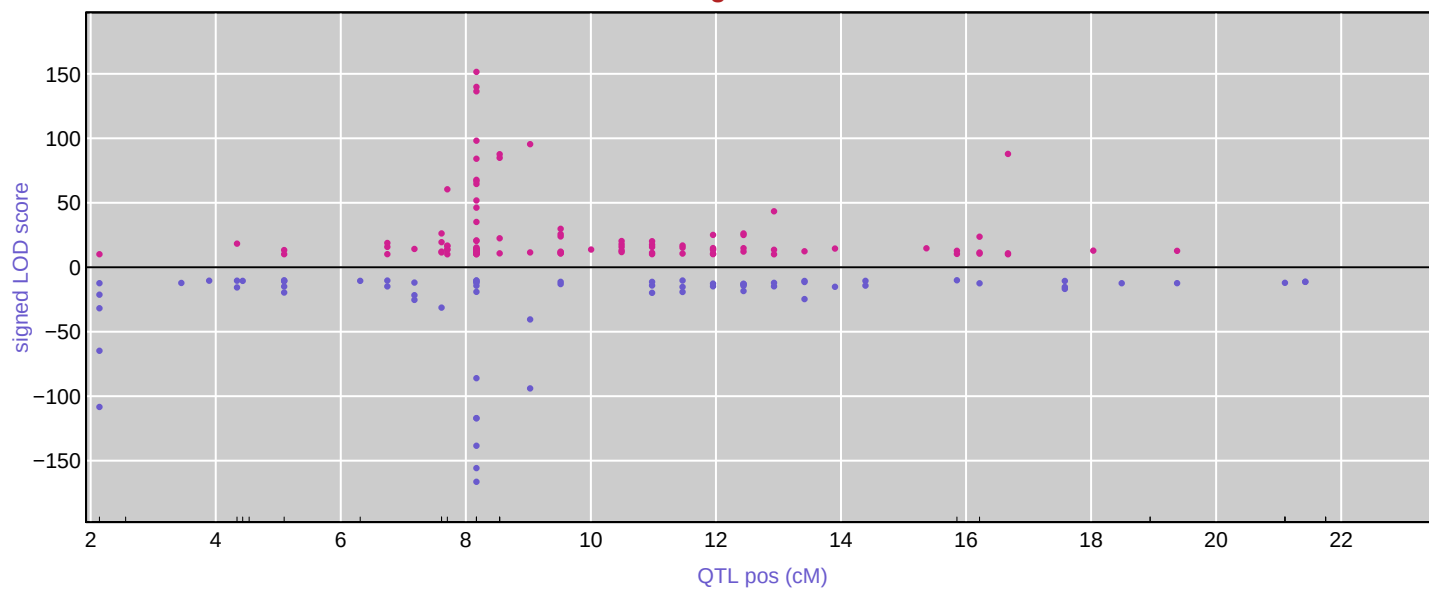


LOD_{2v1} by cut – point

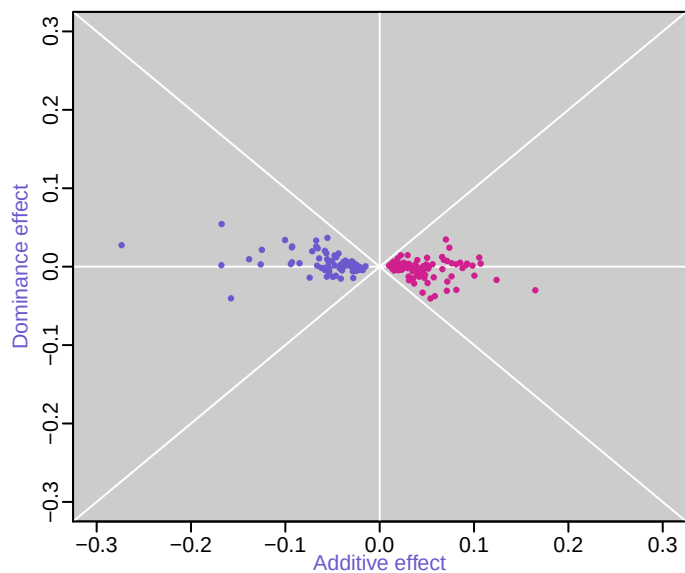


liver chr7 @ 8 p-value=0

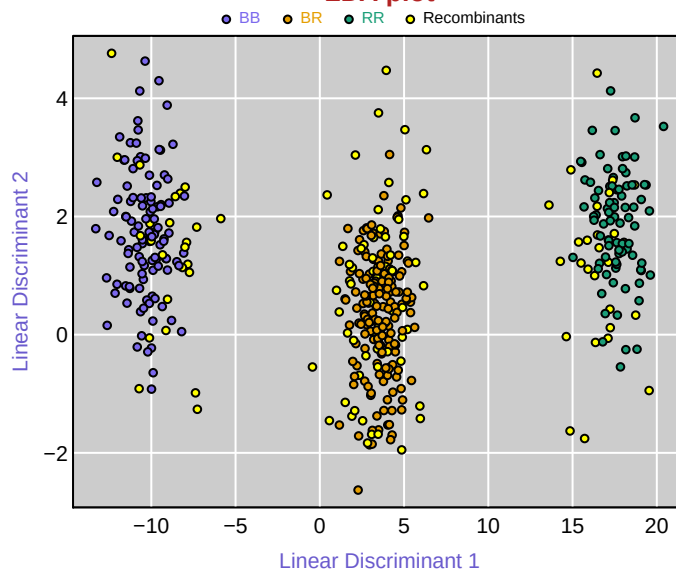
signed LOD



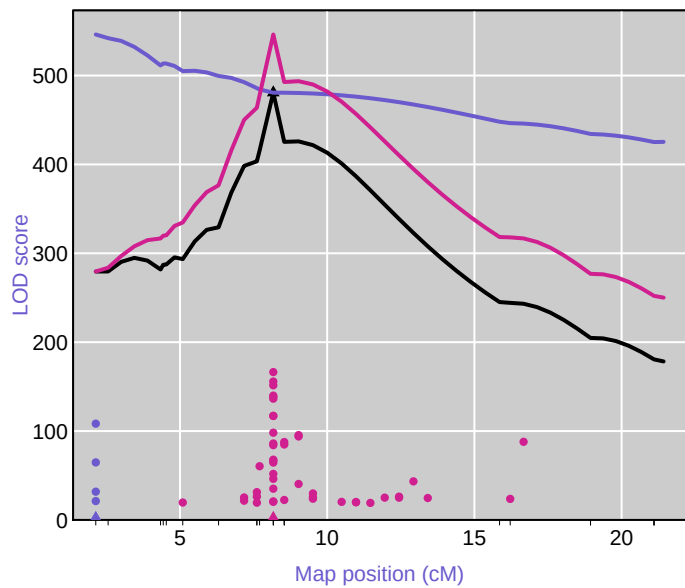
Inheritance Pattern



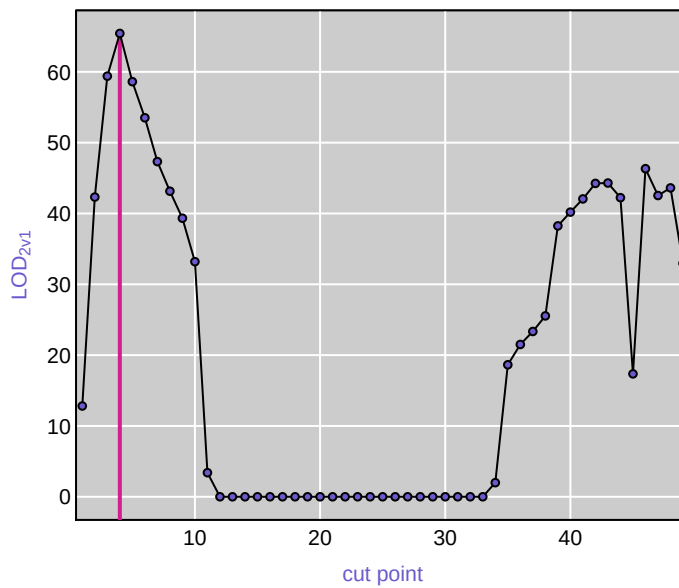
LDA plot



LOD profile

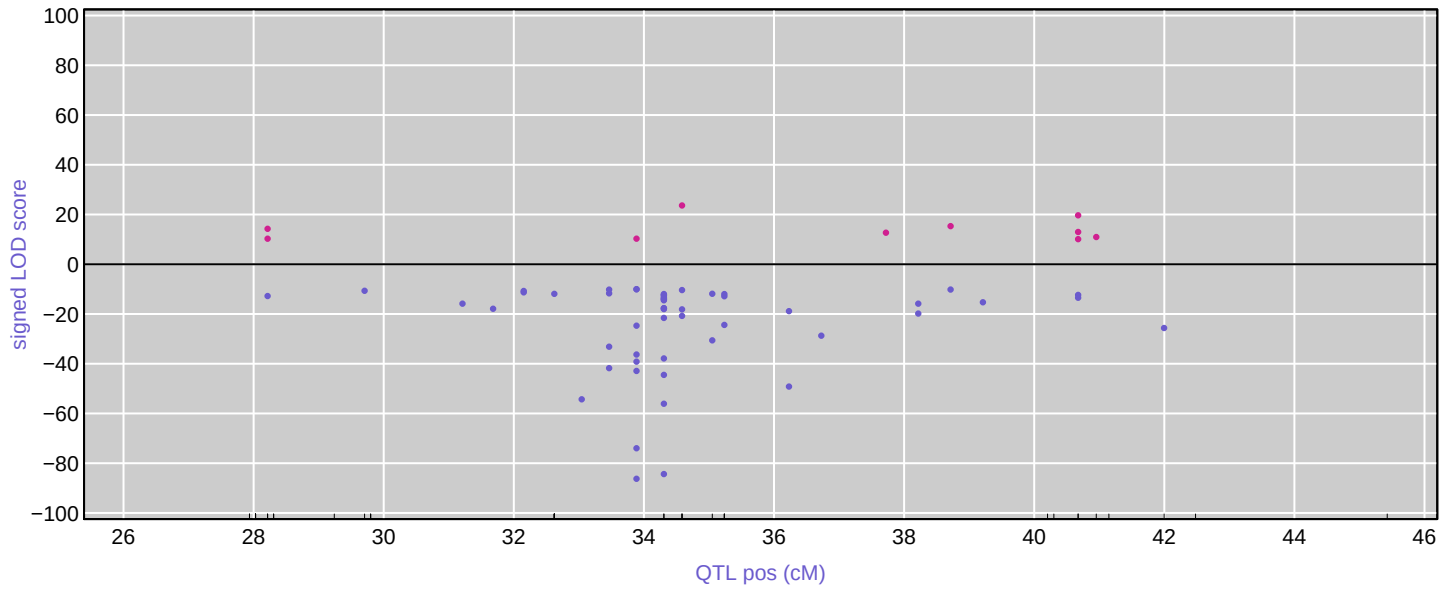


LOD_{2v1} by cut - point

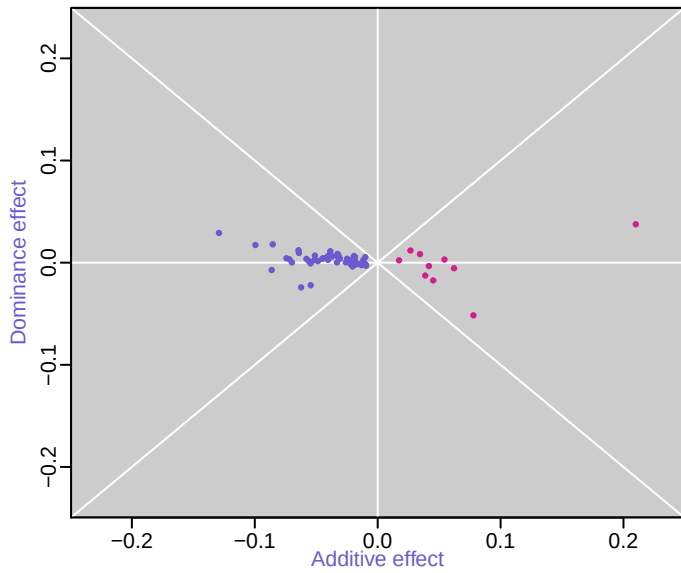


islet chr8 @ 34 p-value=0

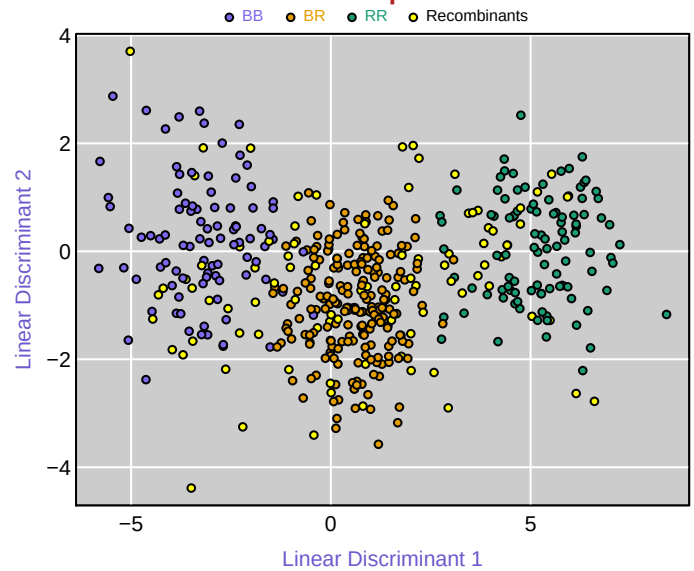
signed LOD



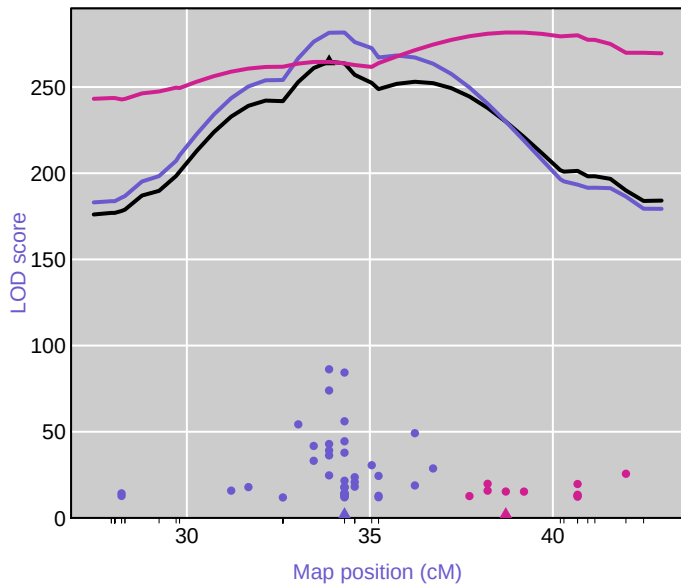
Inheritance Pattern



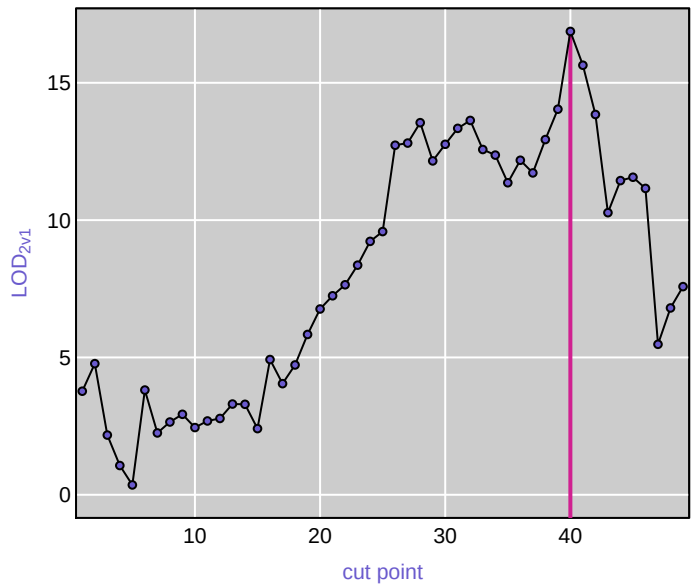
LDA plot



LOD profile

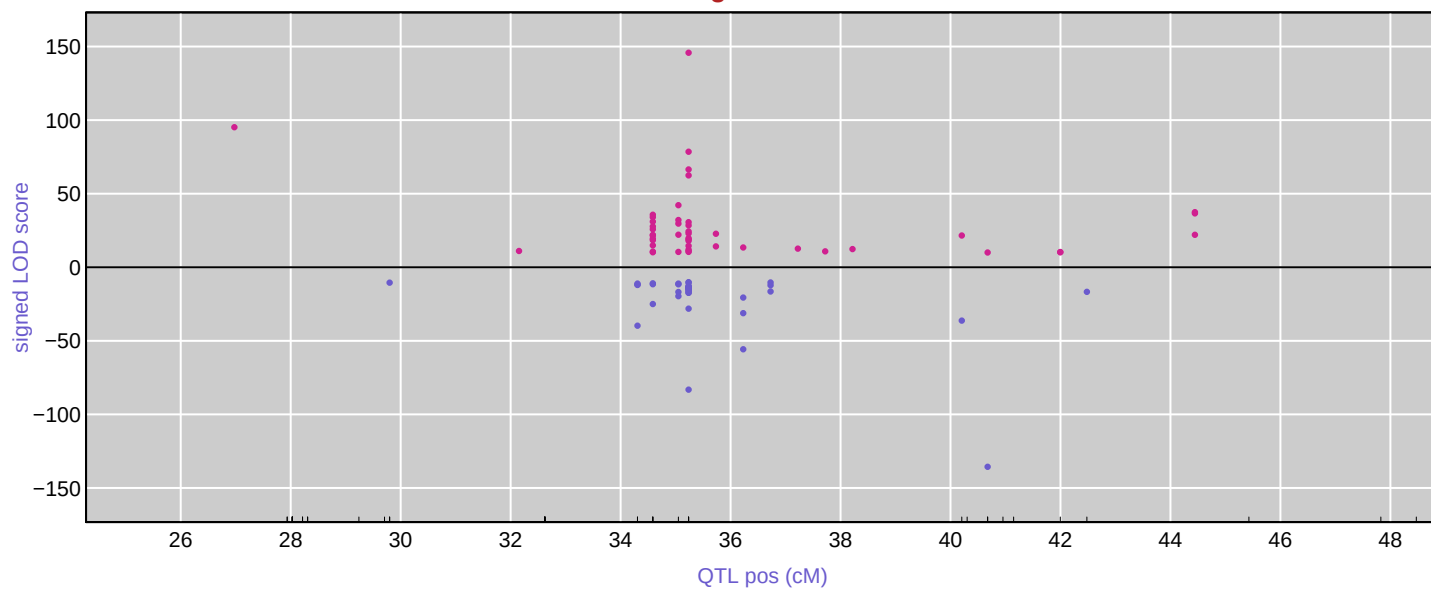


LOD_{2v1} by cut – point

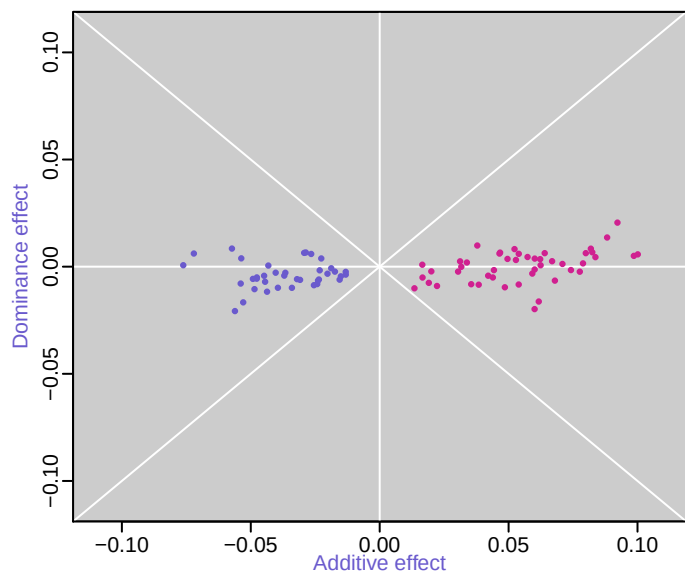


adipose chr8 @ 35 p-value=0

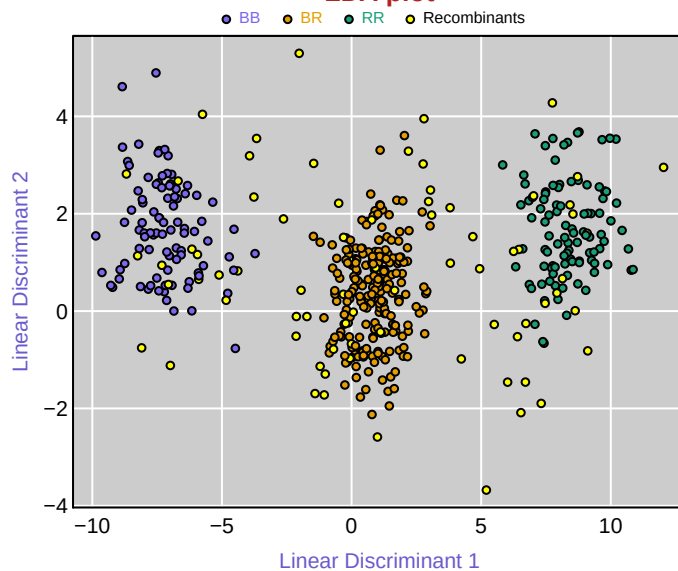
signed LOD



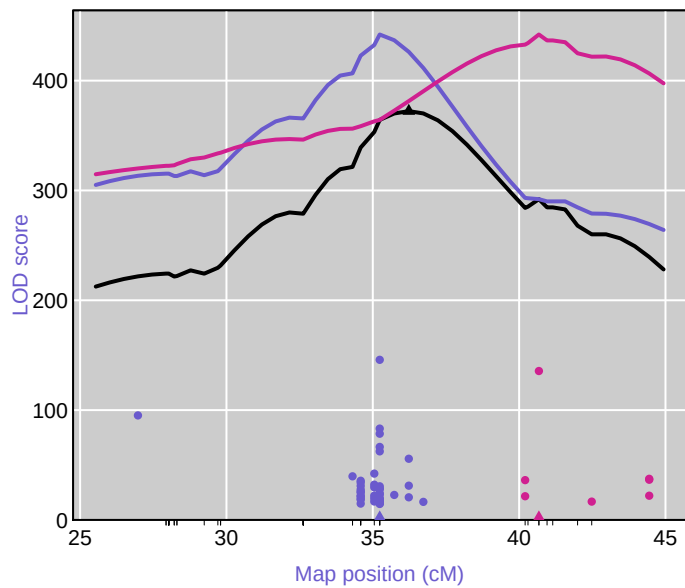
Inheritance Pattern



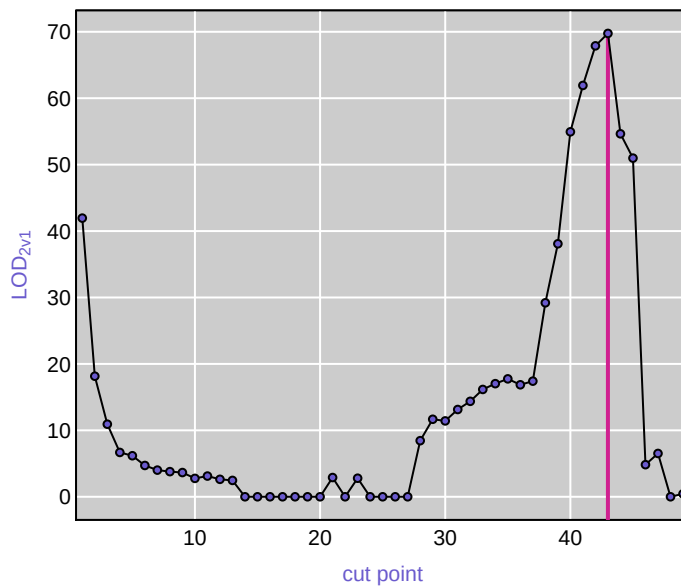
LDA plot



LOD profile

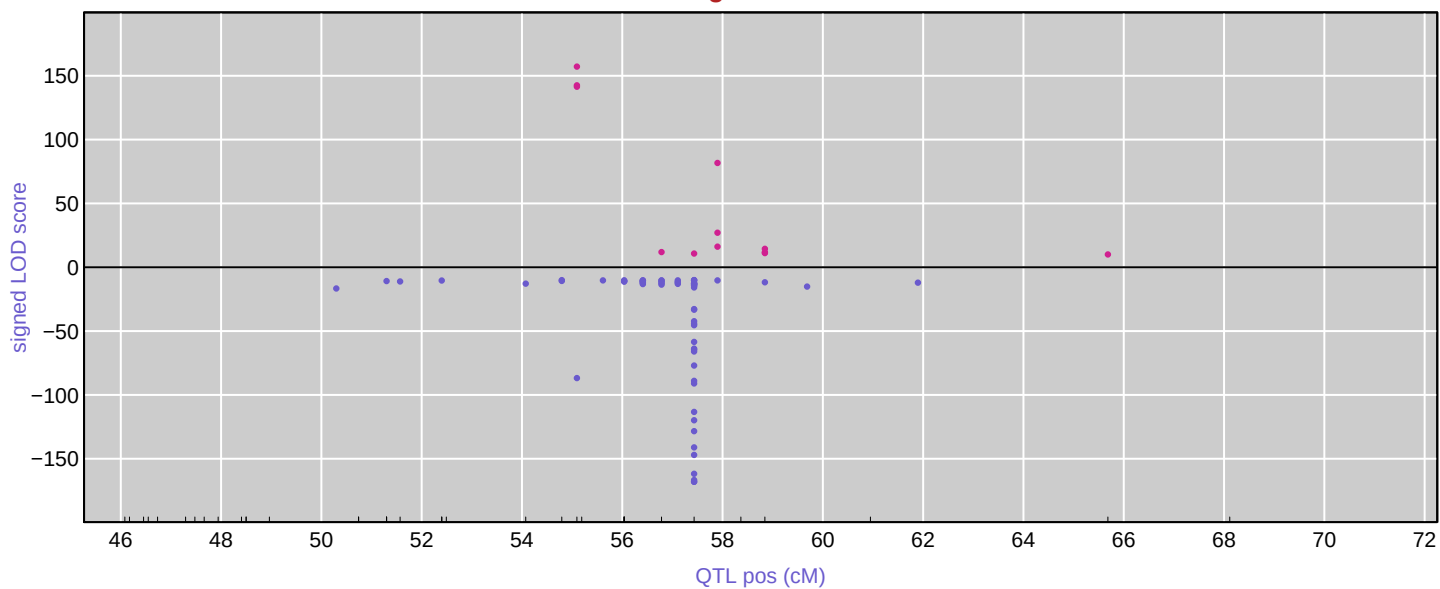


LOD_{2v1} by cut - point

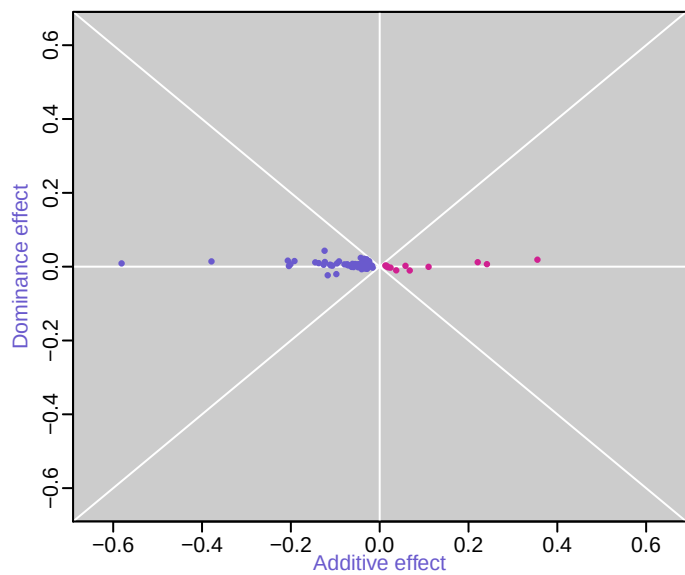


adipose chr9 @ 57 p-value=0

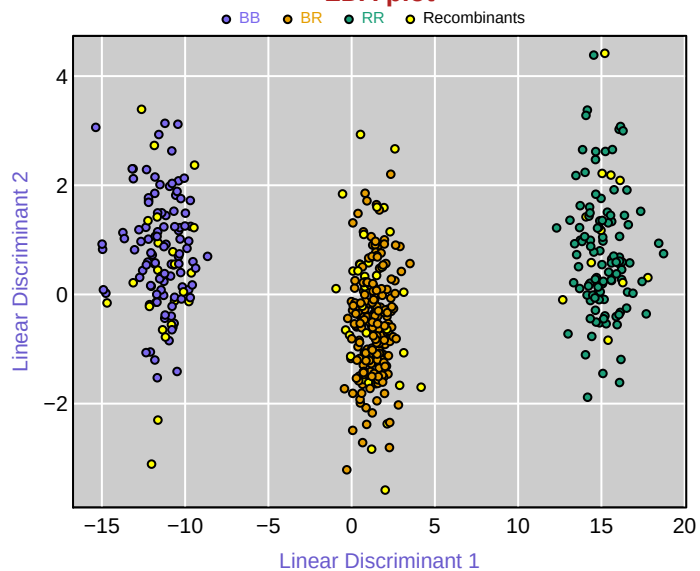
signed LOD



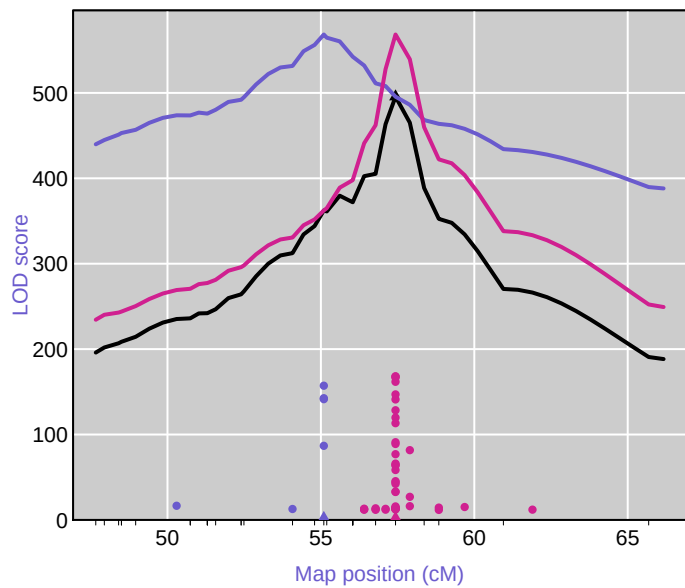
Inheritance Pattern



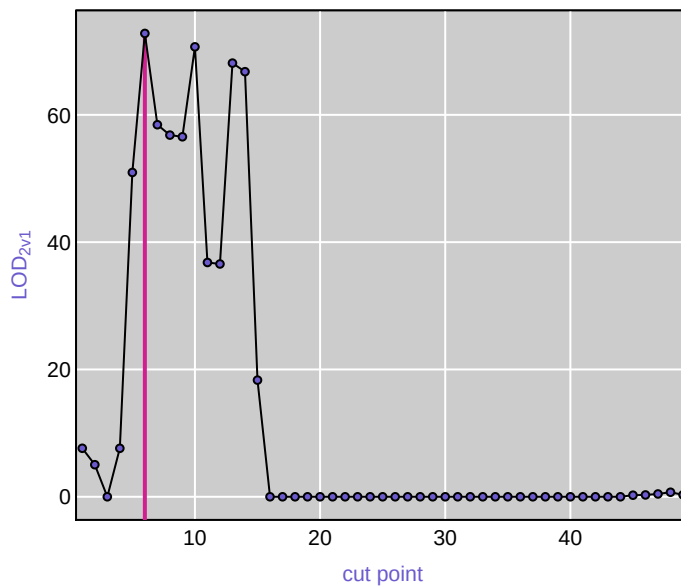
LDA plot



LOD profile

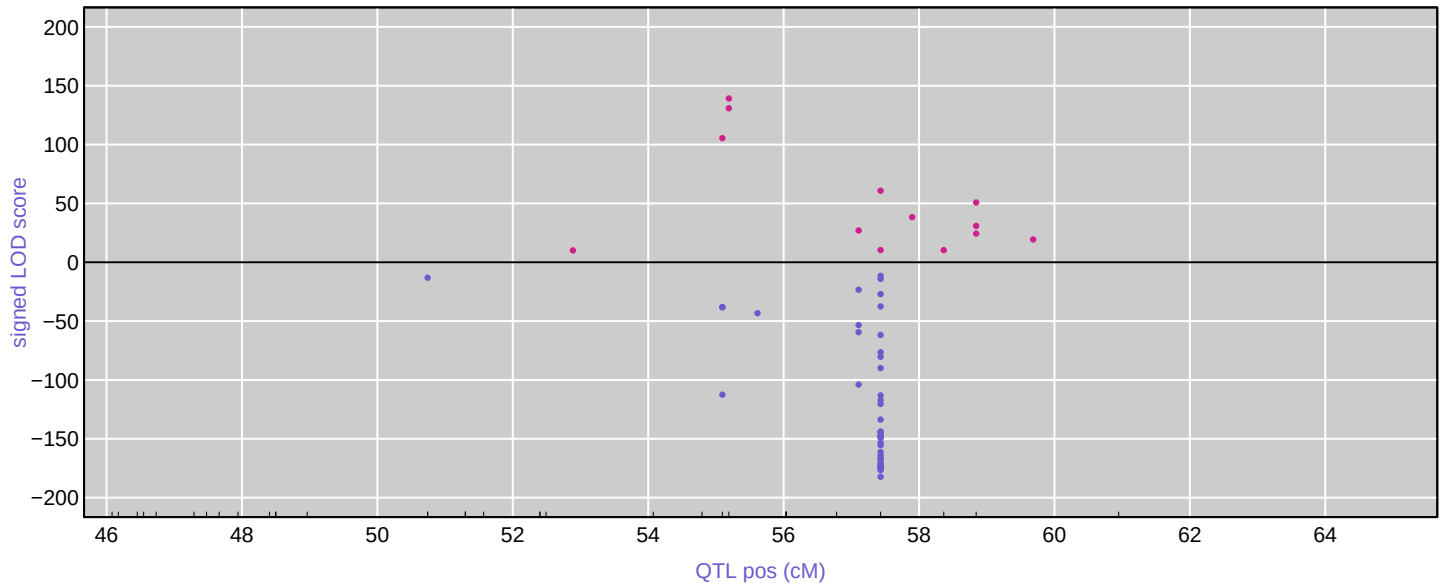


LOD_{2v1} by cut - point

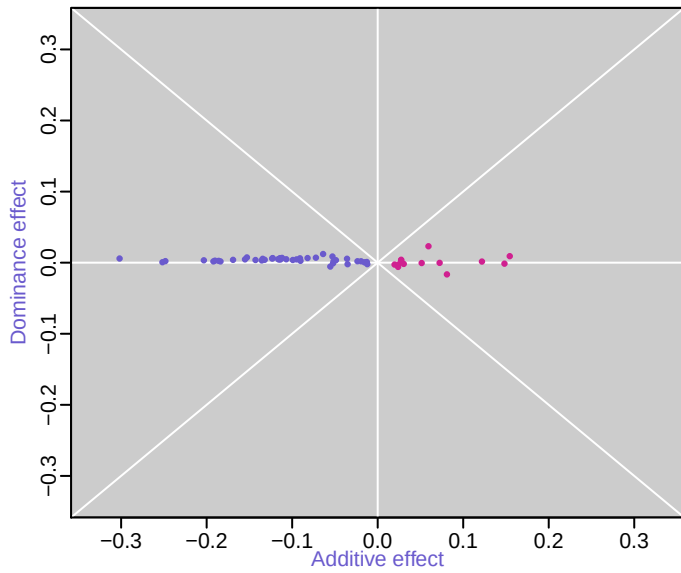


islet chr9 @ 57 p-value=0

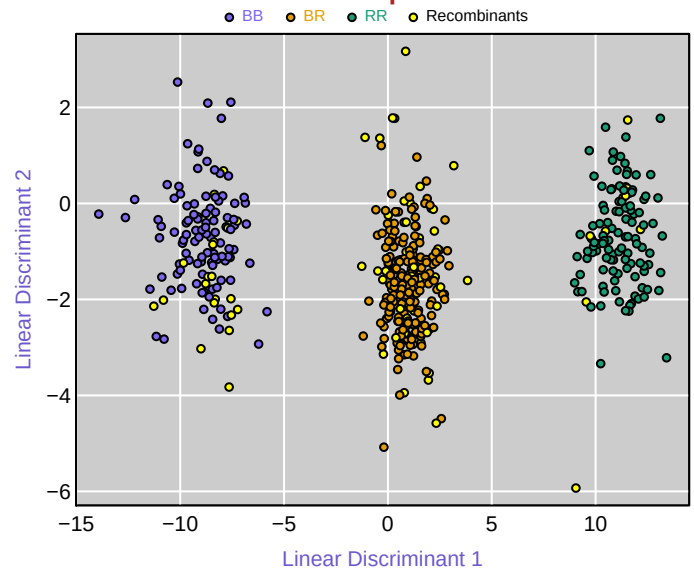
signed LOD



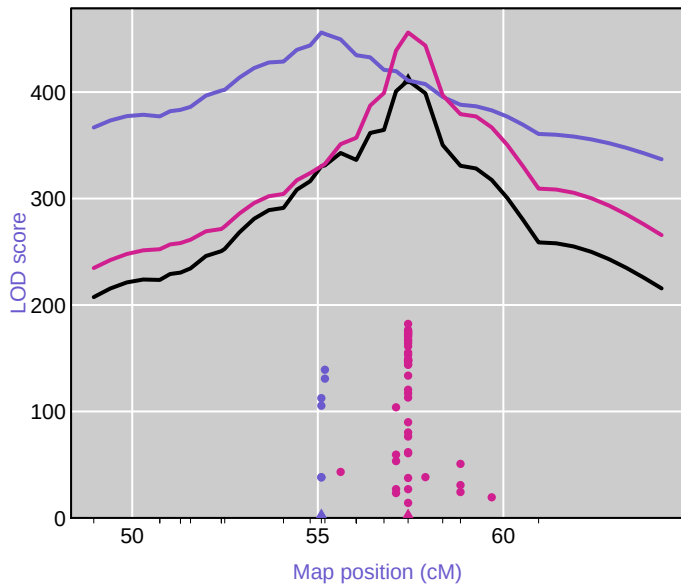
Inheritance Pattern



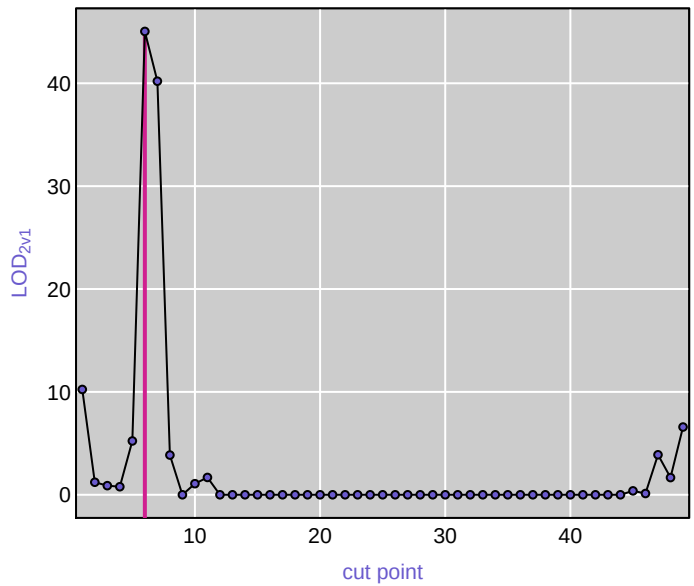
LDA plot



LOD profile

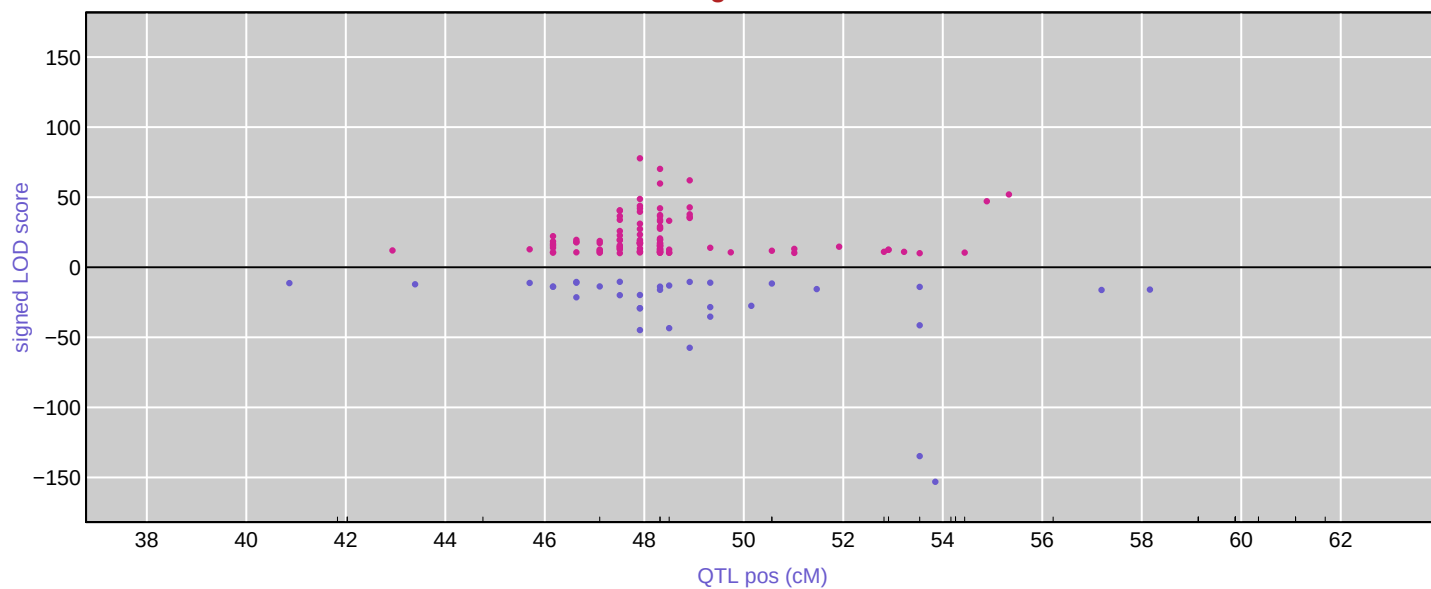


LOD_{2v1} by cut – point

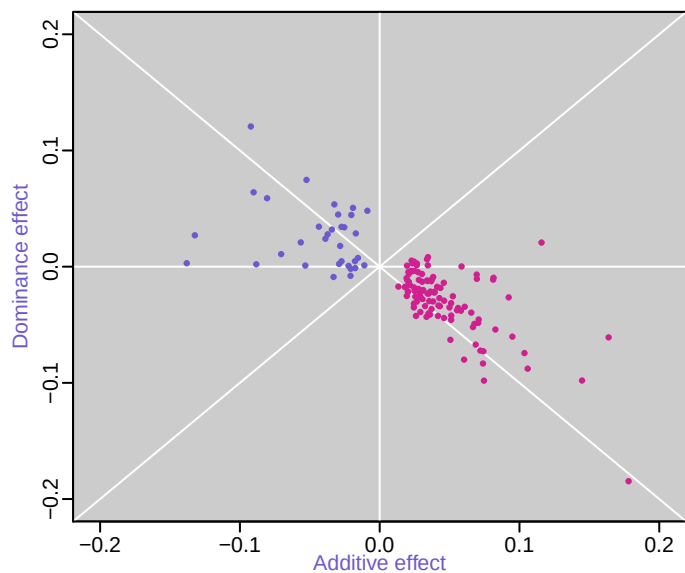


kidney chr10 @ 47 p-value=0

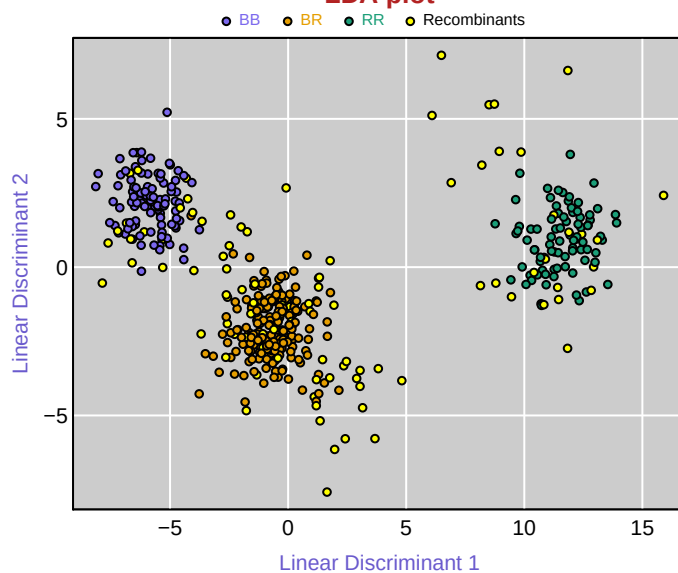
signed LOD



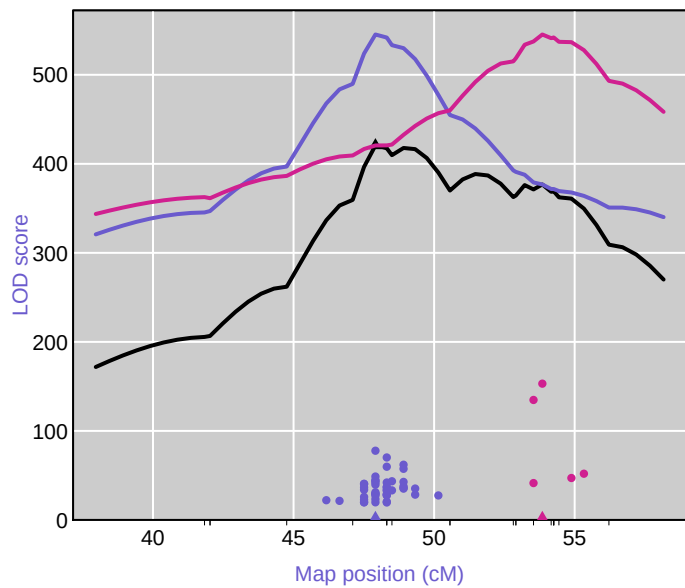
Inheritance Pattern



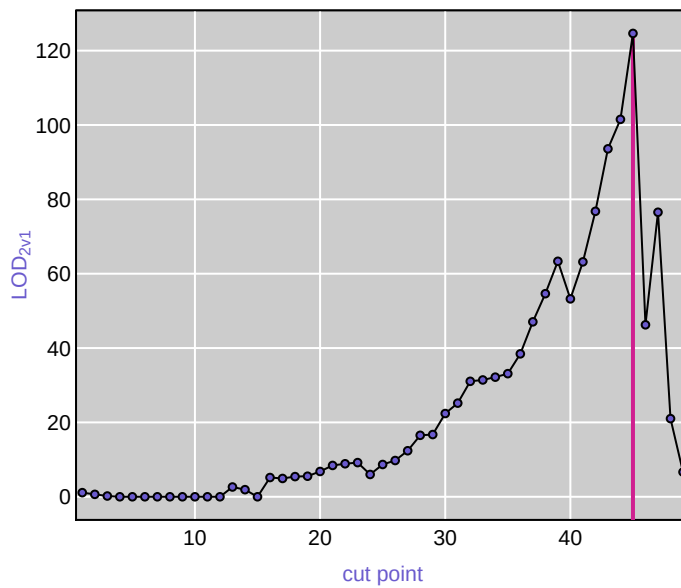
LDA plot



LOD profile

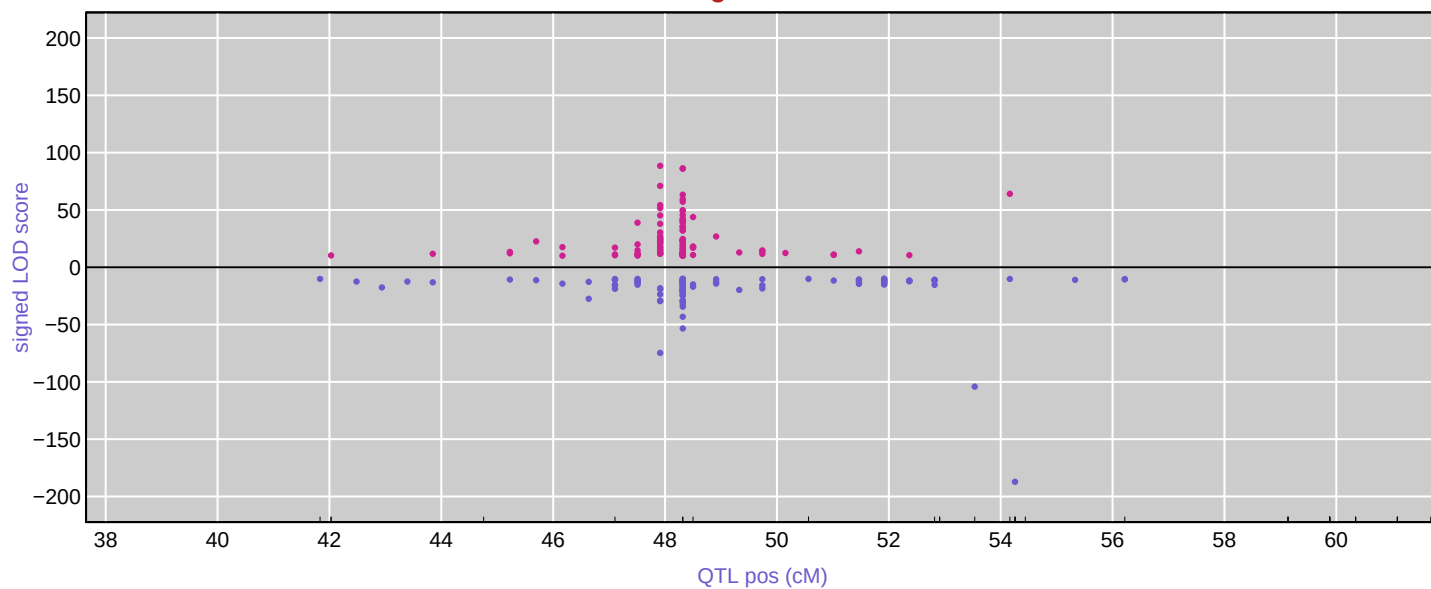


LOD_{2v1} by cut – point

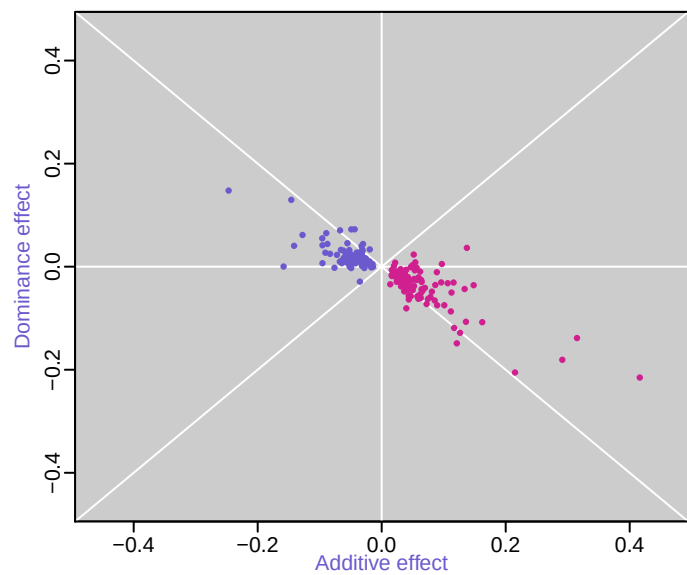


adipose chr10 @ 48 p-value=0

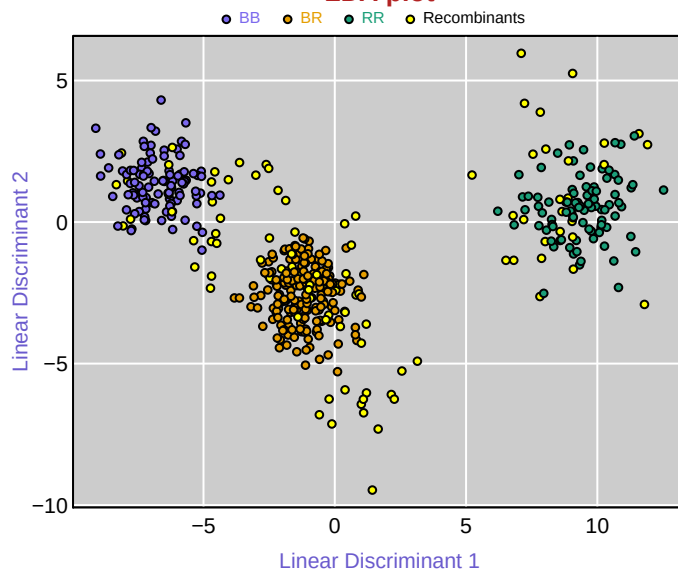
signed LOD



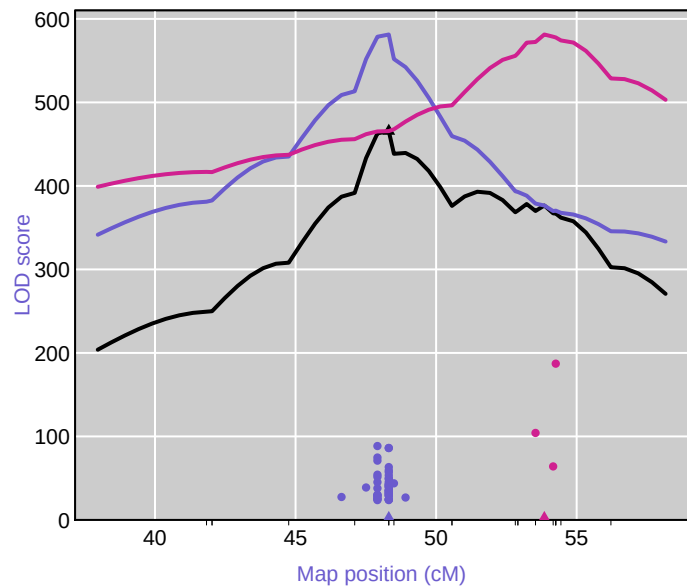
Inheritance Pattern



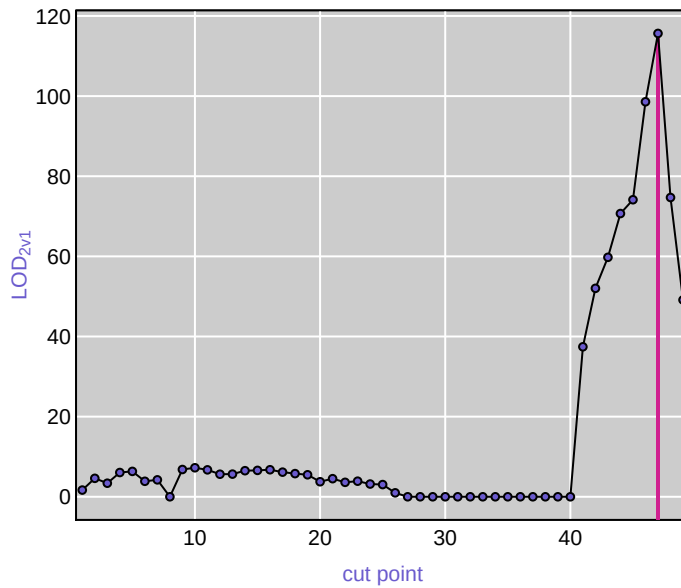
LDA plot



LOD profile

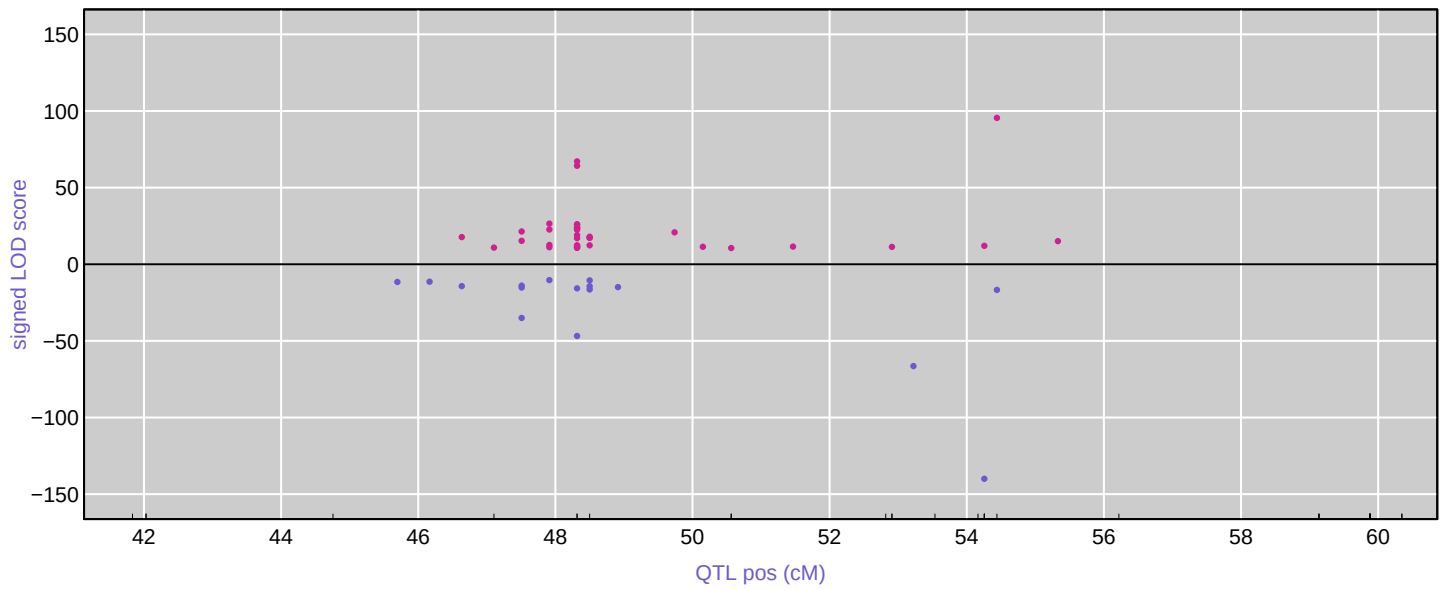


LOD_{2v1} by cut – point

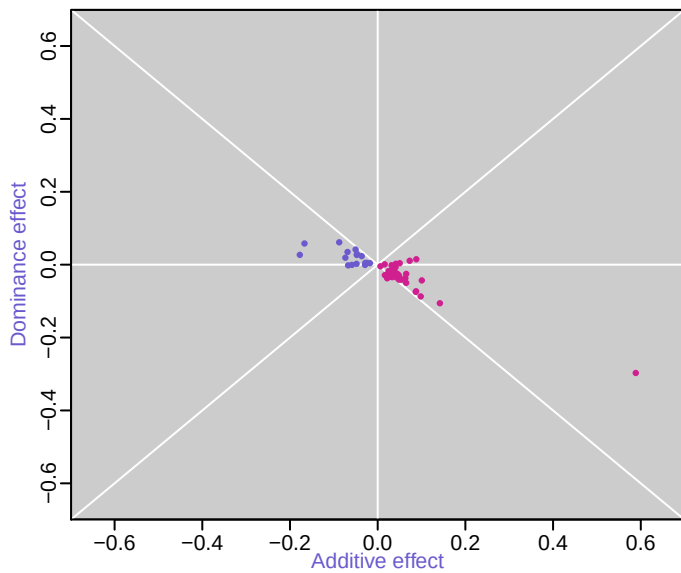


liver chr10 @ 50 p-value=0

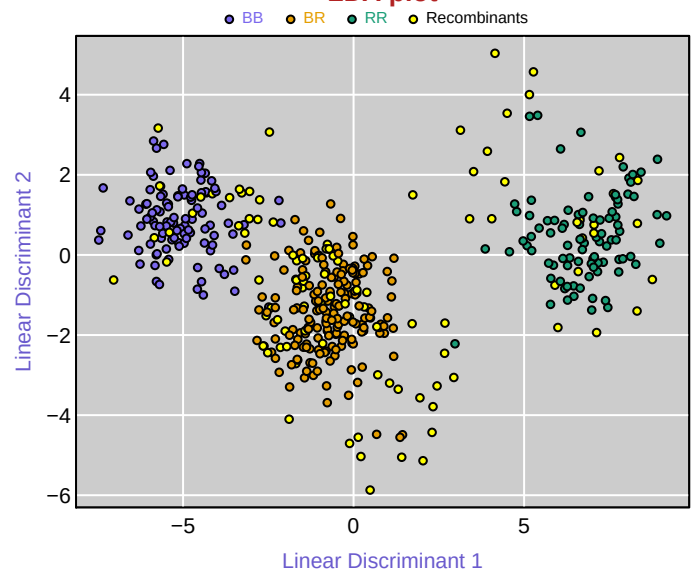
signed LOD



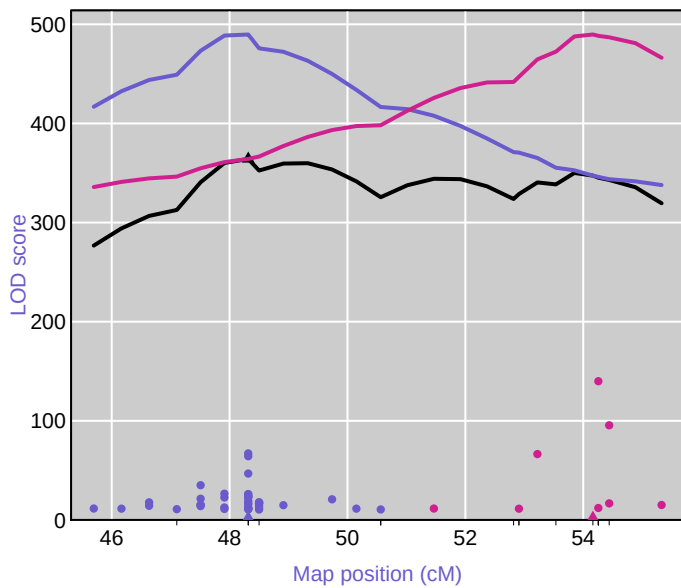
Inheritance Pattern



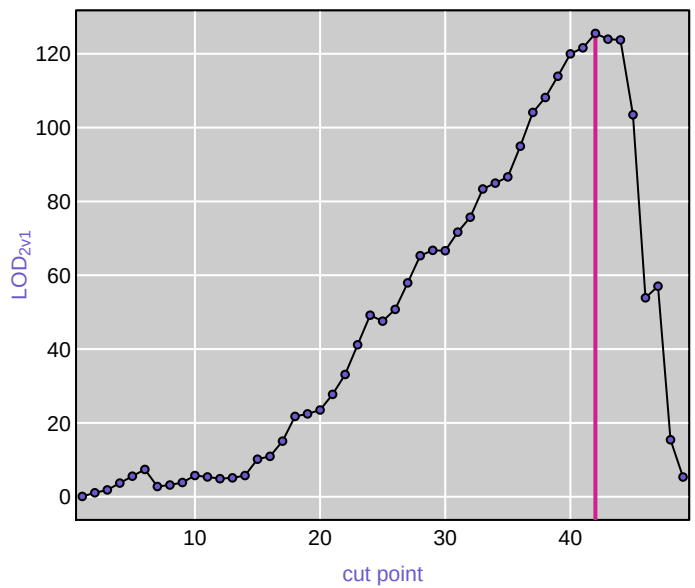
LDA plot



LOD profile

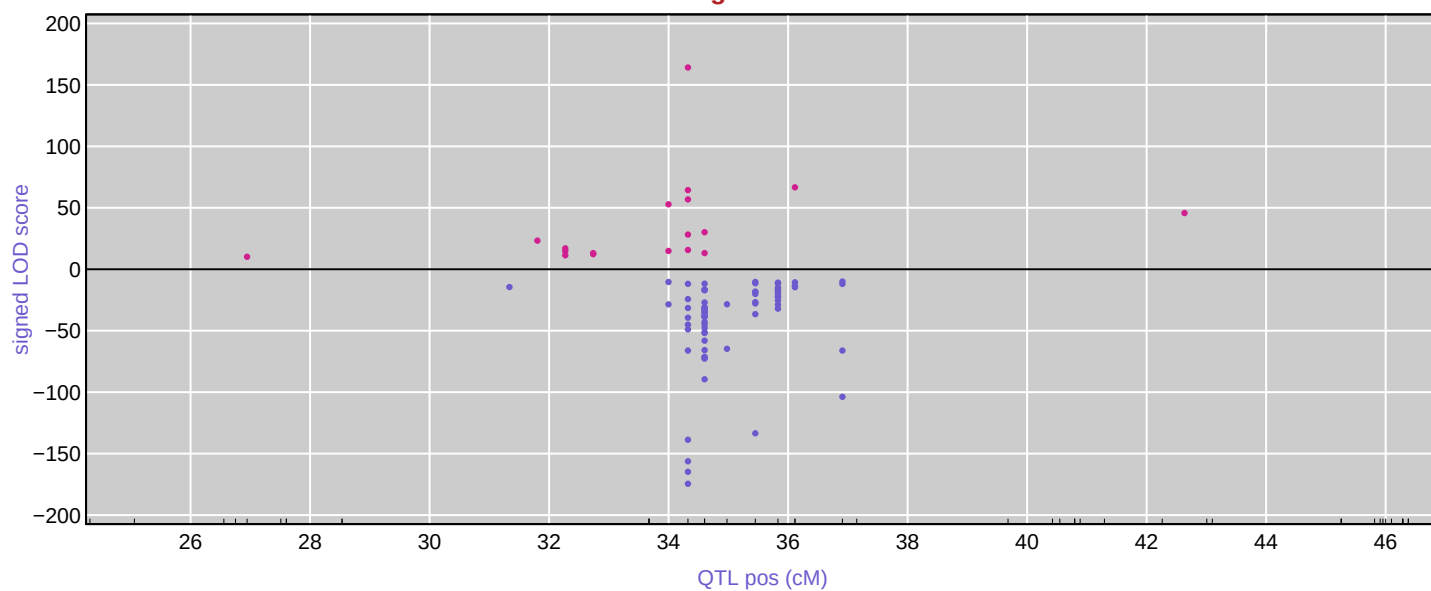


LOD_{2v1} by cut – point

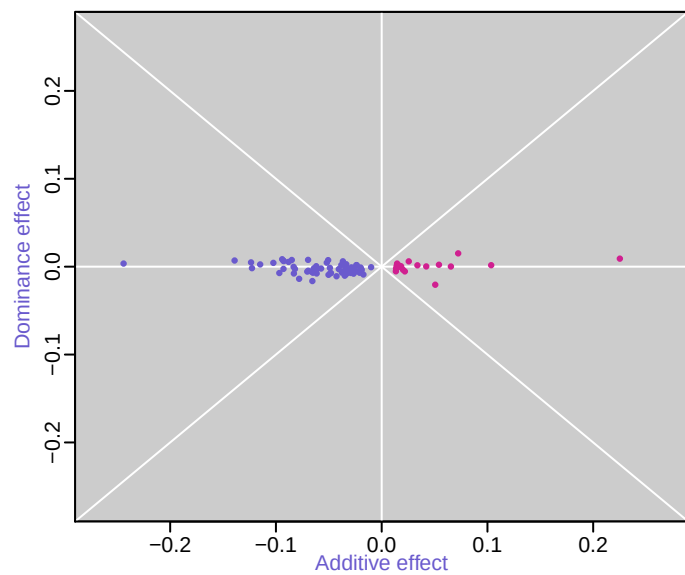


adipose chr11 @ 34 p-value=0

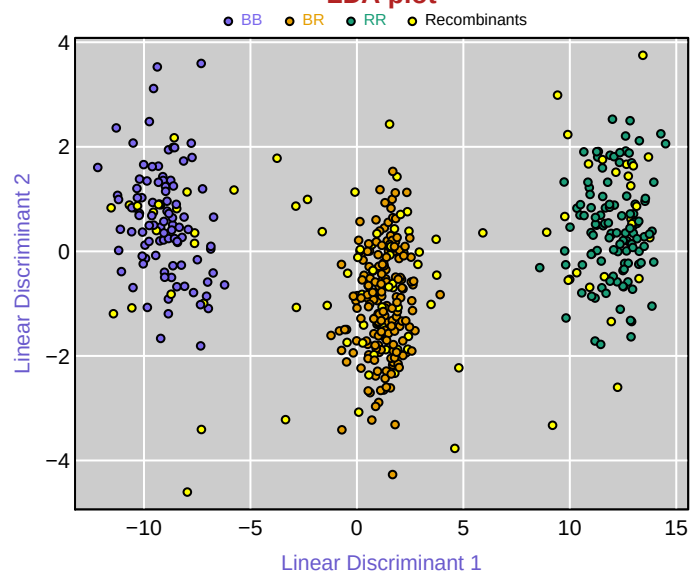
signed LOD



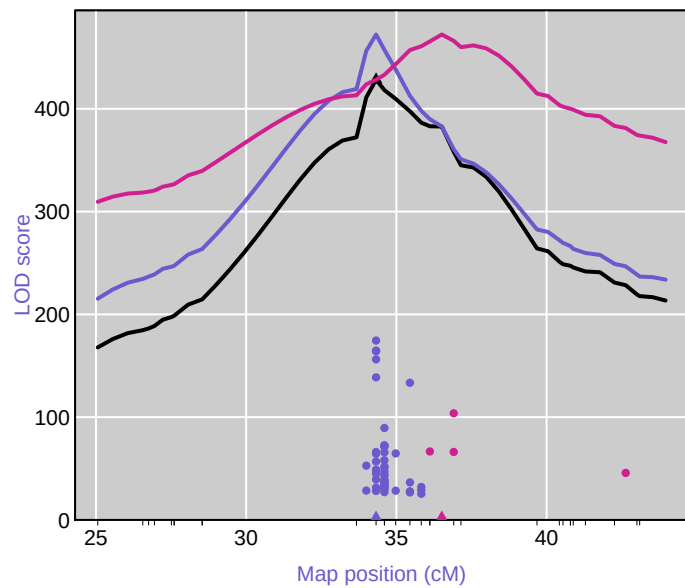
Inheritance Pattern



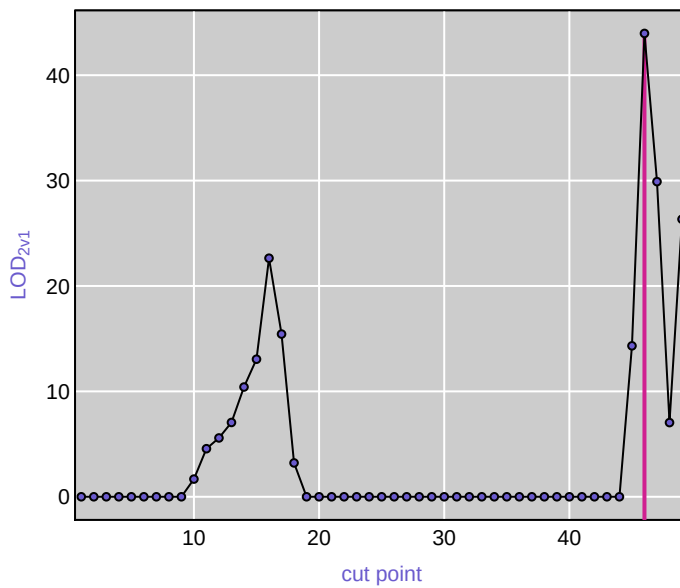
LDA plot



LOD profile

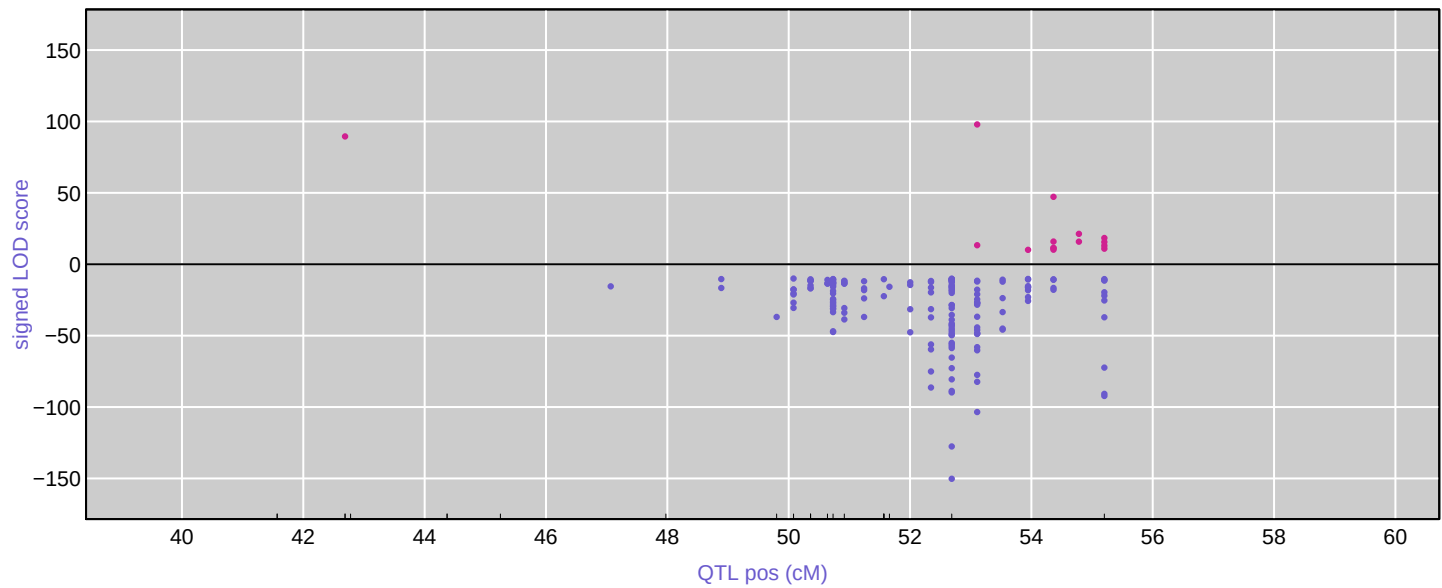


LOD_{2v1} by cut - point

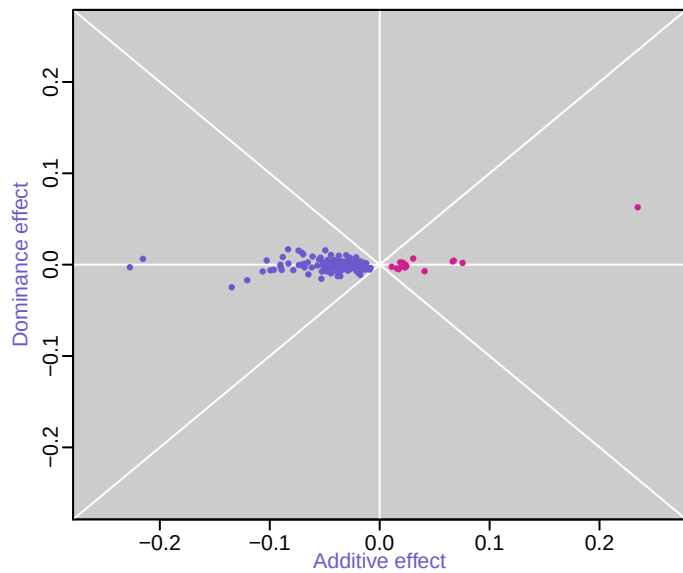


liver chr12 @ 52 p-value=0

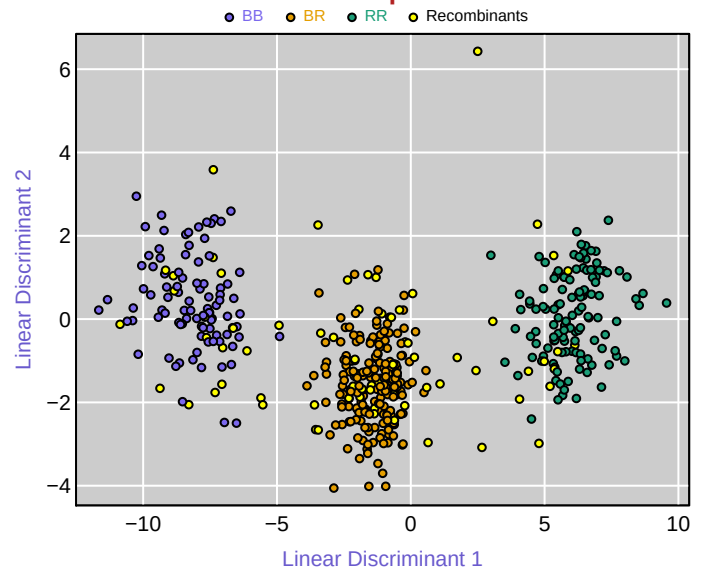
signed LOD



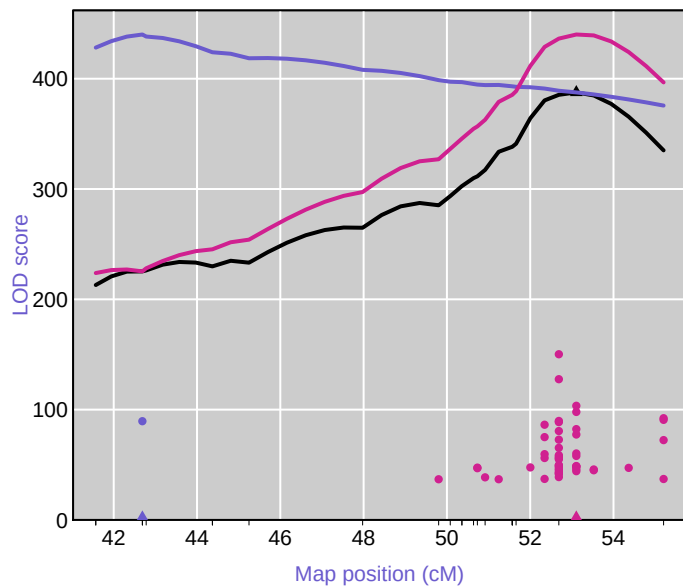
Inheritance Pattern



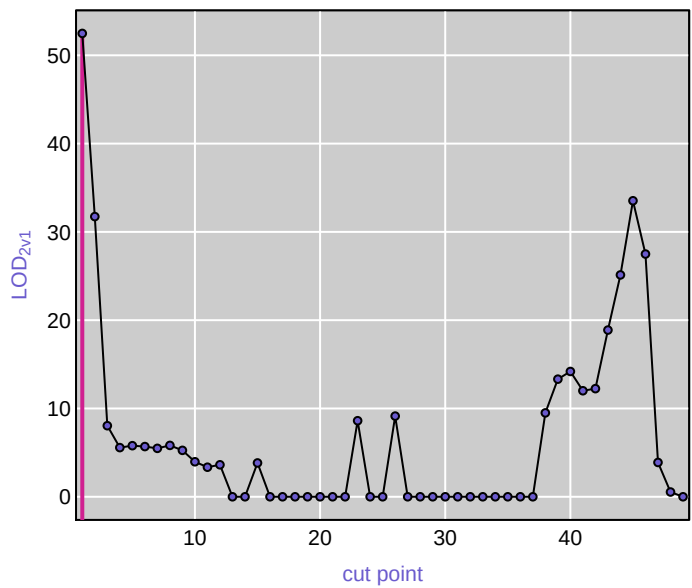
LDA plot



LOD profile

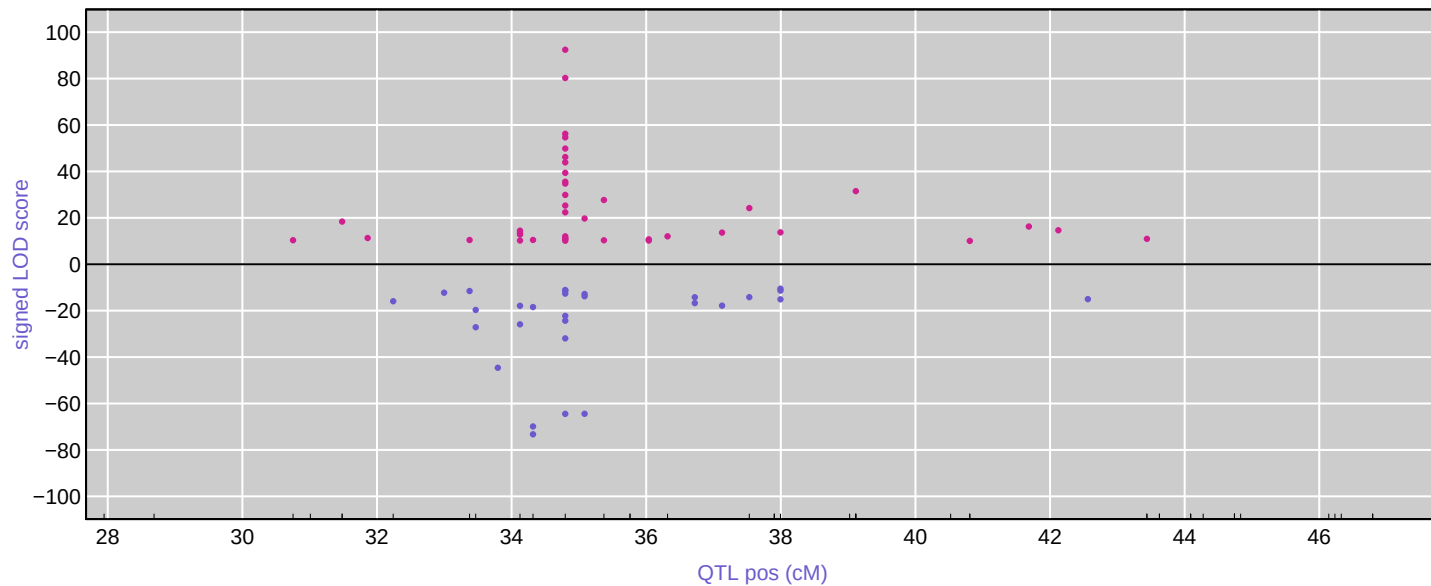


LOD_{2v1} by cut – point

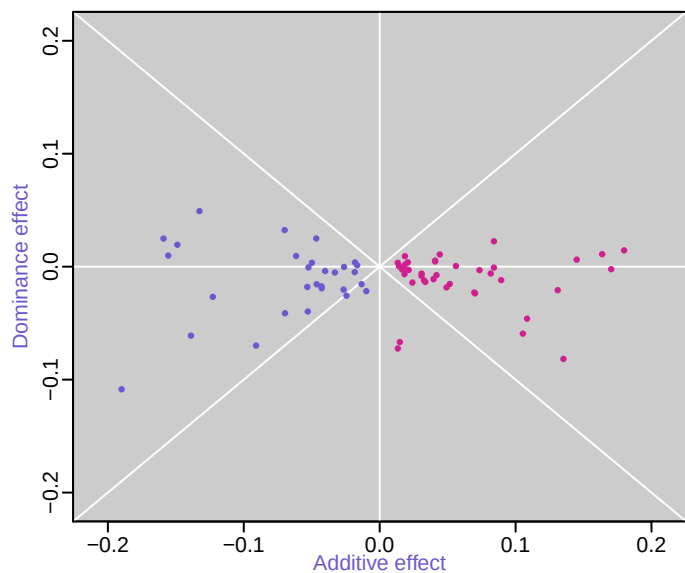


islet chr13 @ 34 p-value=0

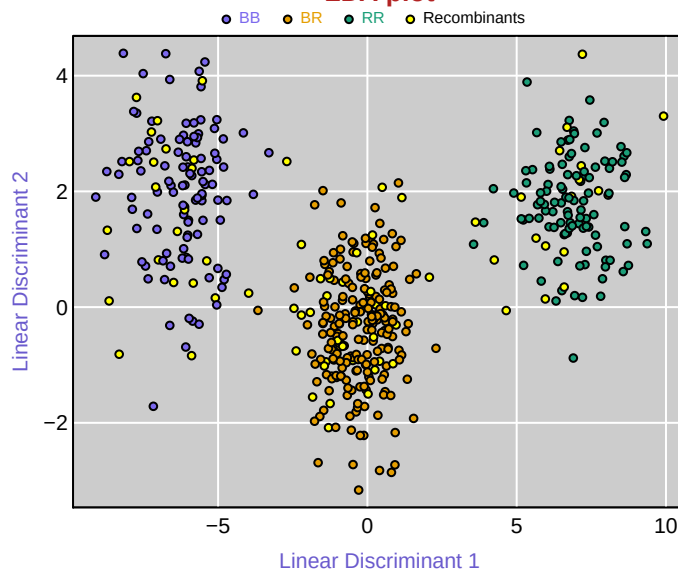
signed LOD



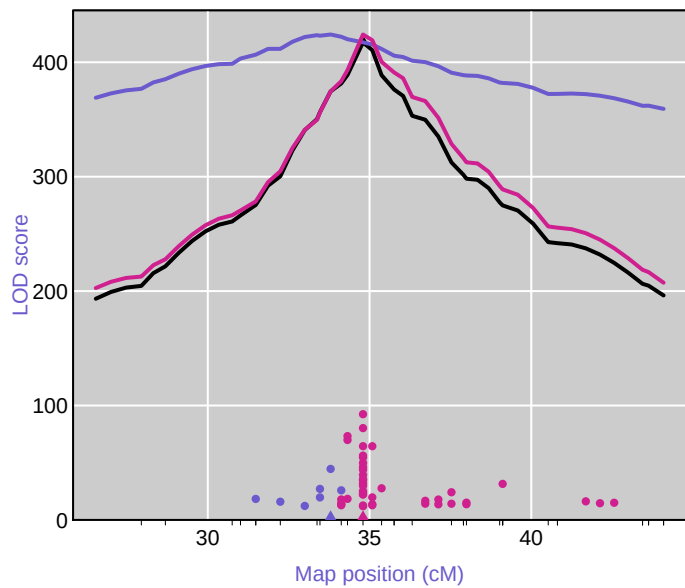
Inheritance Pattern



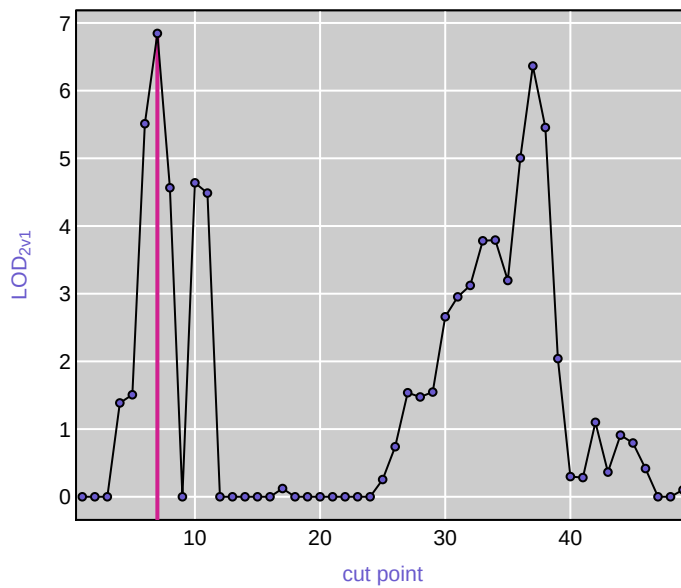
LDA plot



LOD profile

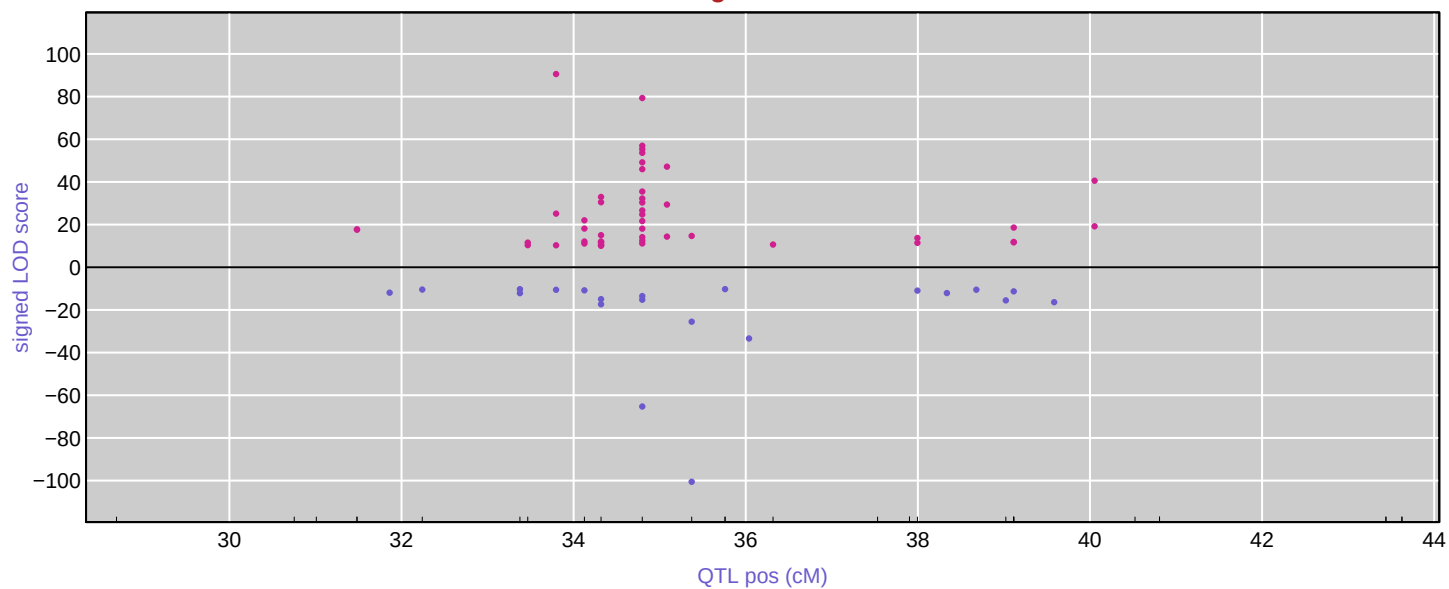


LOD_{2v1} by cut - point

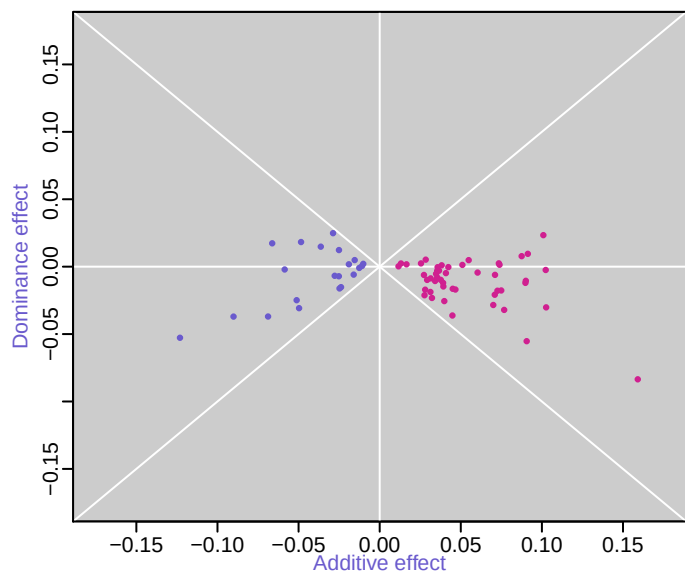


kidney chr13 @ 34 p-value=0

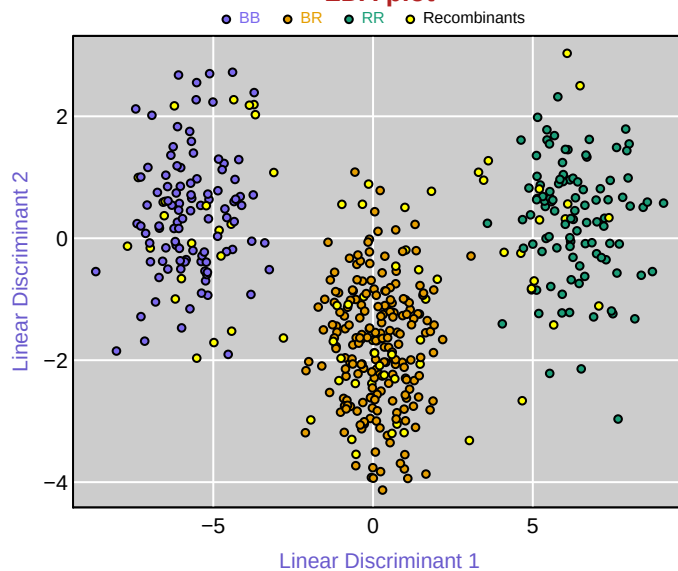
signed LOD



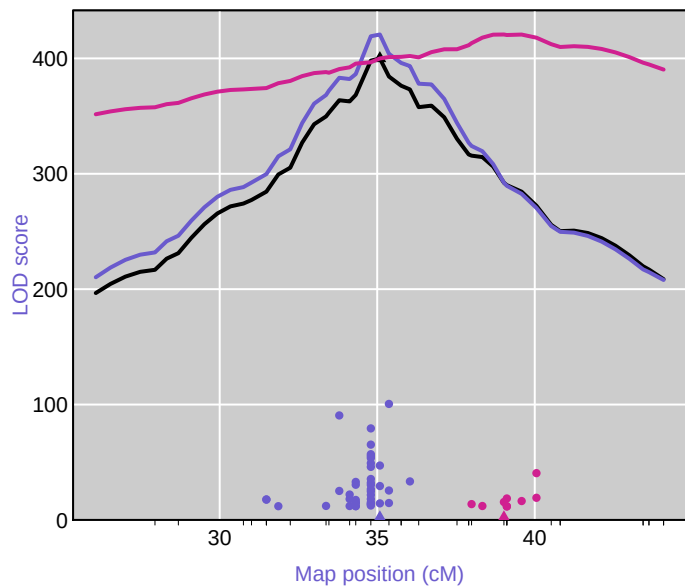
Inheritance Pattern



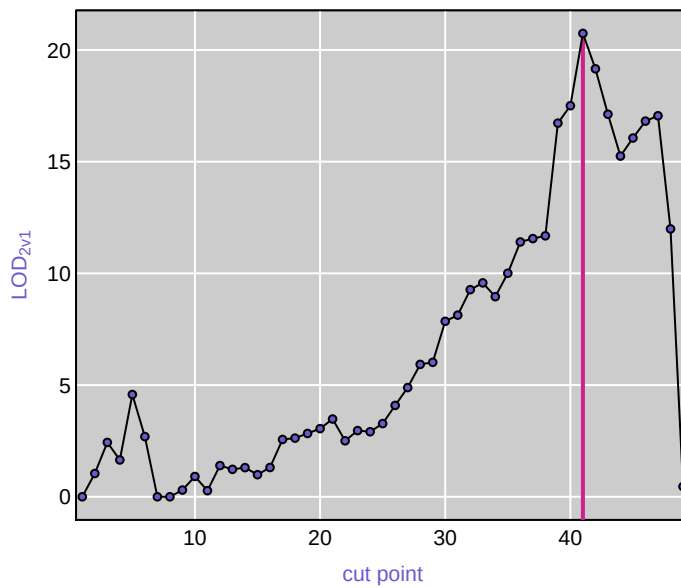
LDA plot



LOD profile

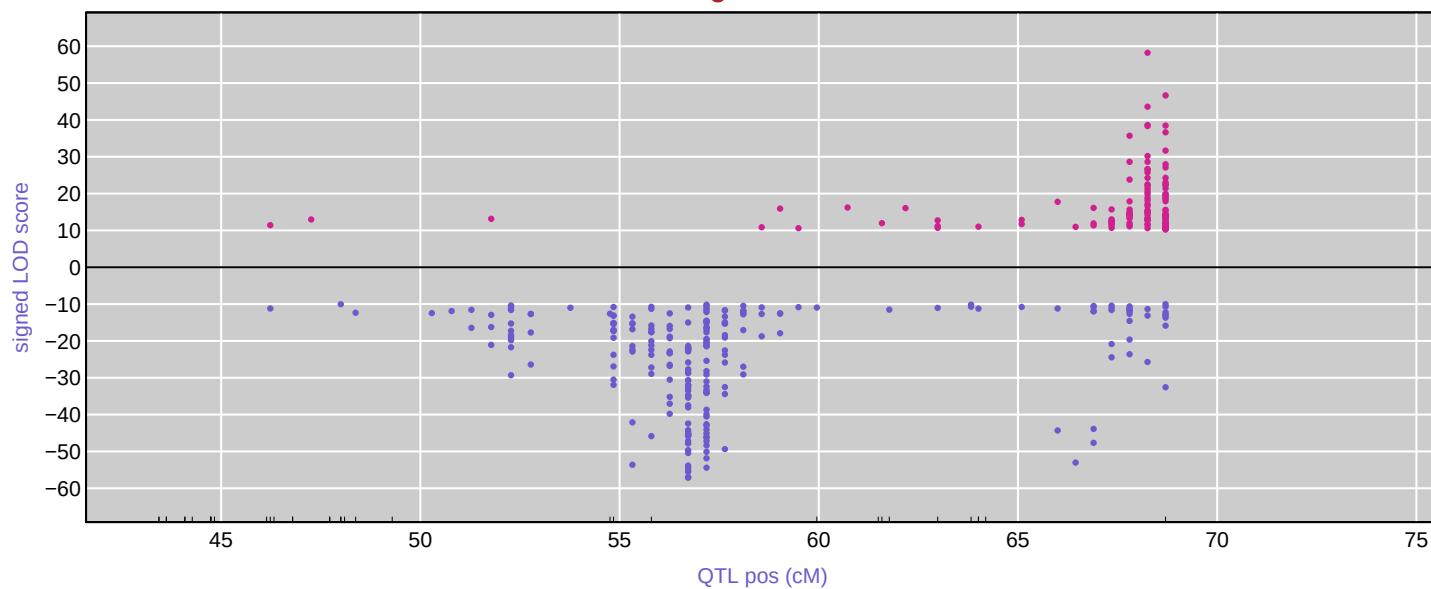


LOD_{2v1} by cut - point

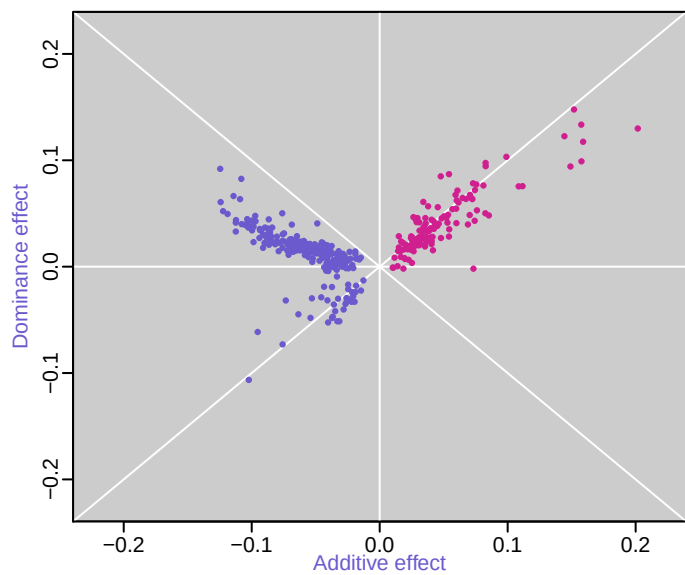


kidney chr13 @ 68 p-value=0

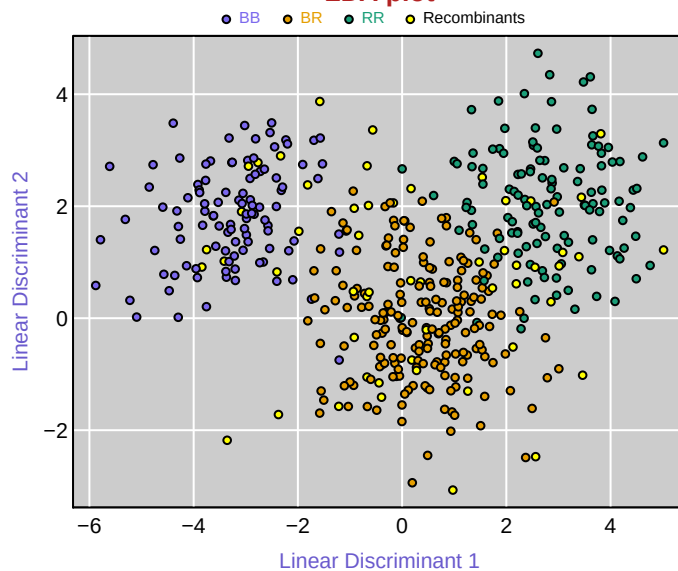
signed LOD



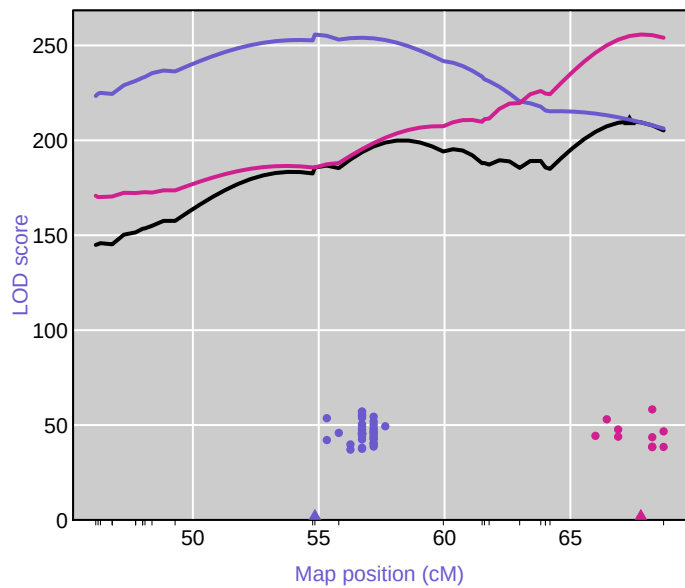
Inheritance Pattern



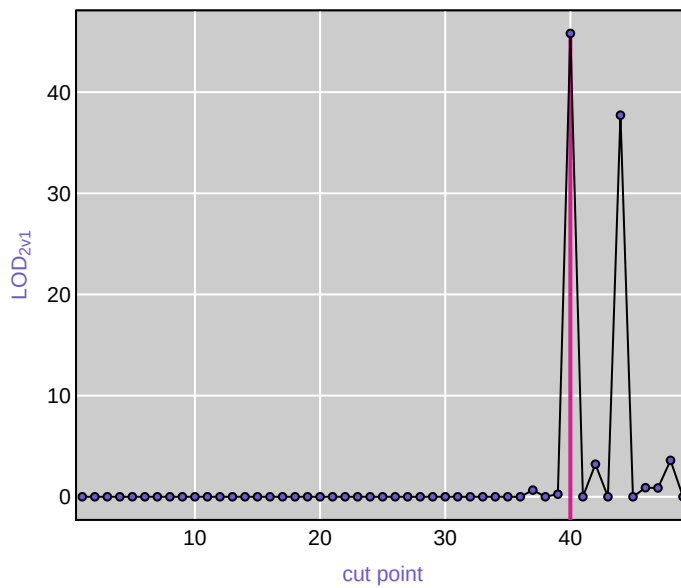
LDA plot



LOD profile

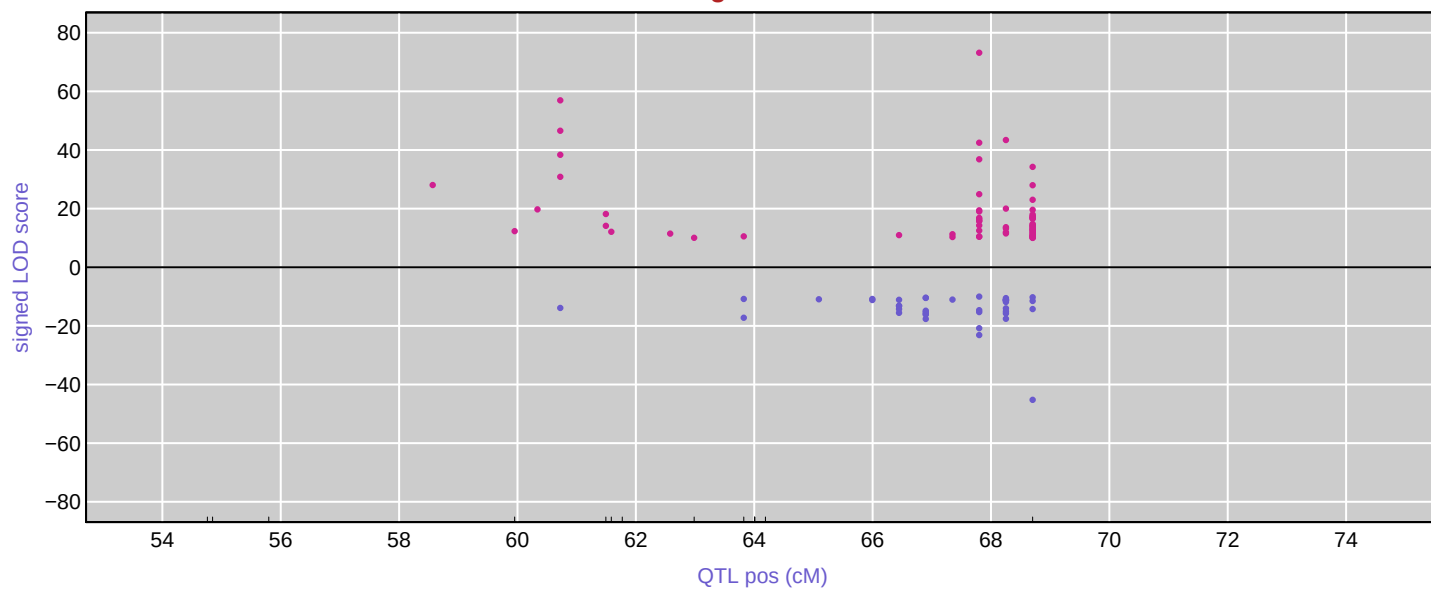


LOD_{2v1} by cut – point

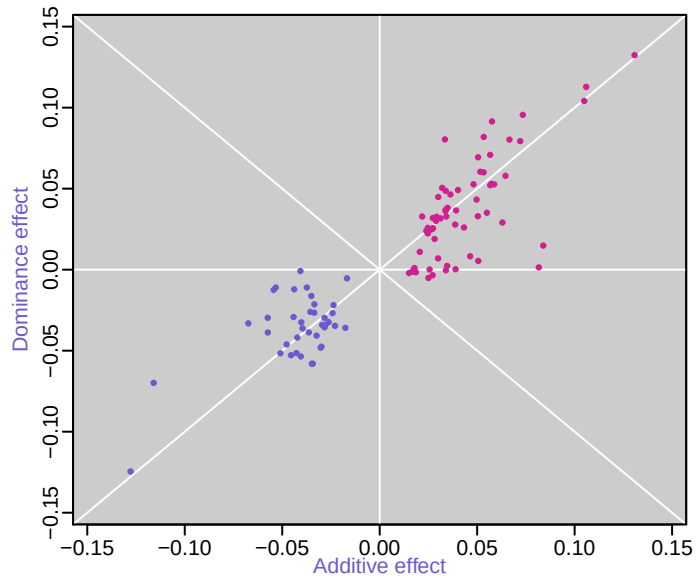


adipose chr13 @ 68 p-value=0

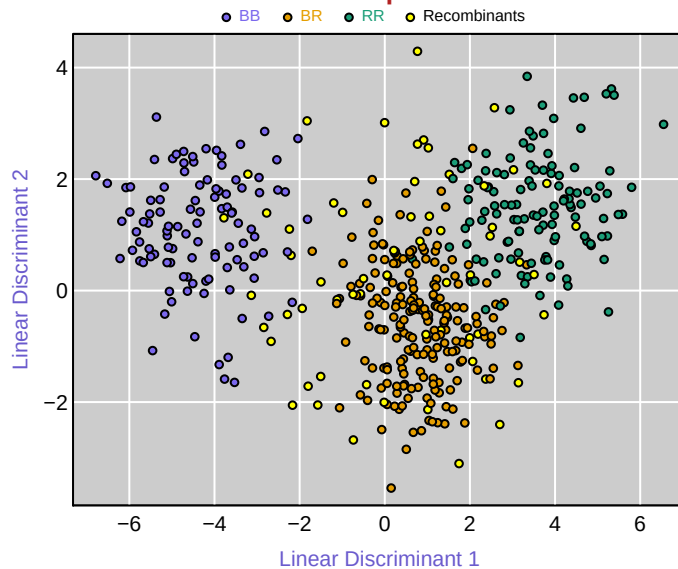
signed LOD



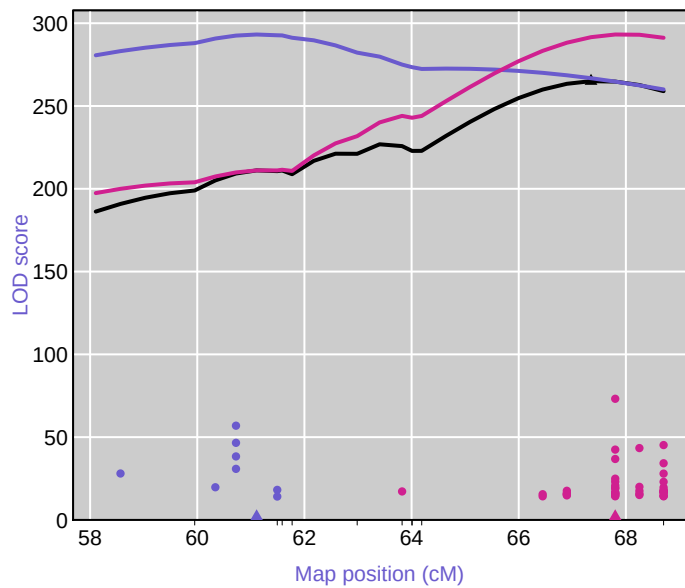
Inheritance Pattern



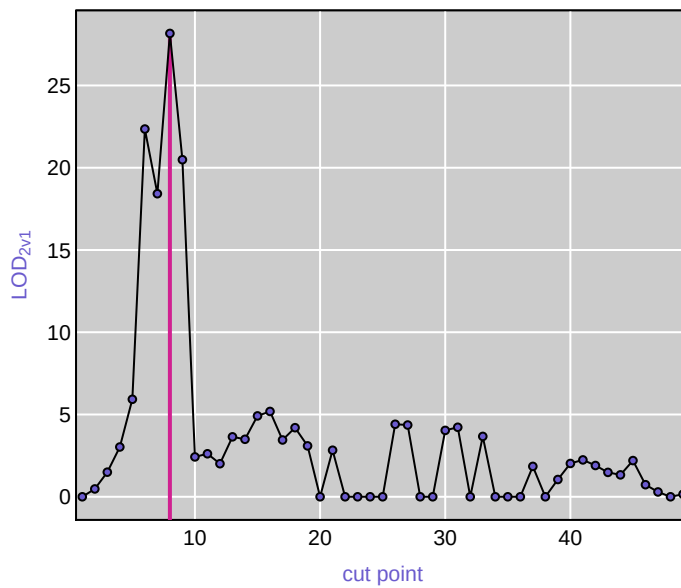
LDA plot



LOD profile

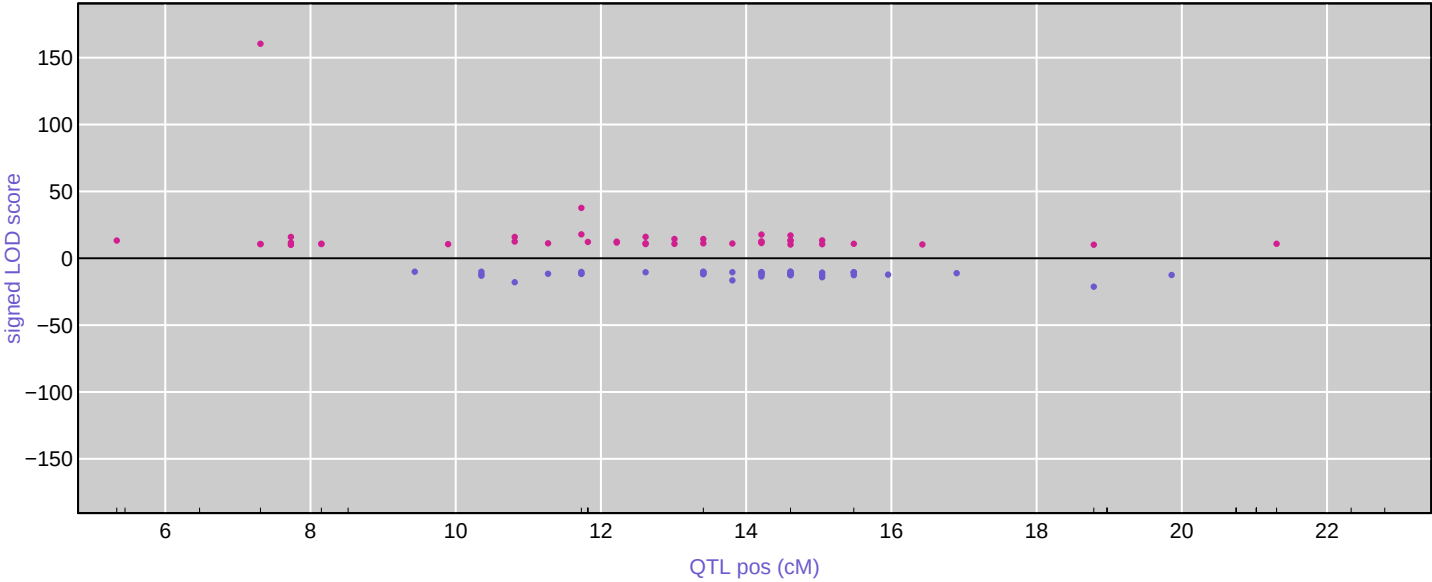


LOD_{2v1} by cut – point

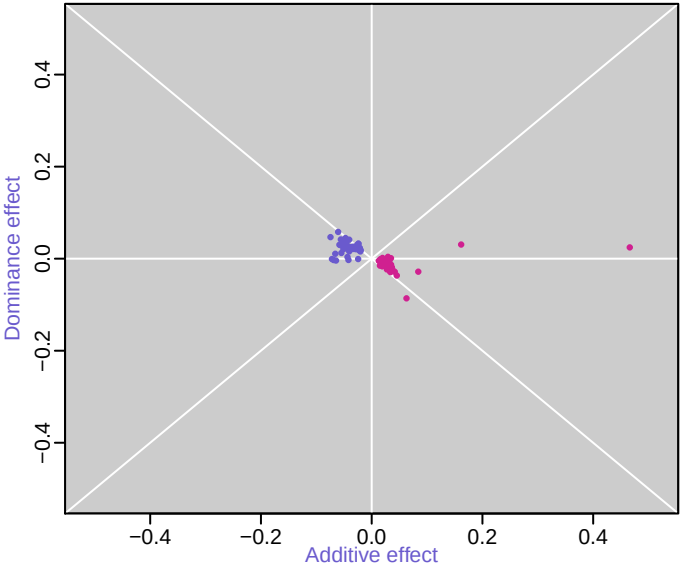


adipose chr17 @ 14 p-value=0

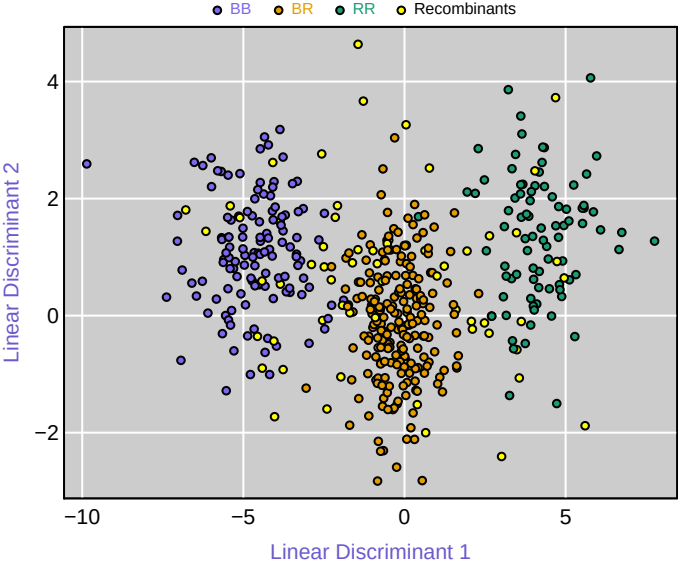
signed LOD



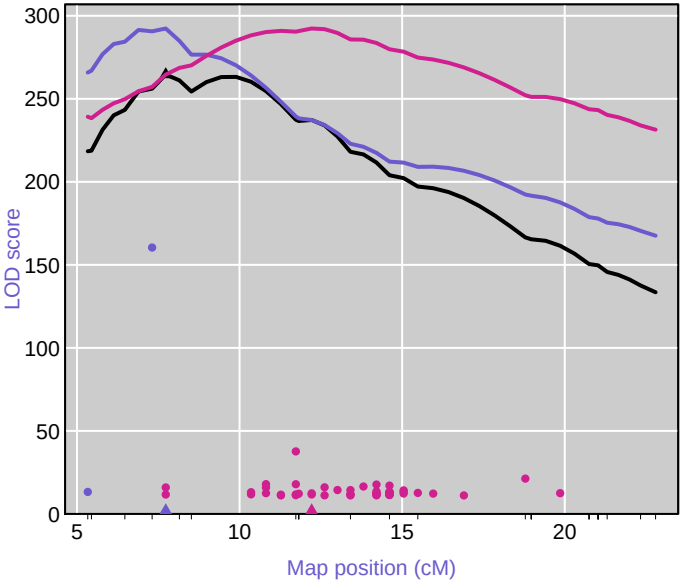
Inheritance Pattern



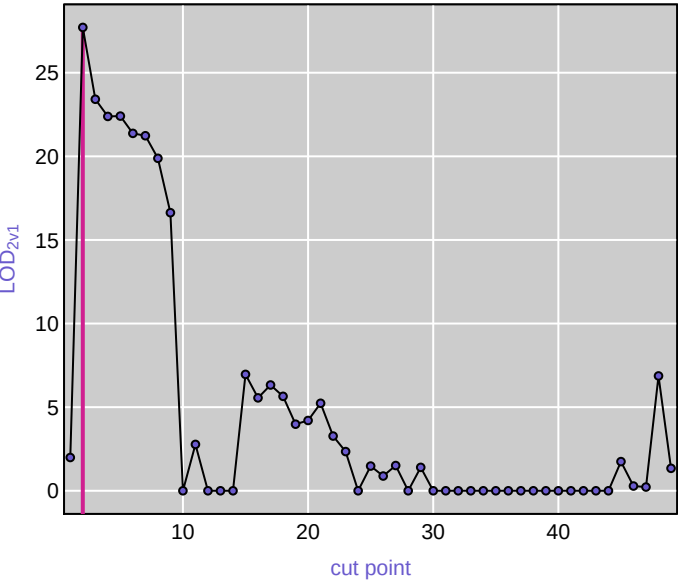
LDA plot



LOD profile

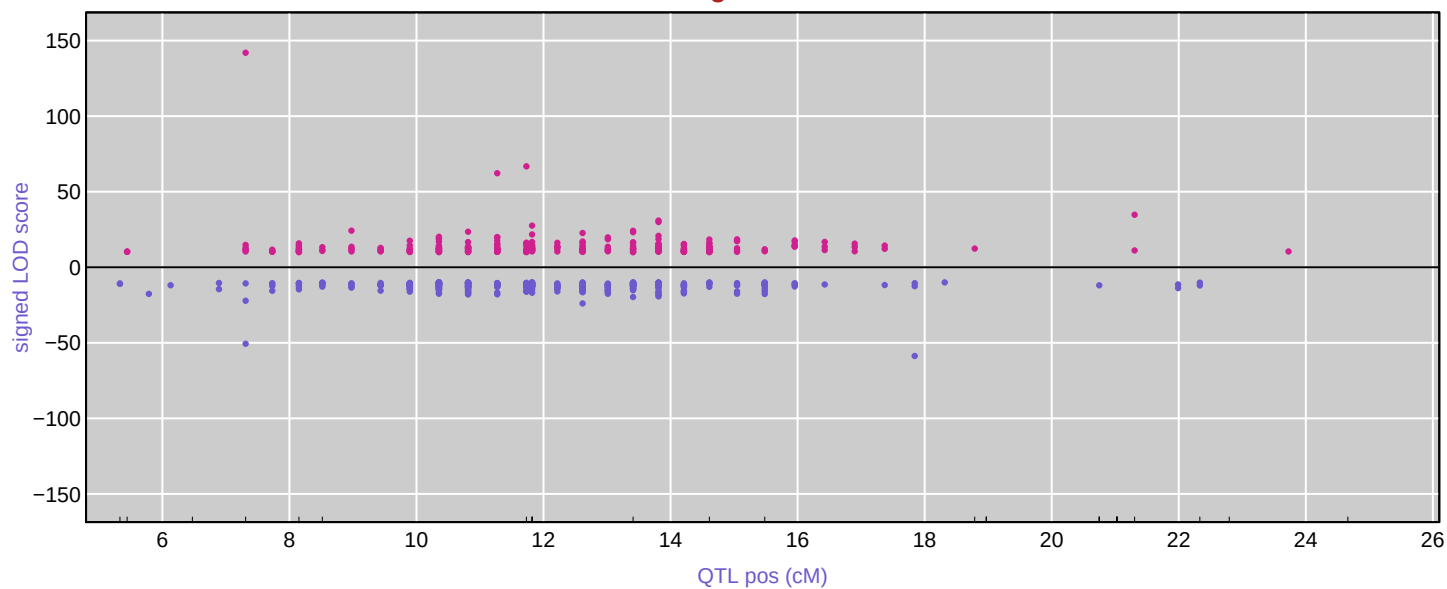


LOD_{2v1} by cut - point

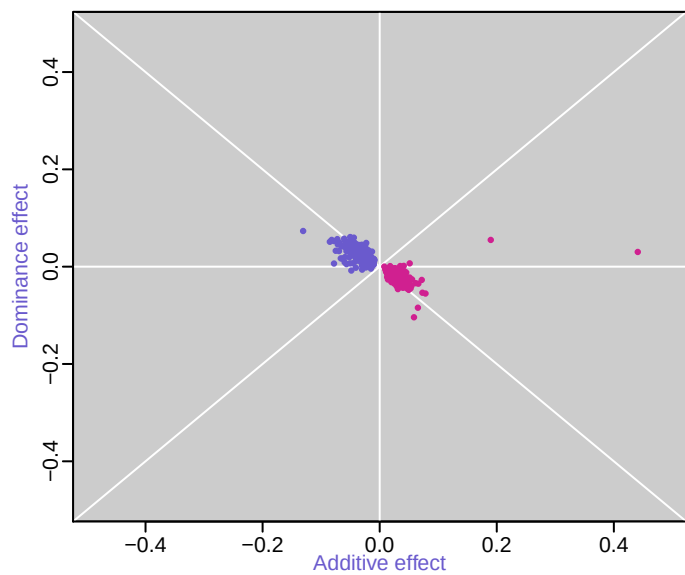


kidney chr17 @ 10 p-value=0

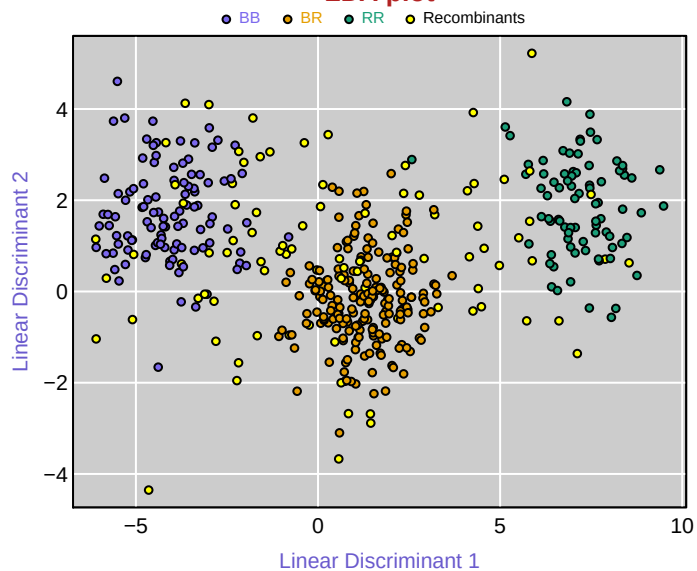
signed LOD



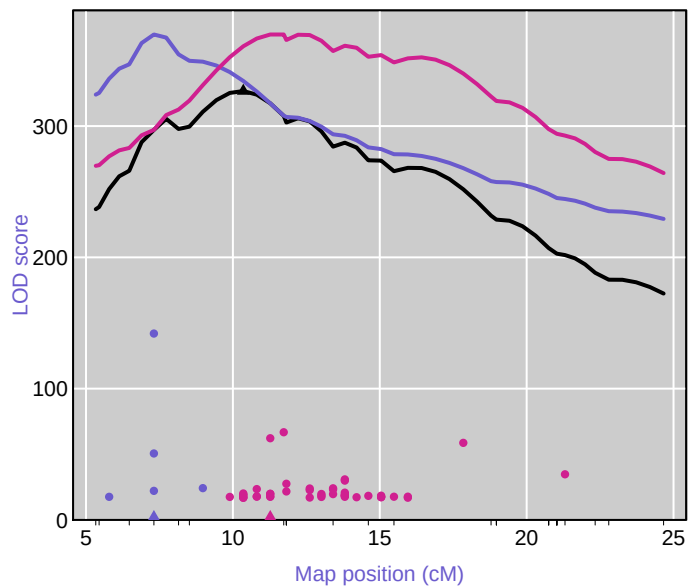
Inheritance Pattern



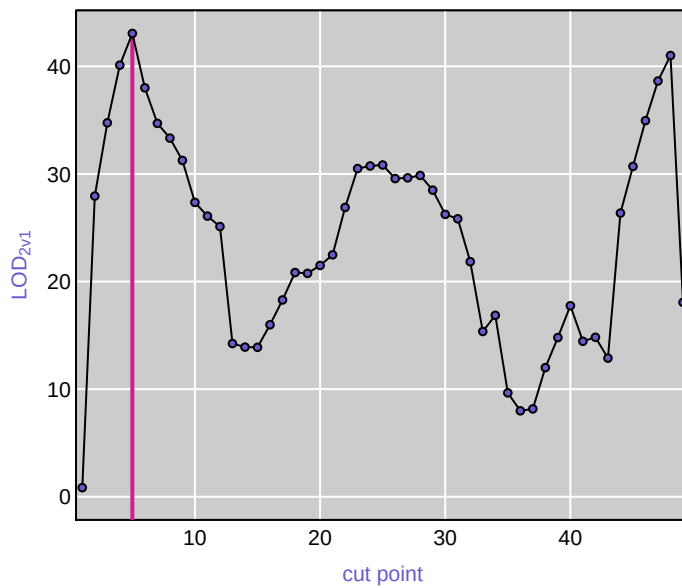
LDA plot



LOD profile

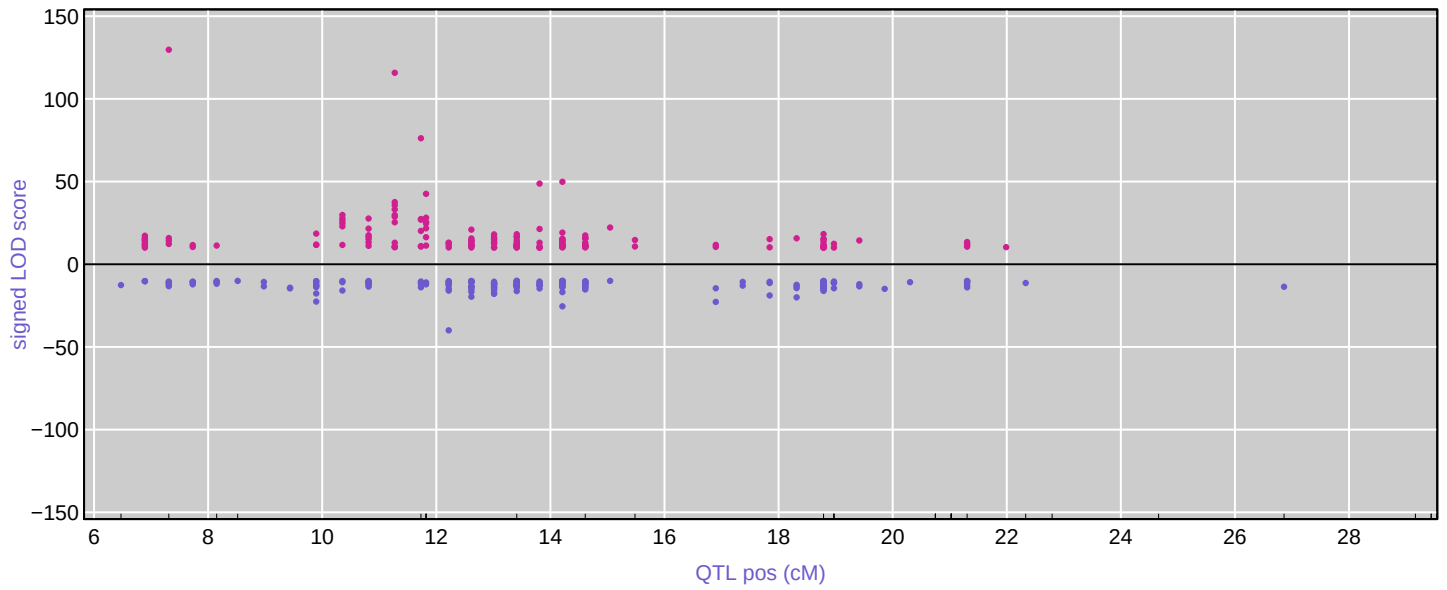


LOD_{2v1} by cut - point

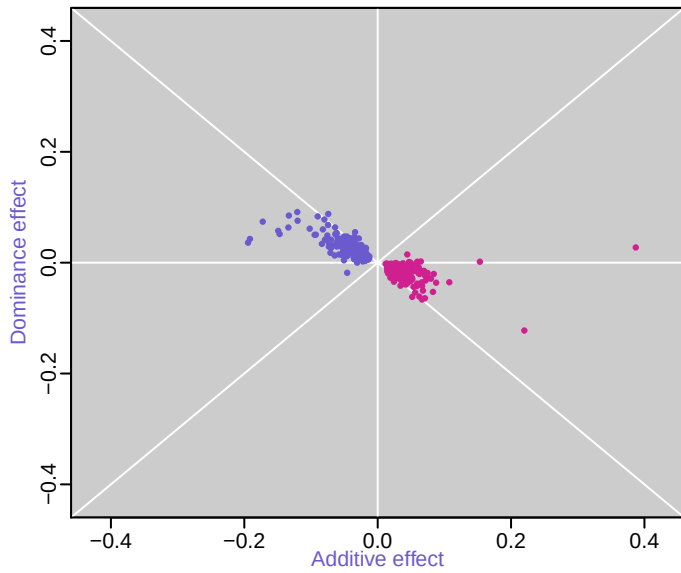


liver chr17 @ 18 p-value=0

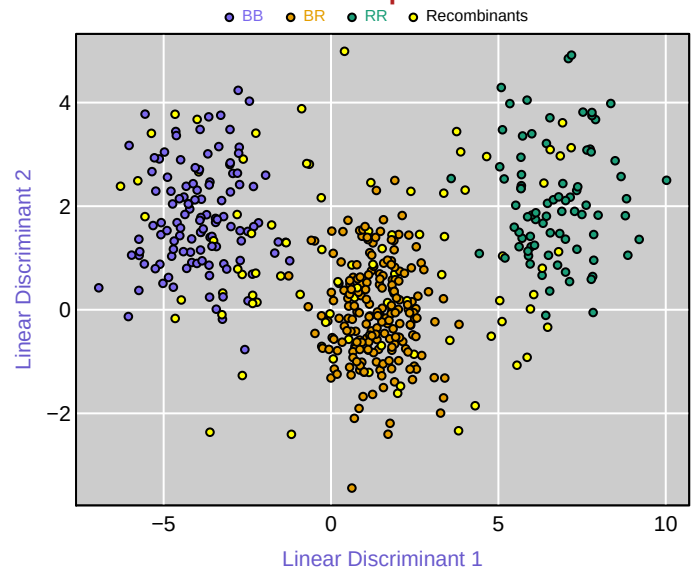
signed LOD



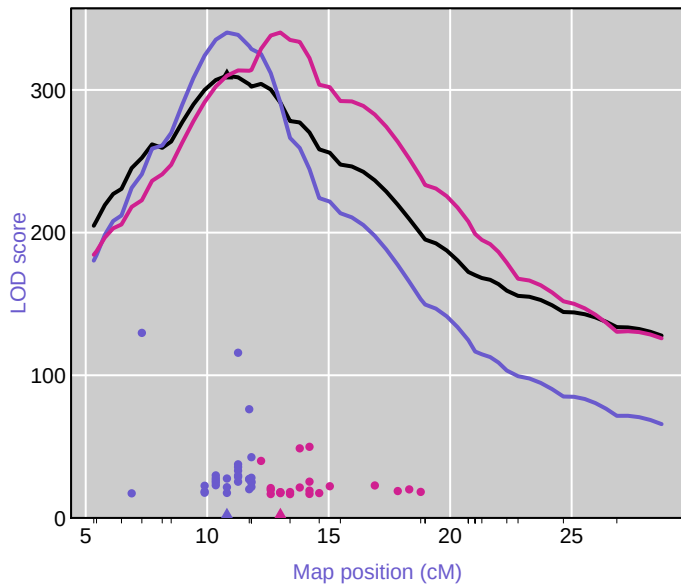
Inheritance Pattern



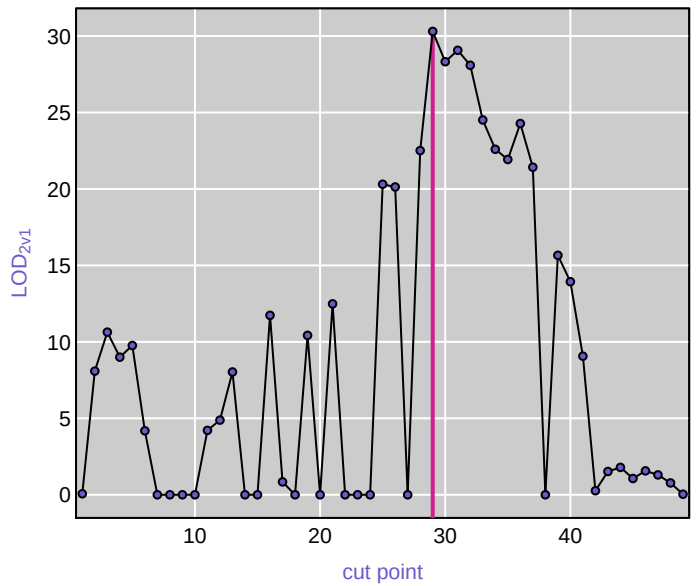
LDA plot



LOD profile

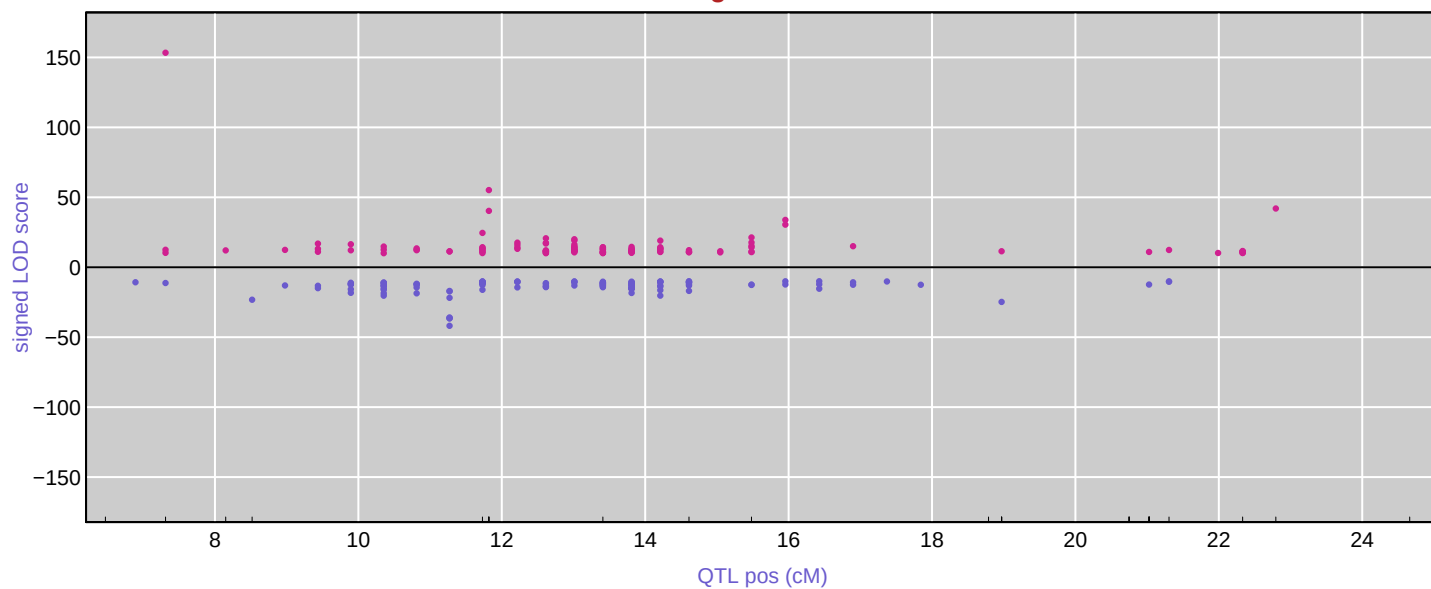


LOD_{2v1} by cut - point

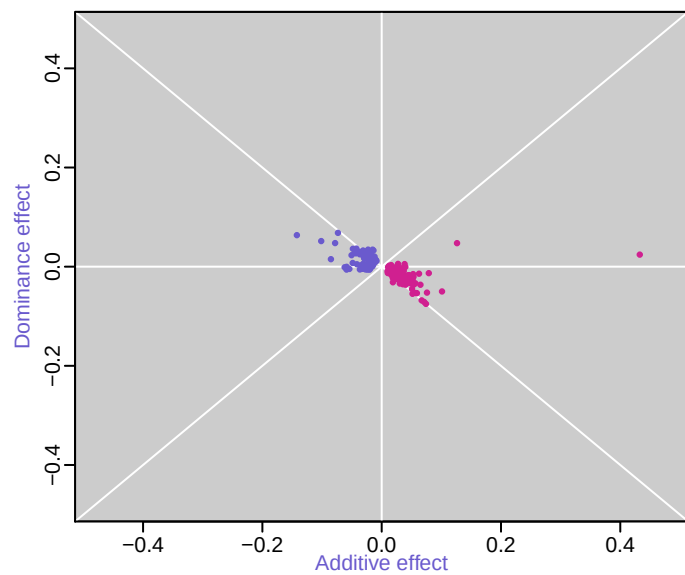


islet chr17 @ 13 p-value=0

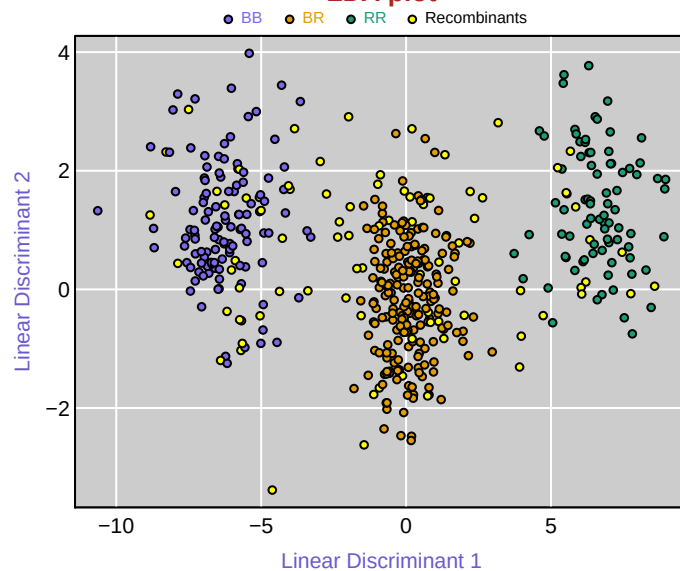
signed LOD



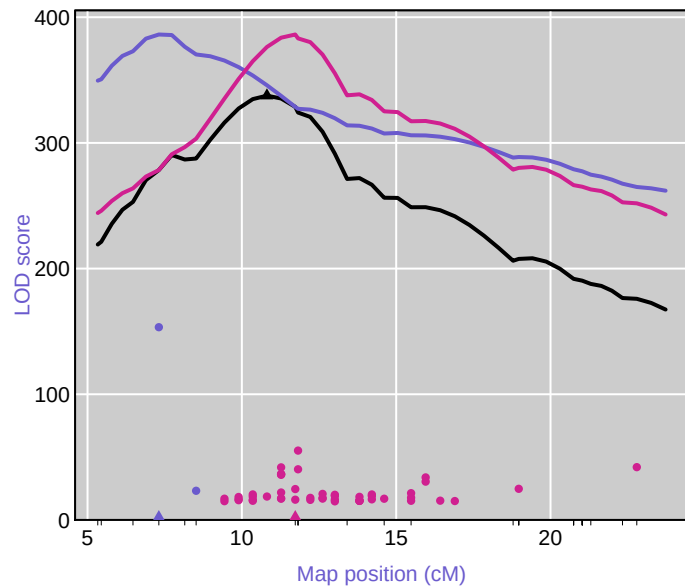
Inheritance Pattern



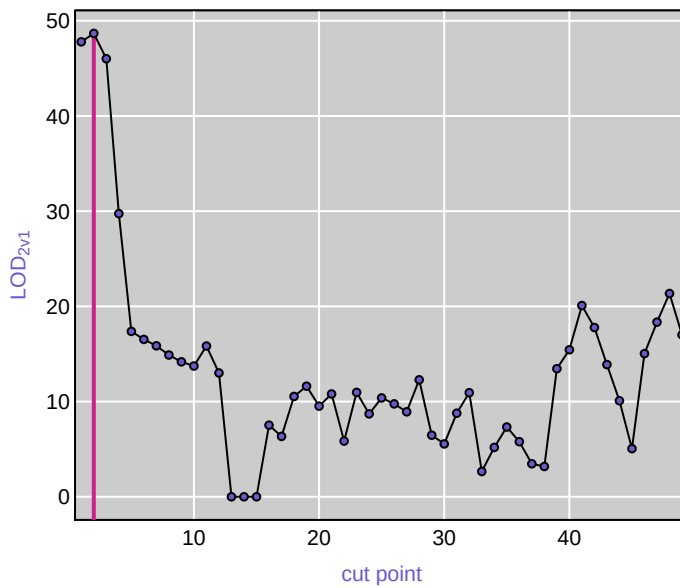
LDA plot



LOD profile

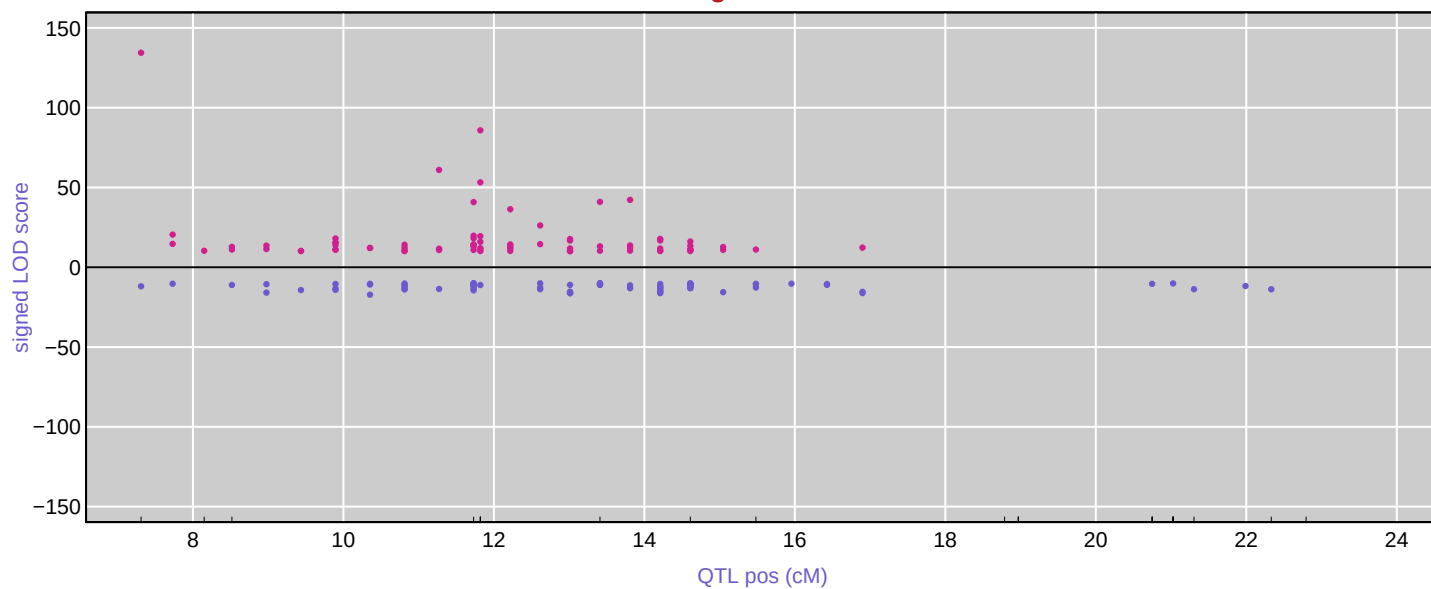


LOD_{2v1} by cut – point

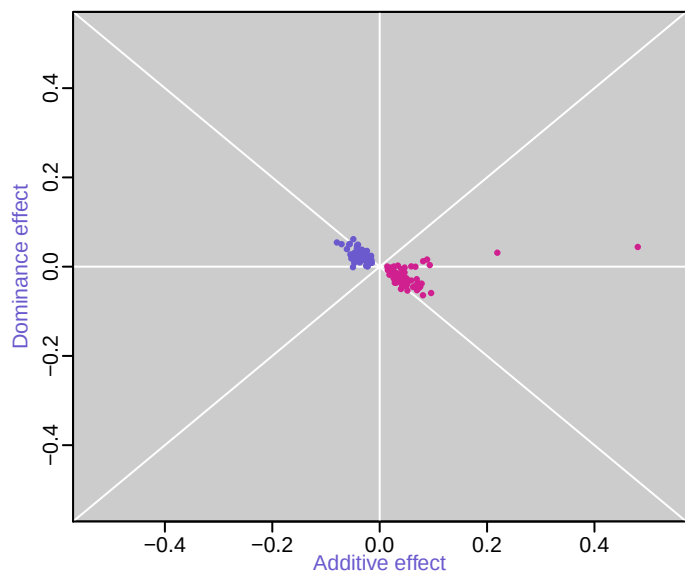


gastroc chr17 @ 14 p-value=0

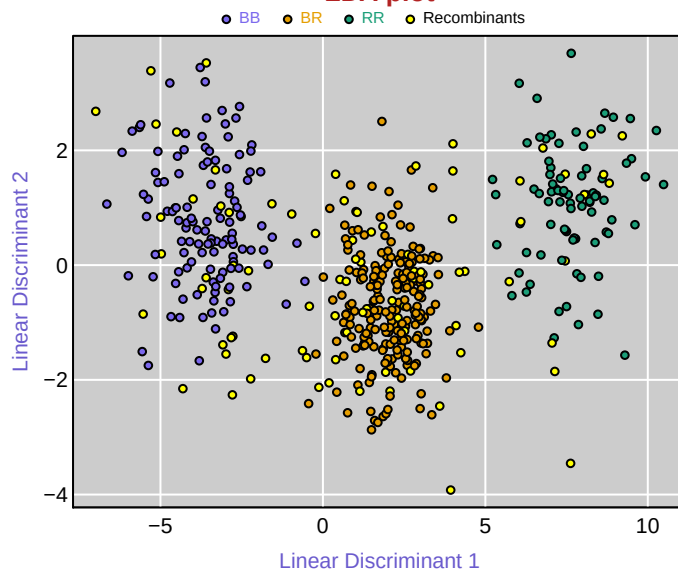
signed LOD



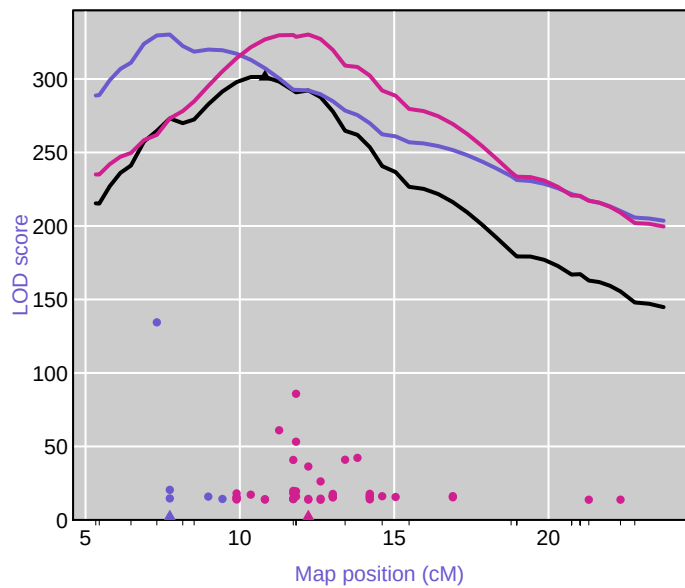
Inheritance Pattern



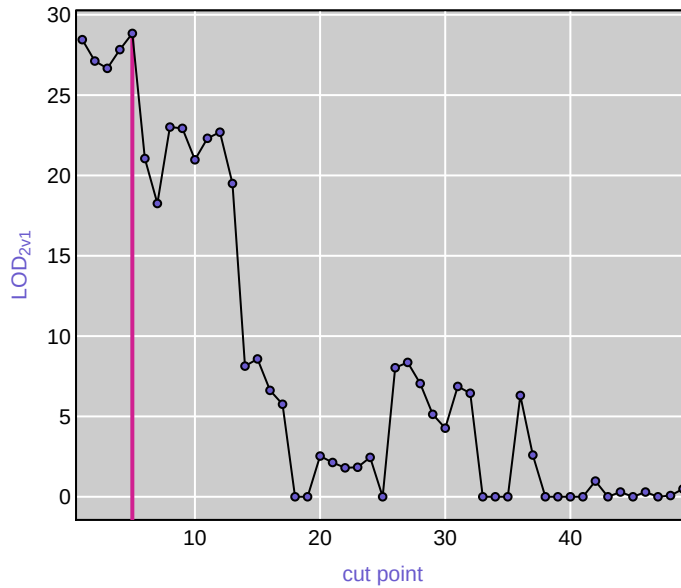
LDA plot



LOD profile

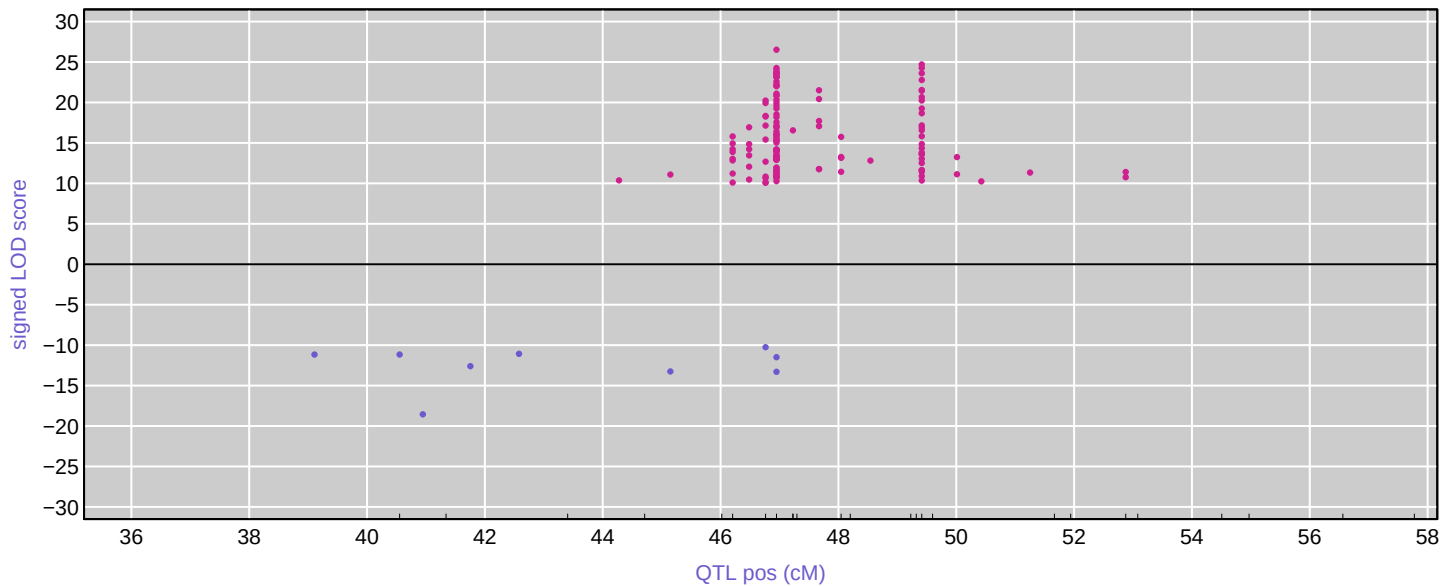


LOD_{2v1} by cut - point

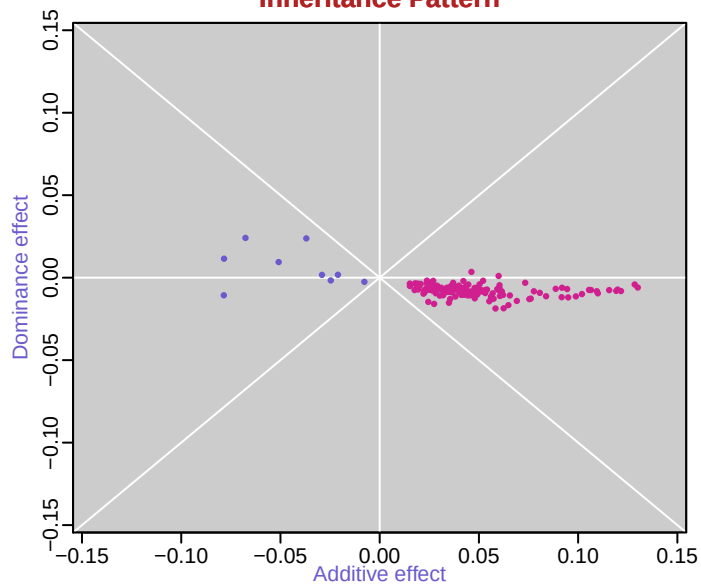


islet chr19 @ 46 p-value=0

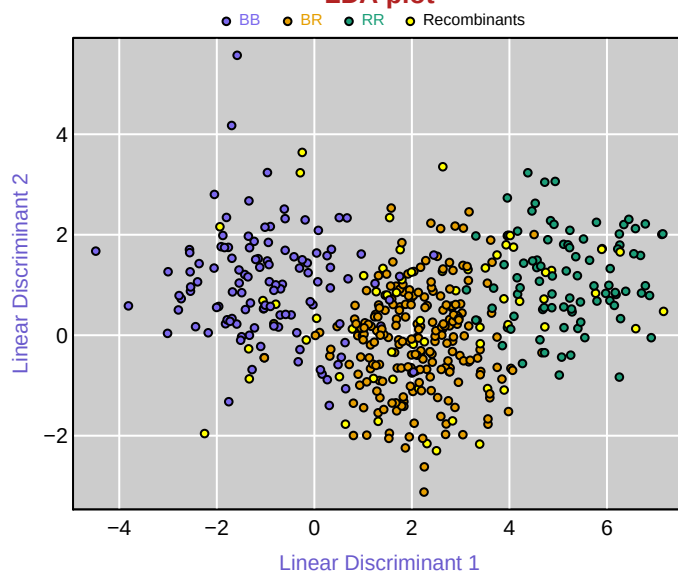
signed LOD



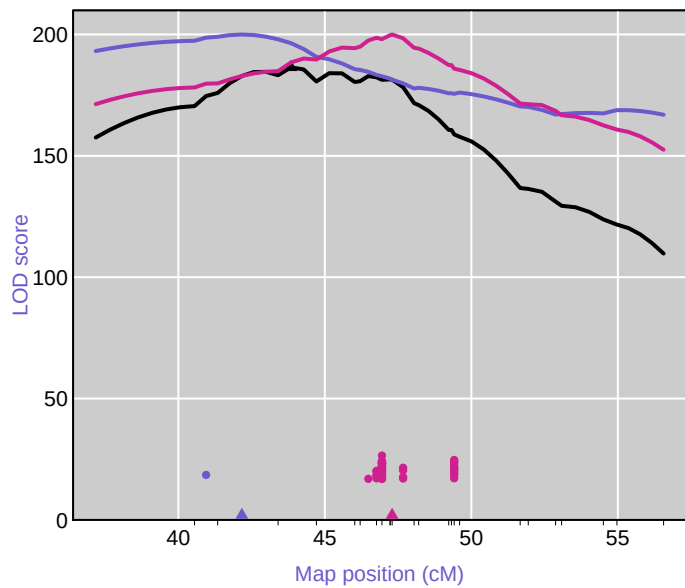
Inheritance Pattern



LDA plot



LOD profile



LOD_{2v1} by cut – point

